

An assessment of five schemes in Karnataka: Report 3

XKDR Forum

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Chapter 1

The project objective

The Government of Karnataka has implemented five welfare schemes, which was started on July 2023. The Schemes are named *Annabhagya*, *Gruhajyothi*, *Gruhalakshmi*, *Shakti* and *Yuvanidhi*. Each scheme is directed towards different citizen audiences. Some are generally applicable, some add to existing schemes, and others target a new and specific audience. The FPI-XKDR Forum project involves the development of an ongoing system of measures to assess the impact of the schemes, using existing datasets such as the CMIE Consumer Pyramids Household Data.

In order to assess the impact, the project seeks to answer three questions about *each* scheme:

- Q1 : What is the intended size of the benefit of the scheme on a household, and has this reached the household?
- Q2 : What is the change in the activities that the household was undertaking before they started receiving benefit, compared to after the benefits from the schemes?
- Q3 : Are the changes *because* of the scheme benefits?

The objective of answering these questions is to provide a feedback loop to the policymaker in terms of both the implementation and the design of the scheme. It helps to answer the following questions: (i) Does the outcome match the purpose of the scheme? (ii) If the observed household behaviour is not what is expected, what are the barriers to achieving the expected outcomes? (iii) Is the pathway to achieving the purpose to review and re-design the scheme, or to identify and remove barriers to achieve the purpose?

The report presents the design of the measurement system used to assess the impact. A detailed description of each scheme was obtained from the Fiscal Policy Institute (FPI), which has coordinated with the relevant departments of the Government of Karnataka in order to make available official documents and material about the design of the five schemes. Additional material has also included details of operationalising the schemes. This material was carefully analysed to infer the *purpose* of each scheme, what is the particulars or the *design* of the scheme, and finally, to identify who among the people or

the household are most likely to benefit from the scheme. Together, these three elements help us to identify what are the likely *benefits* from a specific scheme.¹ Once these likely benefits are identified, it then identifies quantitative or qualitative *measures* to capture the above aspects of well-being of a target household.

In addition, while each scheme is designed to benefit a target household, the impact of the scheme can have indirect effects. Such indirect effects can be at the level of the target households, as well as the community and the overall state. This project seeks to measure both the micro-level response – that of the household – and responses at the level of specific regions of Karnataka – such as northern or southern Karnataka – and the overall state. Put together, the micro and the aggregate effects helps to answer questions Q1–Q3.

This project report presents the preliminary findings for the first two questions listed above for all five schemes. The report is organised as follows: Chapter 2 presents a description of the data used to conduct the assessment of the five schemes, and Chapter 3 describes the approach used to carry out the assessment. The report starts with describing the patterns for government transfers, income and expenditure of households in Karnataka in Chapter 4. This is followed by chapters on each of the five schemes, with a description of the design of the scheme, the possible household responses to the schemes, and an empirical analysis of the household responses using select measures from the CMIE CPHS database. Chapter 5 presents this analysis for the Annabhagya scheme, Chapter 6 has the analysis for the Gruhajyothi scheme, Chapter 7 has the analysis for the Gruhalakshmi scheme, Chapter 8 has the analysis for the Shakti scheme and Chapter 9 has the analysis for the Yuvanidhi scheme.

¹In the ideal case, it is useful to identify the channels through which the benefits accrue to the household. For example, the scheme may serve to increase the well-being and resilience of the target household or community directly, or it may increase the ability of the same to engage in specific economic activities and transactions, which in turn, increases the well-being of the household.

Chapter 2

Measurement of scheme impact: data sources

The development of the system comprises compiling quantitative measures about the well-being and economic activity of the persons who are the intended beneficiaries. Since the system aims to be an ongoing measure, this requires using data sources that provide information about individuals and households in Karnataka, updated on a regular basis.

Such information is available in the Consumer Pyramids Households Survey (CPHS) published by the Centre for Monitoring Indian Economy (CMIE).¹ The CPHS database has a panel dataset of 174,000 households collected by survey, which is conducted once in every four months. CMIE refers to this frequency as a ‘*wave*’. The first wave of information is available from January 2014. The CPHS provides a wide array of socio-economic variables (153 variables in all) that capture information, which is either aggregated at the level of the household or available on the individuals within the household. These are organised as four separate datasets of household information. They cover information related to household income, household expenditure which is available monthly (four times in a wave), and other information such as features of the household such as age, occupation, health and financial status which is available once every wave.

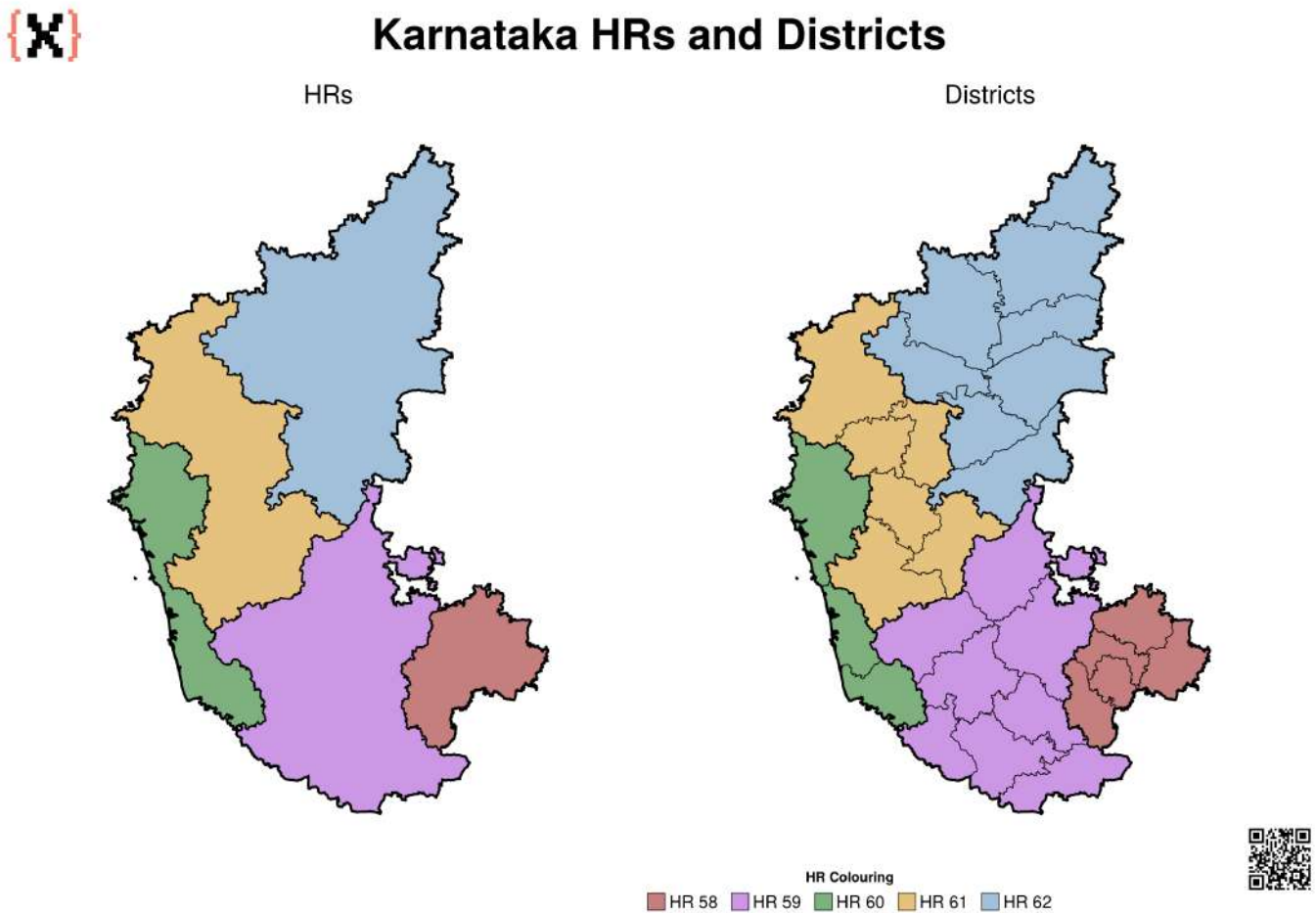
Such a dataset, which is collected independent of policy interventions provide a unique perspective into behaviour of the households. This can help understand how the availability of a scheme to the household shapes changes in the household behaviour. CPHS is the first information source for measures used to understand the impact of the five schemes on households in Karnataka.

¹<https://consumerpyramidsdx.cmie.com/>

2.1 Regions studied: Homogeneous Regions (HR) in Karnataka

The CPHS survey is conducted in all states of the country, other than a few smallest states. It is a longitudinal survey where the broadest stratum is referred to as ‘Homogeneous Regions’ (HRs). Figure 2.1 shows the state of Karnataka and the boundaries of five HRs that makes up the state within the CMIE CPHS dataset. The figure also shows the boundaries of the districts within each HR.

Figure 2.1 Two visualisations of Karnataka: HR and HR with district boundaries



Each HR is organised as a collection of districts. Districts within an HR are similar in agroclimatic features, urbanisation and female literacy. Table 2.1 presents the HRs in

Table 2.1 HRs of Karnataka and the districts covered

HR ID	Reference name	Districts contained
HR 58	Bengaluru HR	Bangalore, Bangalore Rural, Chikkaballapura, Kolar, Ramanagara
HR 59	Mysore HR	Chamarajanagar, Chikmagalur, Chitradurga, Hassan, Kodagu, Mandya, Mysore, Tumkur
HR 60	Coastal HR	Dakshina Kannada, Udupi, Uttara Kannada
HR 61	Dharwad HR	Belgaum, Davanagere, Dharwad, Gadag, Haveri, Shimoga
HR 62	Bellary HR	Bagalkot, Bellary, Bidar, Bijapur, Gulbarga, Koppal, Raichur, Yadgiri

Karnataka and the list of districts that are covered by each. The table shows the HR ID that is used in CPHS, and a ‘Reference name’ by which the HR is referred to throughout this report. The HR ID is an integer from 58 to 62, while the HR reference name which is the name of one of the districts within the HR, picked at random. Within each HR, the CPHS dataset contains data on a set of sample houses which are surveyed three times a year.

Each HR has sub-strata in both rural and urban regions. The primary sampling units are villages and towns of the 2011 census. This provides information about households of Karnataka that are representative across different regions in the state, as well as across rural and urban regions. Further, this survey also provides similar information about households in *neighbouring* states, and can provide counterfactual to indicate causality.

2.2 Sample size: survey and response

Each time that the survey is conducted, there are a certain number of houses where there is no response. The survey ‘non-response’ can change from year to year, as well as within the year. While the magnitude of the non-response varies from wave to wave, there has been a trend in recent times of an increase in the non-response. As an example, Table 2.2 presents the number of houses targeted during the survey period and the number of houses from where there was a response, for Wave 3 2023 and the same wave, one year later, in 2024. This shows that there has been a drop in responses across all HRs except Bengaluru. Further it shows that there is variation in response across different districts, and can be important when interpreting cross-region variation in the analysis.

Table 2.2 Comparing response to survey in Wave 3 2023 vs. 2024

District		Number of households in Karnataka survey			
		Approached In Sep-Dec 2023	Responded	Approached In Sep-Dec 2024	Responded
HR 58	Bangalore	1288	299	1066	656
	Bangalore Rural	96	14	96	79
	Chikkaballapura	409	41	310	256
	Kolar	496	32	416	350
	Ramanagara	64	8	64	4
HR 59	Chamarajanagar	48	40	48	32
	Chikmagalur	64	14	63	15
	Chitradurga	80	0	80	0
	Hassan	448	353	290	153
	Mandya	480	431	352	73
	Mysore	554	532	542	303
	Tumkur	656	374	592	272
HR 60	Dakshina Kannada	670	569	534	299
	Udupi	416	411	308	248
	Uttara Kannada	640	452	450	269
HR 61	Belgaum	952	843	742	553
	Davanagere	48	28	48	0
	Dharwad	564	438	627	379
	Gadag	446	445	446	193
	Haveri	128	127	128	80
	Shimoga	144	115	144	62
HR 62	Bagalkot	64	59	64	8
	Bellary	96	21	96	59
	Bidar	64	0	64	0
	Bijapur	144	64	144	43
	Gulbarga	454	340	400	233
	Koppal	436	276	266	234
	Raichur	504	395	376	338
	Yadgiri	96	60	96	42

Chapter 3

Features of the analysis

In order to observe how the behaviour and choices of the CPHS households respond to the introduction of the schemes, it is useful to use data in the analysis of their behaviour *pre and post* their introduction. For the ‘pre-scheme’ behaviour, data is used from CPHS from January 2022 for the monthly variables (Wave 1 for the wave variables), up to Dec 2022 (Wave 3 2022).

In the current report (Report 3), the ‘post-scheme’ behaviour now covers data from January 2024 (Wave 1 2024) to December 2024 (and Wave 3 2024). With this, there is now a one-year period of observations about the Karnataka households in the pre-scheme and the post-scheme periods. Both the previous reports (Report 1 and Report 2) used data for only a subset of the months in 2024. Report 3 thus presents an update on the state of scheme implementation as well as an update on the responses of households.

In all the Reports, responses of the households include three aspects:

Across time As described above, the analysis presents information about the households in the period *before* the schemes were introduced, and *after*. This allows the analysis of *changes* in households choices around the period of scheme introduction.

Across Regions The analysis presents information about the households aggregated by HR, which captures the cross-region variation in the *changes* in household choices around the introduction of the scheme. The analysis also presents changes in responses separately for *rural* and *urban* households. Wherever possible, the household response for the overall India CPHS sample will be presented as well.

Levels vs. variations: volatility of measures The analysis presents information about changes in the volatility of consumption. Monthly observations about the household’s consumption expenditure is used to calculate consumption volatility as the *standard deviation* of expenditure observations. When interpreting volatility, two features to consider is that volatility is useful as a relative measure, and that it captures the average magnitude of monthly change in consumption. A reason that volatility is not often reported is that it is more sensitive to the availability of data. Therefore, the analysis in this report will include volatility, wherever it is relevant *and* can be calculated.

Chapter 4

Scheme transfers to households in Karnataka

In this chapter, we examine the patterns in government transfers to Karnataka households, as seen in the CPHS data. In addition, we also present some features about the income and expenditure of households in Karnataka, from the CPHS data. The CPHS survey includes questions on sources of household income and household expenditure. Among other sources such as a salaried job or agriculture, the sources of income includes how much government income (both, in cash and in kind) the household receives, at the level of a household. The expenditure includes how much is spent on a wide range of items including food, rent, utilities including electricity, travel, and education, at the level of the household. These are reported at a monthly frequency. This chapter studies:

1. Level or amount of government transfers. These include cash transfers, from schemes of the Union government and the state government. These are observed in rupees at a monthly frequency.
2. Level/amount of household income. These are available both as the total income as well as the value from different sources. The analysis uses the value of total income. These are observed in rupees at a monthly frequency.
3. Level/amount of household expenditure. These are available both as the total expenditure as well as specifics of expenditure allocation. The analysis uses the value of the total consumption expenditure. These are observed in rupees at a monthly frequency.
4. Volatility of monthly household consumption expenditure. These are calculated at two frequencies for each household: (a) Volatility in a wave, as the standard deviation of percentage changes in four observations of household consumption expenditure in a wave. (b) Volatility in a year, as the standard deviation of percentage changes in monthly household consumption expenditure observed for the year.

4.1 Cash income to households from government transfers

Table 4.1 shows that median monthly cash transfers to households in Karnataka in the post-scheme period (2024) remain higher than the levels in the pre-scheme period (2022). The median household reported a transfer of Rs.0 in 2022. In 2024, the median household reported a monthly transfer of Rs.2425. This indicates that the total government transfers reaching all household accounts has gone up significantly.

Table 4.1 Median monthly government transfers to households, cross-region variation

	Median monthly cash income from government transfers (Rs.)		
	2022	2023	2024
HR 58	0.0	85.0	2000.0
HR 59	0.0	0.0	1000.0
HR 60	0.0	190.0	2425.0
HR 61	0.0	255.0	2425.0
HR 62	0.0	255.0	2000.0
KA*	0.0	170.0	2425.0
India [§]	0.0	0.0	0.0

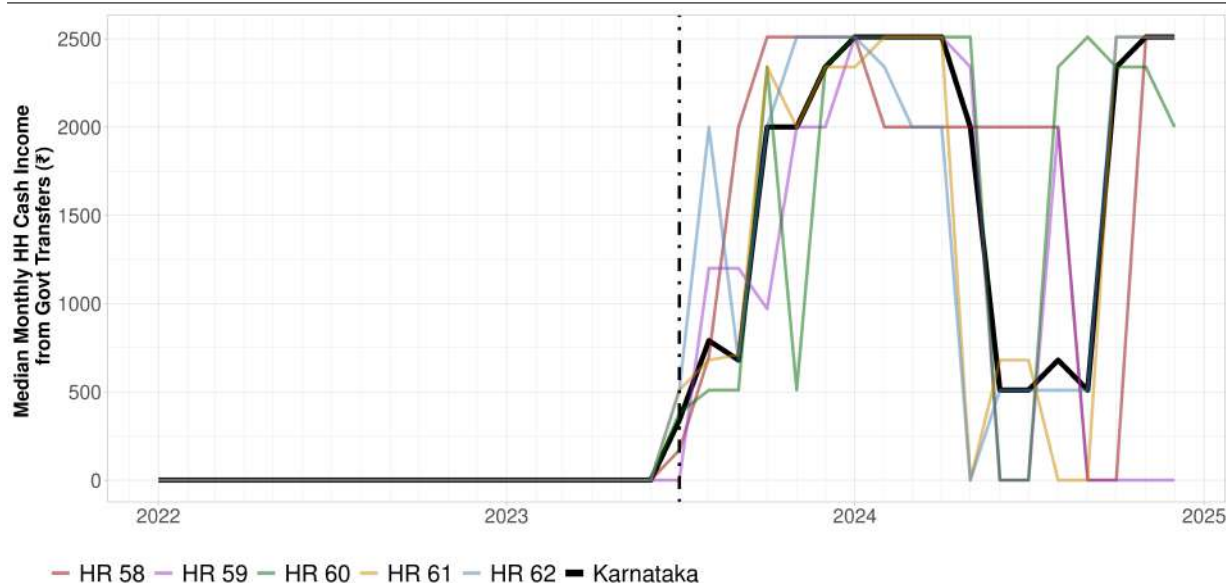
*All households in Karnataka

[§]All households in India including Karnataka

The level of median monthly cash transfers in the analysis covering January to December 2024 has *increased* to Rs.2425, compared with January to August 2024 (Rs.2255). There remains variation in median monthly cash transfers for households in different HRs, and the patterns are also different. The median level for households in HR 59 has fallen, while the levels for households in HR 61 and 62 have risen. These three HRs have reversed the trend that was being observed until last wave of data. This could be because of bunching of transfers happening in some HRs due to operational reasons, leading to lagged effects in the data.

A study of the monthly patterns in consumption in Figure 4.1 shows these changes more explicitly. There was a drop in government transfers to households after May 2024, across all HRs. The median monthly cash transfer had dropped from Rs.2500 in April to Rs.500 in June 2024. This meant that 50% of the households in the sample were receiving Rs.500 per month or less in June, compared to Rs.2500 per month or less in April 2024. However, this drop seems to have reversed post October 2024 for all HRs except HR 59 (Mysore). In December 2024, the entire Karnataka sample was receiving Rs.2500 per month again.

Figure 4.1 Monthly patterns in median government transfers with cross-region variation



What about transfers to rural households when compared to transfers to urban households? Table 4.2 indicates that there are no significant differences between the median government transfers to rural or to urban.¹ There was a drop in government cash transfers in May 2024 across Karnataka but both urban and rural areas seem to have recovered starting October 2024. The median rural and urban households in Karnataka continue to receive higher amounts than the median rural and urban household in India.

Table 4.2 Median monthly government transfers to urban vs. rural households

	Median monthly cash income from government transfers (Rs.)		
	2022	2023	2024
Rural	0.0	255.0	2425.0
Urban	0.0	0.0	2420.0
KA*	0.0	170.0	2425.0
Rural India	0.0	0.0	0.0
Urban India	0.0	0.0	0.0
India [§]	0.0	0.0	0.0

*All households in Karnataka

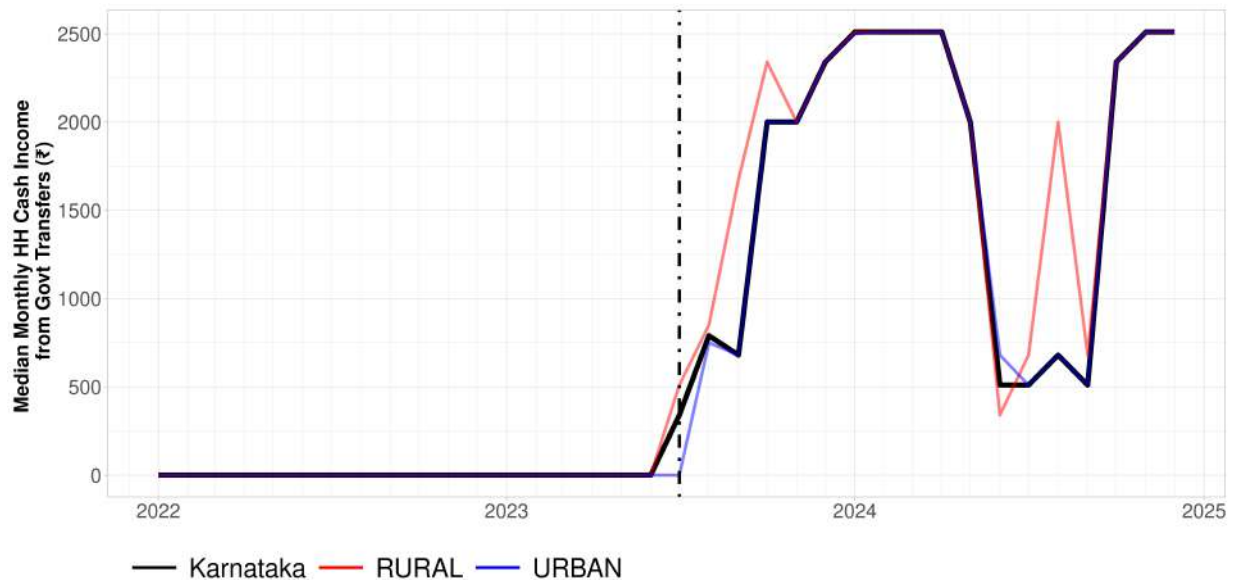
[§]All households in India including Karnataka

Figure 4.2 shows that the growth in the income transfers to rural households has seen an upwards trend since the announcement of the schemes. The decline seen in Figure 4.1

¹Includes government cash transfers from all Union and state government welfare schemes

post May 2024 is also seen for both rural and urban households but it has reversed since October 2024.

Figure 4.2 Patterns in median monthly government transfers, urban vs rural households



In summary, median cash income from government transfers to Karnataka households increased significantly – by $15\times$ – after the schemes were introduced and they remain higher than the amount of cash transfers for the overall India sample of households. The CPHS data does not distinguish between the sources of these transfers from specific schemes/governments.

4.2 Fraction of households with non-zero cash transfers from government

It is useful to ask what fraction of the households report receiving any government transfer at all, particularly across the pre-scheme and post-scheme periods.

Table 4.3 present the fraction of households in the sample who report receiving *any* government transfer. This was 9.3% in 2022 (pre-period) and is 72.7% in the post-scheme period. Karnataka has seen a massive boost in households receiving cash support from the government post scheme implementation. Correspondingly, only $\sim 44\%$ of households in the broader sample of India report receiving any government transfers.

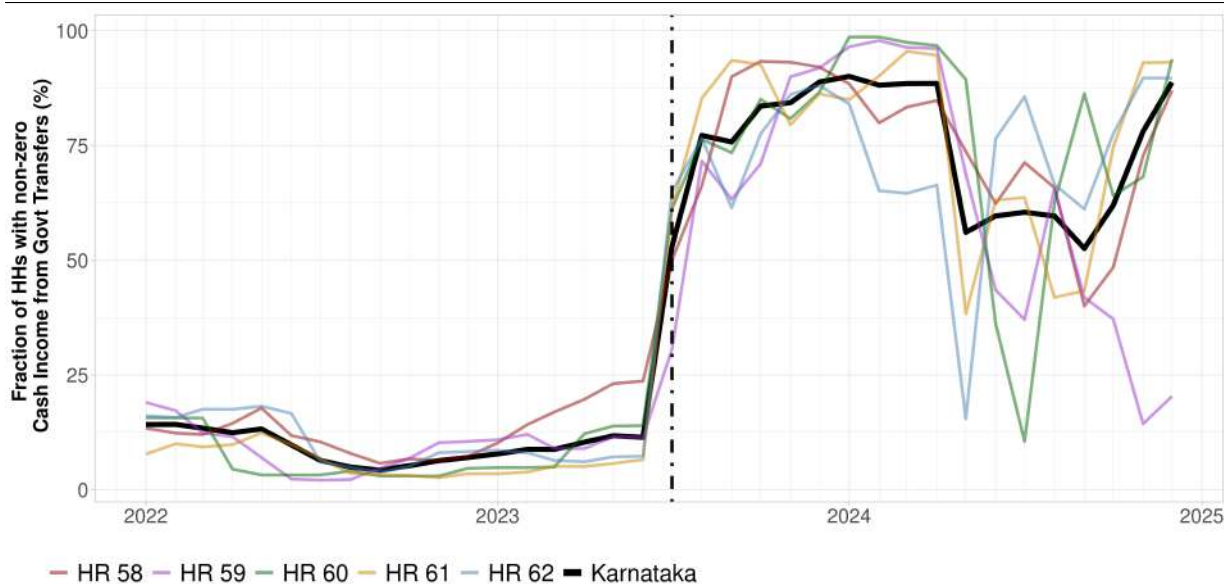
Table 4.3 Fraction of households receiving government transfers

	Fraction of households with non-zero cash transfer from government (Rs.)		
	2022	2023	2024
HR 58	10.5	49.4	71.5
HR 59	8.9	40.2	59.6
HR 60	6.6	43.2	75.2
HR 61	6.9	44.0	73.0
HR 62	11.5	41.5	70.2
KA*	9.3	43.5	72.7
India [§]	29.0	32.3	43.9

*All households in Karnataka

[§]All households in India including Karnataka

More than 70% of households across all HRs, except HR 59 (Mysore) receive some kind of government transfer. In HR 59, this fraction is $\sim 60\%$. Figure 4.3 presents the monthly trends of the fraction of the Karnataka sample receiving transfers, reflecting the patterns seen in the previous table. The monthly patterns also show regional variation. There is much higher coverage of households receiving transfers in the post-scheme period in all the CPHS HRs, compared with the pre-scheme period. All HRs faced a decline in cash transfers post May 2024 but recovered starting October 2024, except Mysore. This pattern is aligned with the decreasing amounts of median monthly cash transfer shown by Figure 4.1. Mysore's downward trend has continued post May 2024. This needs to be investigated further.

Figure 4.3 Monthly patterns of the fraction of households receiving government transfers

These patterns are similar for rural and urban households in the state. Table 4.4 shows the fraction of rural and urban households receiving cash transfer. There is no rural bias in government transfers across all the schemes put together.

Table 4.4 Fraction of rural vs. urban households receiving government transfers

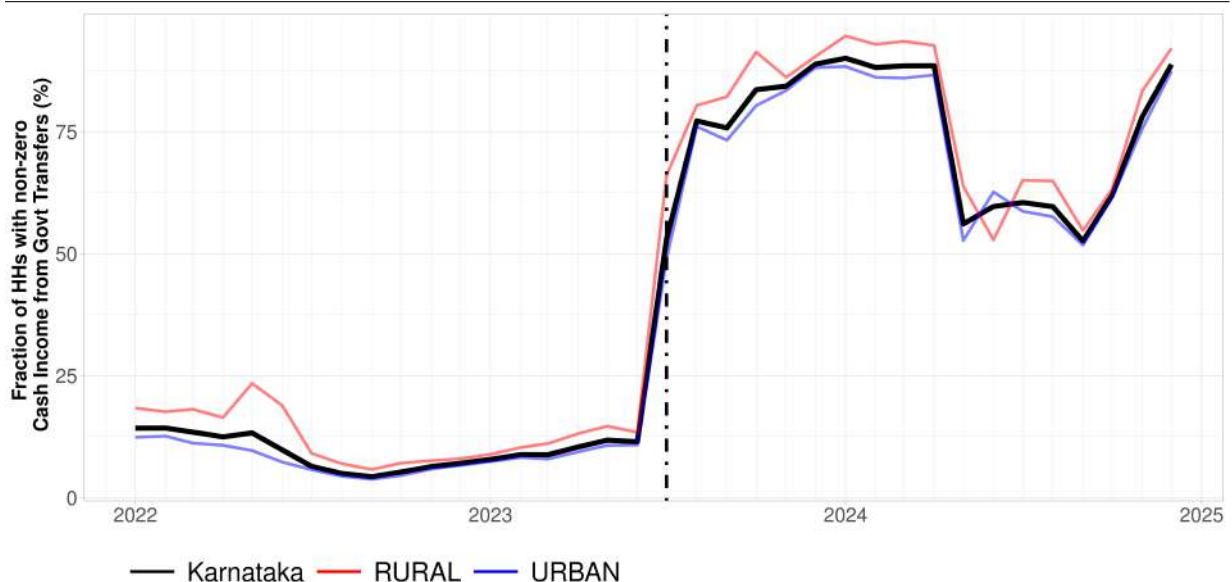
	Fraction of households with non-zero cash transfer from government (Rs.)		
	2022	2023	2024
Rural	13.1	47.3	76.1
Urban	7.9	42.1	71.2
KA*	9.3	43.5	72.7
Rural India	31.7	34.1	48.3
Urban India	27.8	31.5	41.4
India [§]	29.0	32.3	43.9

*All households in Karnataka

[§]All households in India including Karnataka

Figure 4.4 shows that more rural households report receiving government transfers, than urban households. The difference between the fraction of rural compared to urban households is positive but small. This has remained the same in the period before the recent schemes, and after. The decline post April 2024 and the recovery starting October 2024 has been seen in both rural and urban households.

Figure 4.4 Monthly patterns of the fraction of rural vs. urban households receiving government transfers



4.3 Income of households

Table 4.5 presents the median monthly income of households in Karnataka. There appears to be a rise in median monthly household income² in 2024 compared to 2022, a change to Rs.27,080 from Rs.22,100 for the overall sample. The median monthly household income in Karnataka is ~2K higher than the corresponding India number in 2024.

Table 4.5 Median monthly household income, across regions

	Median monthly household income (Rs.)		
	2022	2023	2024
HR 58	29000.0	25395.0	30510.0
HR 59	23100.0	25000.0	28825.0
HR 60	20000.0	23182.5	28510.0
HR 61	18500.0	21307.5	25000.0
HR 62	18675.0	19765.0	23837.5
KA*	22100.0	22755.0	27080.0
India [§]	18425.0	21558.0	25000.0

*All households in Karnataka

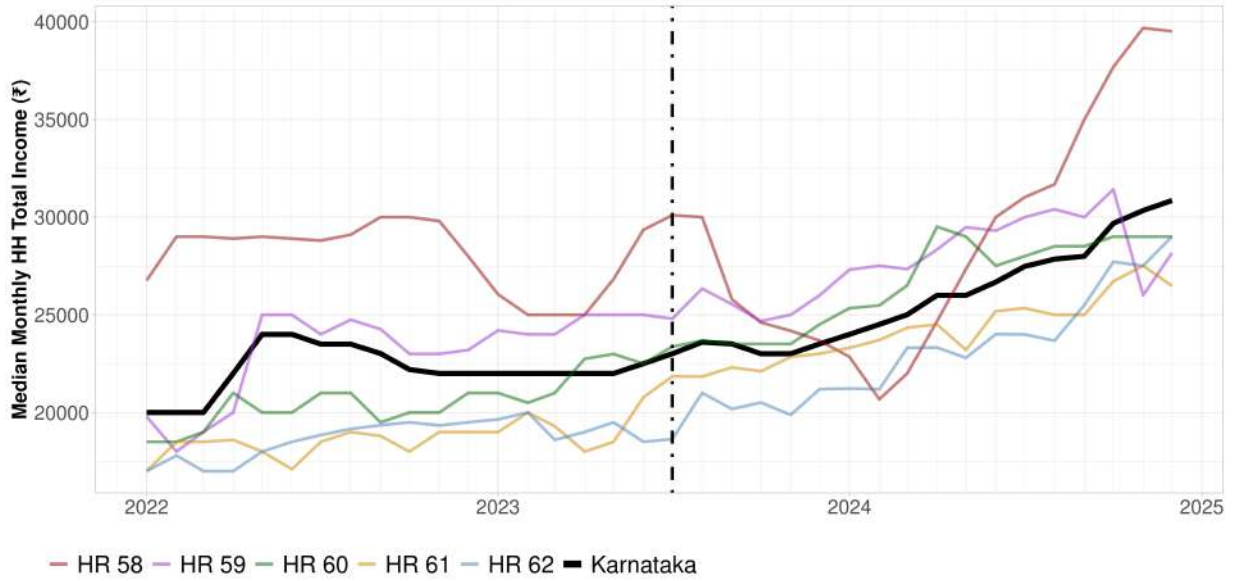
[§]All households in India including Karnataka

There is an increase of more than Rs.6500 per month for the median overall Indian households between the pre-scheme and post-scheme periods, compared to Rs.5000 per month for the Karnataka households. Household incomes in Karnataka are growing slower than those of households in the broader India sample.

Table 4.5 shows that there remains variation in the median monthly household income across regions in Karnataka. The median monthly income for households in HR 58 (which is the HR containing Bengaluru) only slightly increased from the pre-scheme to the post-scheme period, but it has increased at a much higher gradient for households in HRs 60, 61 and 62.

²All income and expenditure numbers in this report are nominal

Figure 4.5 Monthly patterns in median household income across regions



The monthly patterns in Figure 4.5 shows a consistent increase in the median monthly household income for all the HRs, except for households in HR 58. This is the HR containing Bengaluru city. The median monthly income reported by households in HR 58 shows a drop after the introduction of the schemes until February 2024. But there has been a steady increase in income post February 2024 and now the income in Bengaluru HR is much higher than other HRs.

These changes are consistent for rural and urban households. Table 4.6 shows that the median monthly household income has increased for both urban and rural households. Changes in household income has no rural-urban biases.

Table 4.6 Median monthly income in rural vs. urban households

	Median monthly household income (Rs.)		
	2022	2023	2024
Rural	18000.0	20020.0	24595.0
Urban	24100.0	23620.0	27755.0
KA*	22100.0	22755.0	27080.0
Rural India	14980.0	16675.0	19015.0
Urban India	20000.0	23505.0	27750.0
India [§]	18425.0	21558.0	25000.0

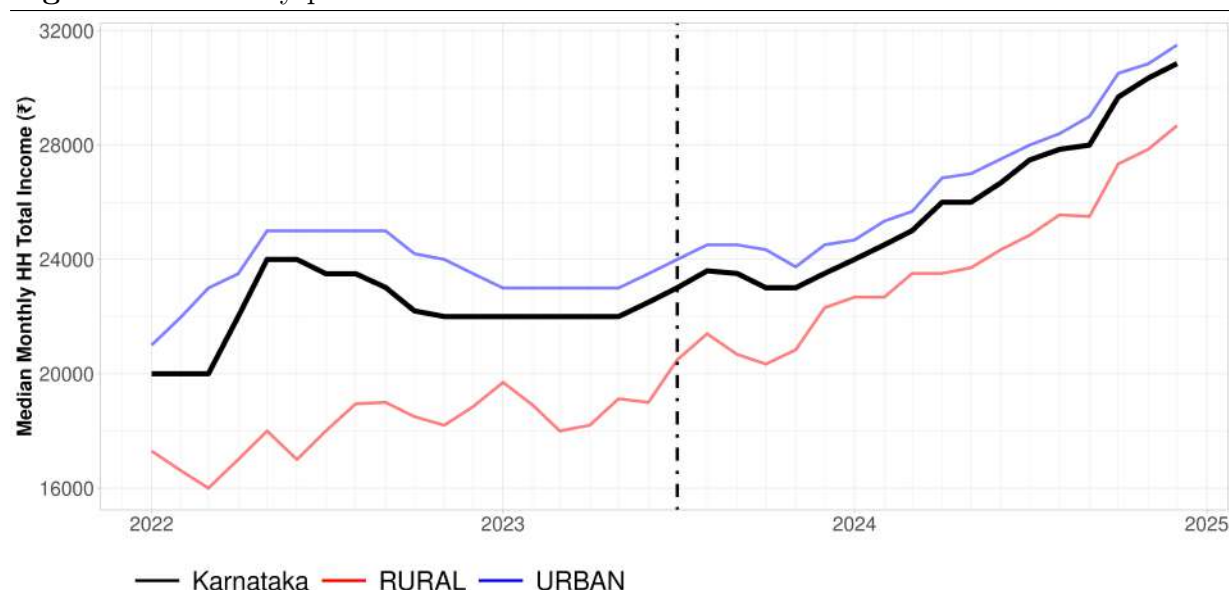
*All households in Karnataka

[§]All households in India including Karnataka

However, a comparison of the median monthly incomes for Karnataka and the overall

India samples in the pre-scheme and the post-scheme periods, show a change in income growth for rural and urban households. Households in urban Karnataka have a lower growth in income than urban households in the rest of India. There is an increase of about Rs.3600 in the median monthly income for urban Karnataka households, compared to Rs.7750 for urban overall India households. Rural Karnataka households median monthly incomes rose by more than Rs.6500 from the pre-sample to the post-sample period, while this increase was Rs.4000 in rural households in the overall India sample. Therefore, while there does not appear to be a rural-urban bias in the level of incomes in Karnataka, the growth of income is higher for rural Karnataka households than urban Karnataka households.

Figure 4.6 Monthly patterns of rural vs. urban household income



The monthly patterns in Figure 4.6 indicates that the gap between the median monthly incomes of rural and urban households appear to be consistent in the past year.

4.4 Expenditure of households

Table 4.7 shows that median monthly household expenditure in Karnataka increased from Rs.13,856.5 in the pre-scheme period to Rs.14,329.8 in the post-scheme period. In comparison, the median monthly expenditure of households in the overall India sample increased from Rs.12,406 in 2022 to Rs.15,684 in 2024.

As with income, there is significant variation across different regions. Households in the southern regions (HRs 58 and 59) report a *decrease* in expenditure, while households in the northern HRs (HRs 61 and 62) report an *increase* in expenditure.

Table 4.7 Median monthly household expenditure, across regions and time

	Median monthly household expenditure (Rs.)		
	2022	2023	2024
HR 58	16017.5	19228.8	15386.0
HR 59	17598.5	18914.0	11042.5
HR 60	12382.0	15055.8	12656.5
HR 61	9730.8	12638.0	14949.2
HR 62	9389.0	12497.5	15034.5
KA*	13856.5	14714.5	14329.8
India§	12406.0	14414.5	15684.0

*All households in Karnataka

§All households in India including Karnataka

The monthly pattern of differences across regions can be seen in Figure 4.7. There is an increase in median monthly household expenditure of HR 61 and 62 throughout the pre-scheme and post-scheme period. Households in HR 60 have maintained a steady trend. We observe a downward trend across all HRs immediately after the scheme implementation. Bengaluru and Mysore start at the top before the scheme but fall down lower than other HRs after scheme implementation. We start seeing a recovery from January 2024 and Bengaluru recovers to highest median monthly household expenditure compared to other HRs. Mysore has not seen full recovery after the fall.

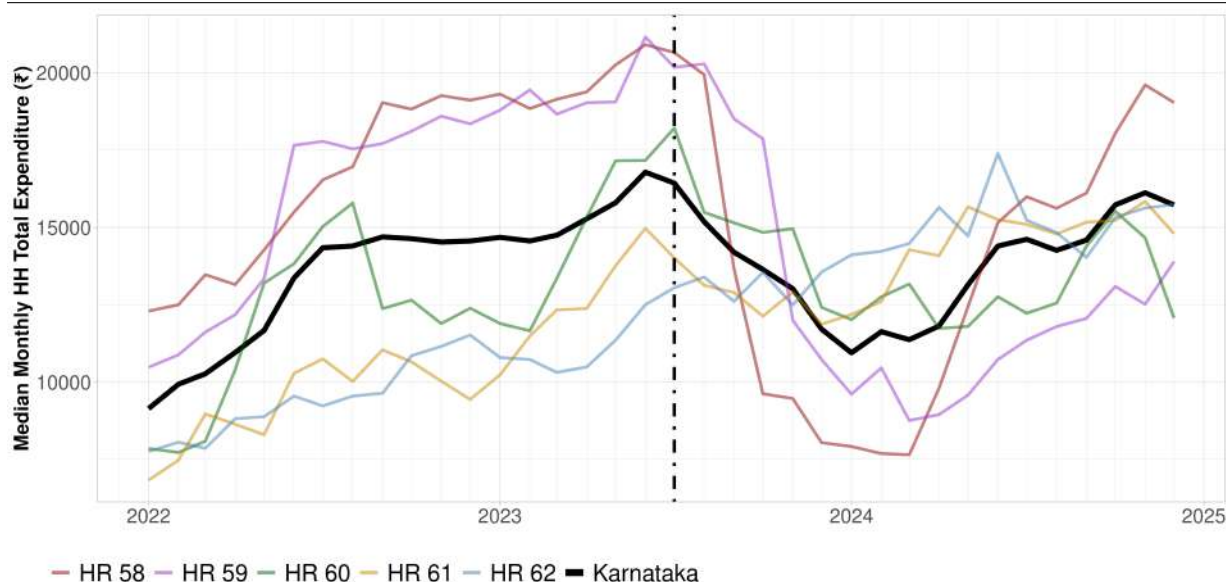
Figure 4.7 Monthly patterns of median household expenditure, across regions

Table 4.8 shows that median monthly household expenditure increased for rural areas

and remained constant for urban areas in Karnataka. We see higher increase for urban areas compared to rural areas for the India households sample. We observed a decrease in expenditure in Karnataka immediately after scheme implementation as seen in Figure 4.8. This decline reversed in 2024.

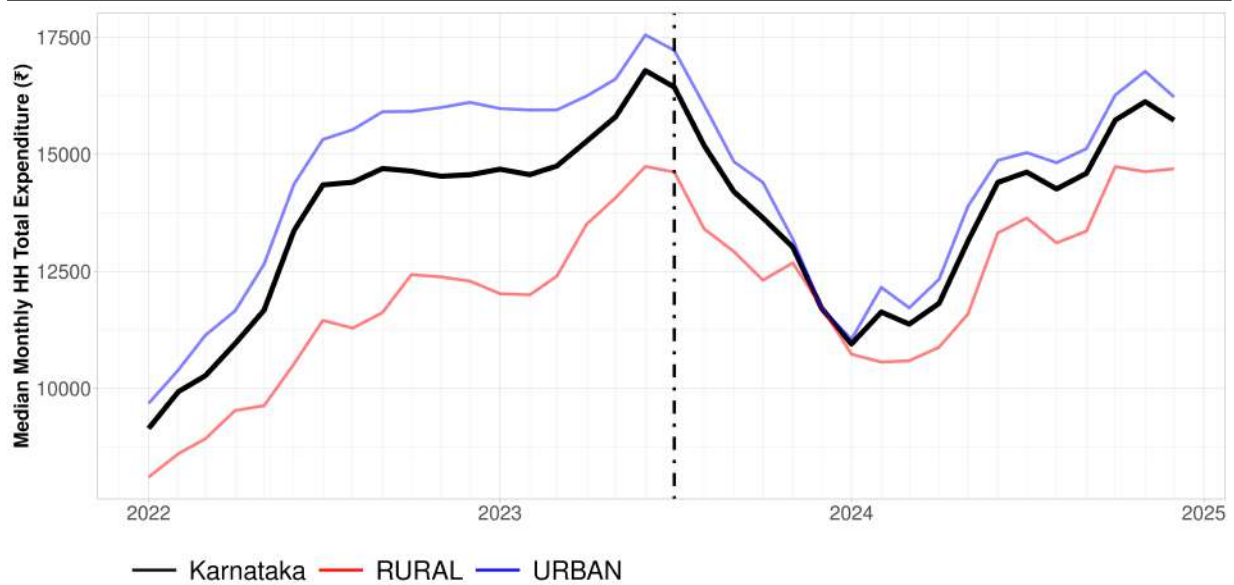
Table 4.8 Median monthly expenditure in urban vs. rural households

	Median monthly household expenditure (Rs.)		
	2022	2023	2024
Rural	10903.0	12800.0	13213.5
Urban	14838.5	15961.8	14844.8
KA*	13856.5	14714.5	14329.8
Rural India	11196.5	12898.2	13690.0
Urban India	12970.8	15173.8	16963.5
India [§]	12406.0	14414.5	15684.0

*All households in Karnataka

[§]All households in India including Karnataka

Figure 4.8 Monthly patterns of median urban vs. rural household expenditure in Karnataka



4.5 Volatility of household consumption expenditure

The changes in Tables 4.5 and 4.7, and Figures 4.5 and 4.7 in median household income and household expenditure show up as higher monthly volatility in the post-scheme period

compared to the pre-scheme period. Table 4.9 shows that consumption expenditure volatility for Karnataka households has risen from a pre-scheme period level of 21.9% to 28.4% in the post-scheme period. The increase in consumption expenditure volatility observed in Report 1 continues at a higher growth rate, compared to the growth in consumption expenditure volatility for the overall India sample households.

Table 4.9 Volatility of monthly household expenditure in Karnataka

	Median volatility of monthly expenditure (%)		
	2022	2023	2024
HR 58	22.4	19.4	37.8
HR 59	16.0	21.0	26.1
HR 60	24.0	24.4	26.5
HR 61	29.6	27.4	27.5
HR 62	23.8	24.7	31.4
KA*	21.9	22.4	28.4
India [§]	17.4	19.7	18.8

*All households in Karnataka

[§]All households in India including Karnataka

There is significant regional variation in the pre-scheme and post-scheme behaviour of households observed in Table 4.9. In four out of the five HRs, there is an increase in volatility of household expenditure in the post-scheme period, while there is a decrease in the post-scheme expenditure volatility for one HR. Figure 4.9 shows the same patterns persist in the monthly data as well. The households of the south Karnataka HRs (58 and 59) and HR 62 (Bellary) have sharply increased volatility in their monthly expenditure, HR 60 (coastal) has slightly increased volatility while expenditure volatility drops for households in Dharwad HR. However, these trends cannot be attributed to the introduction of the schemes.

Figure 4.9 Patterns in consumption expenditure volatility of Karnataka households

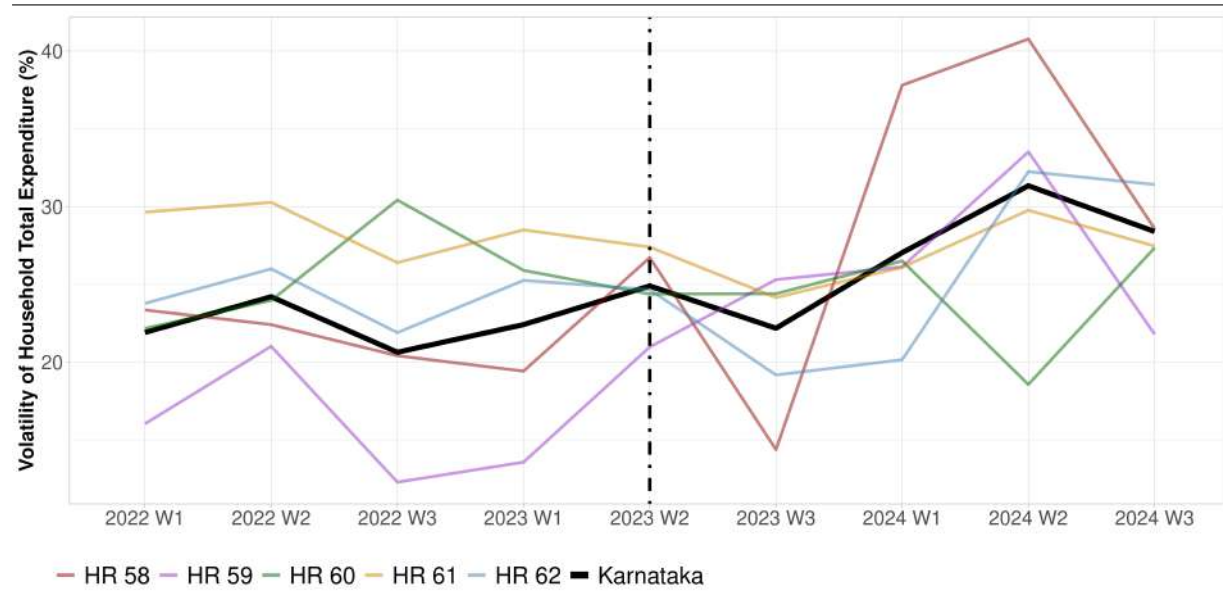


Table 4.10 presents the differences in the volatility of rural and urban household expenditures between the pre-scheme and post-scheme periods. Volatility has increased in rural and urban Karnataka. In comparison, the volatility of household expenditure for the larger sample of households in India, the level has remained similar throughout this period from 2022 to 2024. This indicates that there were larger changes in month on month median household expenditure in Karnataka compared to the broader sample of India.

Table 4.10 Volatility of monthly expenditure in rural vs. urban households

	Median volatility of monthly expenditure (%)		
	2022	2023	2024
Rural	22.2	23.4	28.2
Urban	21.7	23.1	28.5
KA*	21.9	22.4	28.4
Rural India	16.7	19.4	18.5
Urban India	17.8	19.5	19.2
India [§]	17.4	19.7	18.8

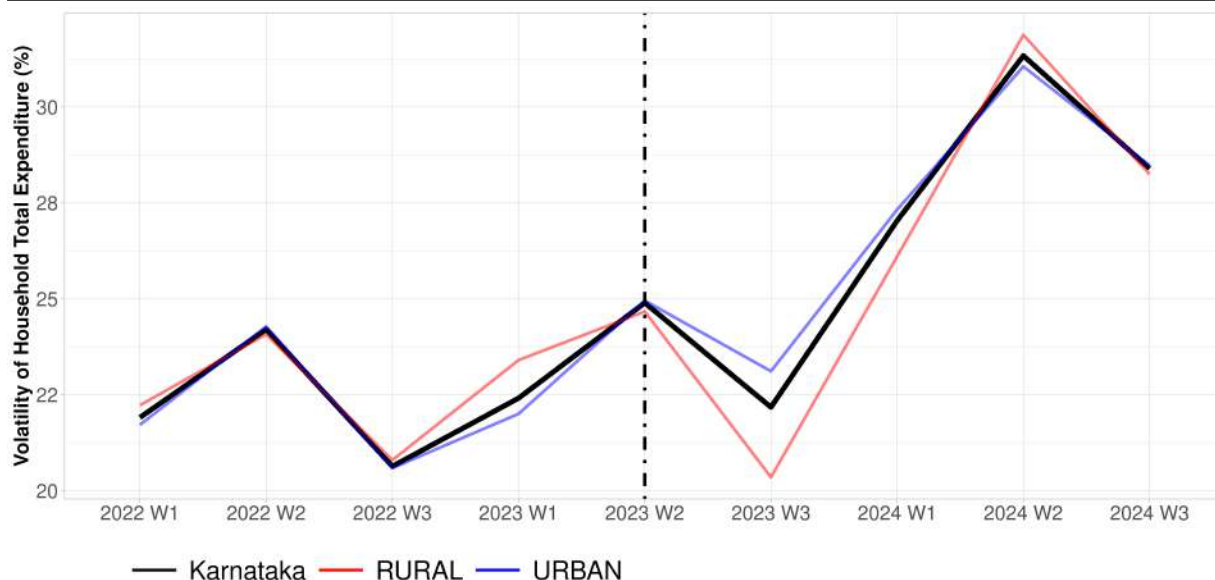
*All households in Karnataka

[§]All households in India including Karnataka

Figure 4.10 indicates that rural and urban households responded slightly differently in monthly volatility of their expenditure. While household expenditure dropped for households in both these regions immediate after scheme implementation, the decrease in expenditure volatility was higher for rural households. However, expenditure volatility rose from

the first wave of 2024. For the entire 2024 year, expenditure volatility for urban households increased more than for rural households.

Figure 4.10 Monthly pattern in volatility of rural vs. urban household expenditure



Volatility in household expenditure, like the level of household expenditure, tends to move with household income. Table 4.11 shows the volatility of monthly household income in Karnataka has increased in the post-scheme period. The increase in income volatility is smaller in magnitude compared with the increase in consumption expenditure volatility. In comparison, there was a increase in the volatility of the overall India sample also. But this increase is smaller than the increase for the Karnataka households.

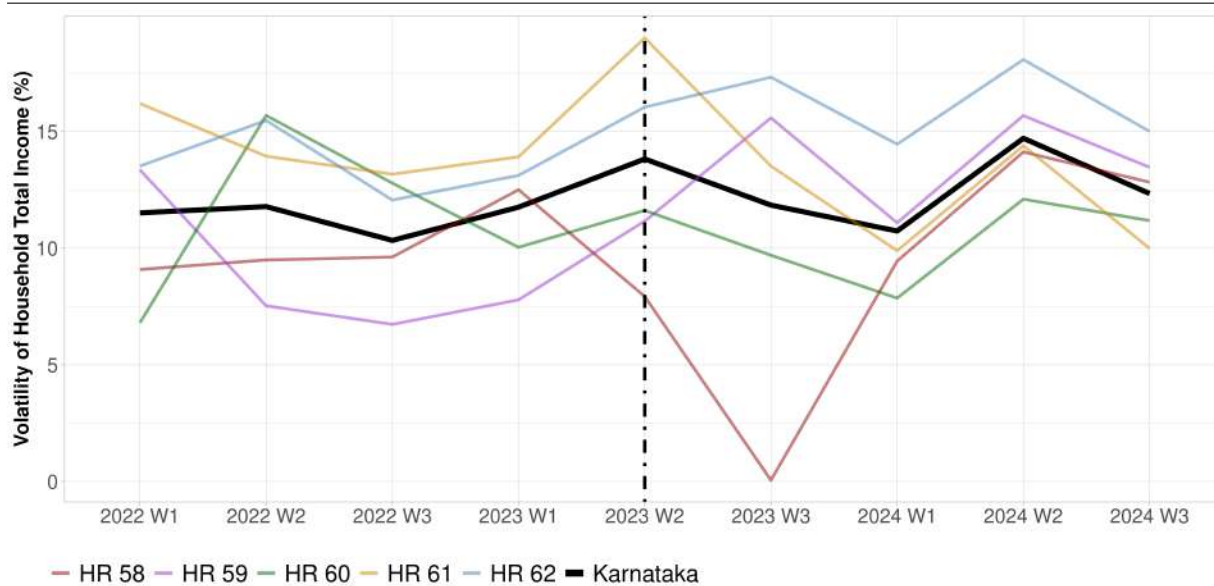
Figure 4.11 also shows a similar pattern in monthly observations of household income volatility. We also observe variation between the HRs. While there was a sharp drop in the income volatility of households in HR 58 (the region with Bengaluru city) immediately after the scheme, it rose back up to similar levels as other HRs.

Table 4.11 Volatility of median monthly income, across regions

	Median volatility of monthly income (%)		
	2022	2023	2024
HR 58	9.5	7.9	12.8
HR 59	7.5	11.2	13.5
HR 60	12.8	10.0	11.2
HR 61	13.9	13.9	10.0
HR 62	13.5	16.0	15.0
KA*	11.5	11.8	12.3
India§	11.3	11.6	11.8

*All households in Karnataka

§All households in India including Karnataka

Figure 4.11 Monthly patterns of volatility of median household income across regions

4.6 Summary about transfers, and household income and expenses in Karnataka

This report update incorporates fresh household observations for September to December 2024 (Wave 3), compared to Report 2 where the post-scheme period included observations only up to August 2024. The present analysis shows that:

1. While there was a drop in government cash income transfers to households in Karnataka in Wave 2 2024, there has been a recovery in these transfers across Karnataka, except HR 59 - Mysore (Table 4.1, Figure 4.1). This is also reflected in a larger fraction of households in the sample receiving any amount of government income transfers at all (Tables 4.3, 4.4, Figures 4.3, 4.4).
2. For the overall Karnataka sample, the median monthly household *income has consistently increased*, but the median monthly *expenditure has only slightly increased* compared to the pre-scheme period (Tables 4.5, 4.7, Figures 4.5, 4.7). Median monthly household expenditure dropped immediately post scheme implementation but has risen since 2024.
3. The *changes* in income and expenditure between the pre-scheme and post-scheme periods are different for rural and urban Karnataka households. The changes are *smaller* for urban households compared to rural households. These are counter to the patterns of changes between rural and urban households in the larger India sample. Urban Karnataka households have smaller growth in income compared to both rural Karnataka as well as the all India urban households sample (Tables 4.6, 4.8, Figures 4.6, 4.8).
4. Monthly expenditure volatility has *increased* in the post-scheme period for Karnataka households (Table 4.9, Figure 4.9). The month-on-month changes in household expenditure is higher in the post-scheme period, relative to the pre-scheme period in Karnataka. This is unlike the households in the full India sample where the volatility remains similar between the two periods.
5. There is high cross-regional variation in the responses of households between the pre-scheme and post-scheme periods.

There is variation across the median household income across HRs (Table 4.11, Figure 4.11). HR 58 households report a *small increase* in the median monthly income, while households in the remaining HRs report a *higher increase* in household income in the post-scheme period. Such variation is even higher when examining the monthly median household expenditure, where HRs 61 and 62 show an *increase*, HR 60 is nearly constant, while the remaining show a *decrease*. Household incomes and expenditures have *increased* for India households over the years.

There is some commonality among households in the southern Karnataka regions (HR 58 and 59) in their response to the schemes. Similarly, there is a greater commonality of responses among the households in the northern Karnataka regions (HR 61 and 62). These cross-regional variations need to be analysed and discussed more closely in order to identify how the scheme influenced household behaviour.

Chapter 5

The Annabhagya scheme

The Annabhagya scheme involves the grant of a defined amount of rice to families in Karnataka, with eligibility criteria that is related to their income. This scheme was announced on 1st July 2023, and was immediately effective.

5.1 Scheme design

Scheme eligibility - Inclusion criteria:

1. Priority households (PHH) ration card holders are households in rural areas.
2. Antyodaya Anna Yojana (AAY) ration card holders. These ration cards are granted to families with *annual income less than Rs.15,000*, or *monthly income less than Rs.1250*.

Scheme eligibility – Exclusion criteria:

1. Ration card families with no/multiple Head of Households
2. Tax payer households
3. Permanent employees of Govt. and allied institutions;
4. Households owning > 3 acres of land/pucca house in rural areas, or > 1000sq ft in urban areas;
5. Households owning 4-wheeler except one commercial vehicle like tractor;
6. Households with annual family income > Rs 1.2 lakhs

Benefits to eligible households:

- Additional 5kg of rice *per person, per month*, above an existing 5 kg allocation under the National Food Security Act (NFSA).
- DBT at Rs.34 per kg of food grain to eligible households, if the grain cannot be provided.

Mode of delivery: payment into the bank account of the *head* of the eligible household.

5.2 Measures to assess responses to the scheme

Annabhagya is targeted to deliver benefits to low income households, and those in the rural areas. The scheme is to deliver additional quantities of food grains to target households. This leads to the following expected responses in households behaviour:

Hypothesis 1: Since the scheme delivers additional food grains, eligible households will *reduce* expenditure on food.

Hypothesis 2: If the scheme delivers additional monthly income in lieu of monthly delivery of food grains, then there may be *no change* in the household expenditure on food since the household is likely to be consuming the amount of food grain required already.

Hypothesis 3: Some eligible households may use the income transfers to purchase other types of food items such as vegetables, fruits, dairy and meat products. In this case, eligible households may *increase* expenditure on food.

Hypothesis 4: Eligible household may utilise the increase in income to *increase* expenditure on other consumption items. This may be seen as *little or no change* in food expenditure and an *increase* in household total expenditure.

Hypothesis 5: If household expenditure on food *decreases* in response to the introduction of the Annabhagya scheme, this will be observed as an *increase* in volatility of expenditure in the period immediately after the scheme introduction. If household total expenditure *increases*, it will also be observed as an *increase* in expenditure volatility immediately after the scheme introduction. In both cases, volatility of either increase or decrease in household expenditure, after the scheme, should appear as *increased* volatility. Subsequent volatility should be low.

In contrast, if there is no change in the level of food consumption expenditure, volatility of expenditure will be close to zero.

Who are the eligible households that can benefit from the Annabhagya scheme? These are the PHH and the AAY ration card holder households. AAY ration card holder households are those whose annual income is less than Rs.15,000. This is an average monthly less than Rs.1250. After the introduction of the schemes, the median monthly government transfers to households have risen to Rs.2500 (Table 4.1). Households receiving these government transfers will not be eligible to be AAY ration card holders, after the introduction of the schemes. Since PHH are rural households, these may still be eligible for benefits under the Annabhagya scheme.

5.3 Annabhagya assessment

In the following assessment, the changes as listed in the above hypotheses will be examined using household expenditure (median monthly levels and volatility) for the CPHS sample, by the overall state as well as for the HR samples. This will also be done for the sample of rural and urban households.

5.3.1 Median monthly household expenditure on food and cooking fuel

We examine the median monthly household expenditure on food and cooking fuel in Table 5.1. We only focus on the household expenditure for the pre-scheme period (2022) and the post-scheme introduction period (2024) since the 2023 period is a mix of observations before and after the introduction of the scheme. In this table, both the pre-scheme period values and the post-scheme period values cover 12 months of observations per household in 2022 and 2024 respectively.

Table 5.1 Median monthly household expenditure on food and cooking fuel, across regions

	Median monthly basic expenditure (Rs.)		
	2022	2023	2024
HR 58	7142.5	8732.5	5648.8
HR 59	7877.5	7555.0	4628.5
HR 60	5492.5	6762.2	5893.8
HR 61	4872.5	6152.5	6115.0
HR 62	4811.0	5805.8	6220.0
KA*	6176.2	6743.2	5667.0
India [§]	6485.0	7082.5	7560.0

* All Karnataka sample

[§]All India sample including Karnataka

Table 5.1 shows that the monthly household expenditure for the overall Karnataka sample has *decreased* from Rs.6,176 in the pre-scheme period to Rs.5,667 in the post-scheme period. This suggests that the households now spend less on food grain after the start of the scheme. The monthly pattern in Figure 5.1 also shows a decline in household expenditure on food as compared to the start of the scheme.

Figure 5.1 Patterns in household expenditure on food and cooking fuel, across regions

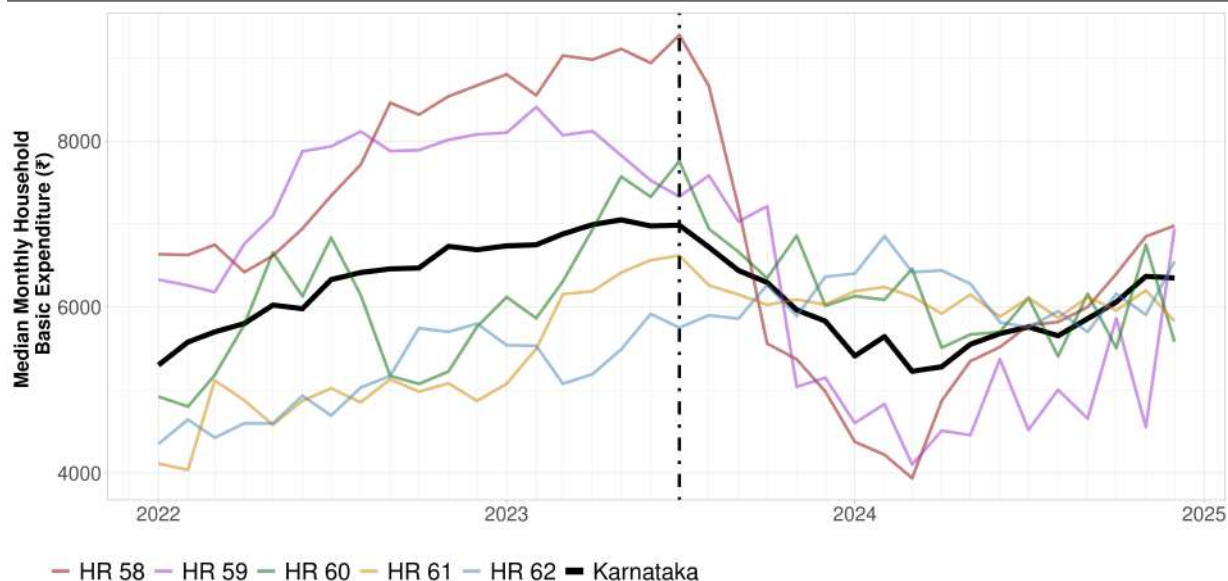


Table 5.1 shows that households responded *differently* to the scheme in different Karnataka HRs. Household expenditure *decreased* in the two southern HRs (HR 58 and 59). For instance, households in HR 58 *decreased* their expenditure from Rs.7142.5 per month in the pre-scheme period to Rs.5,648.8 in the post-scheme period. We are observing some recovery in expenditure in the past 8 months (May to December 2024) for HR 58 and HR 59.

But households in the coastal and northern HRs *increased* monthly expenditure. For example, households in HR 61 increased their expenditure from Rs.4,872.5 in the pre-scheme period to Rs.6,115 in the post-scheme period. Thus, the southern Karnataka households appear to have reduced expenditure on food after the schemes' introduction, while the others appear to have increased food expenditure.

One potential explanation could be that the scheme provided food grain to households in the southern HRs (which reduced expenditure on food as expected in hypothesis 1) while providing cash income transfers to the households in the northern and coastal HRs (where the income transfers may have been used to purchase higher priced non-food grain items as expected in hypothesis 3). Another potential explanation is that grain/cash were distributed at different times in the month in different regions, affecting the consumption pattern of households. Further work is required to test whether households in the different regions are responding differently because of differences in the *form* or *timing* of the government transfers they each receive.

The above analysis does not differentiate between eligible households and those which are not, since there is no clear identification in the CPHS database of whether the household is an AAY ration card holder, and the household incomes cannot be used as a proxy indicator for this, since all household incomes have risen after the introduction of the schemes to greater than Rs.15,000 annually.

5.3.2 Median monthly rural and urban household expenditure on food and cooking fuel

Table 5.2 presents a comparison of the expenditure of rural and urban households. Both sets of households have a *decrease* in expenditure, consistent with the expenditure decrease for the overall Karnataka sample. There is still a distinction: there is a *lower* decrease in expenditure by rural households (from Rs.5,417.5 in the pre-scheme period to Rs.5,375 in the post-scheme period), compared to a *higher* decrease for urban households (from Rs.6465 in the pre-scheme period to Rs.5667 in the post-scheme period).

Table 5.2 Median monthly expenditure on food and cooking fuel by rural vs. urban households

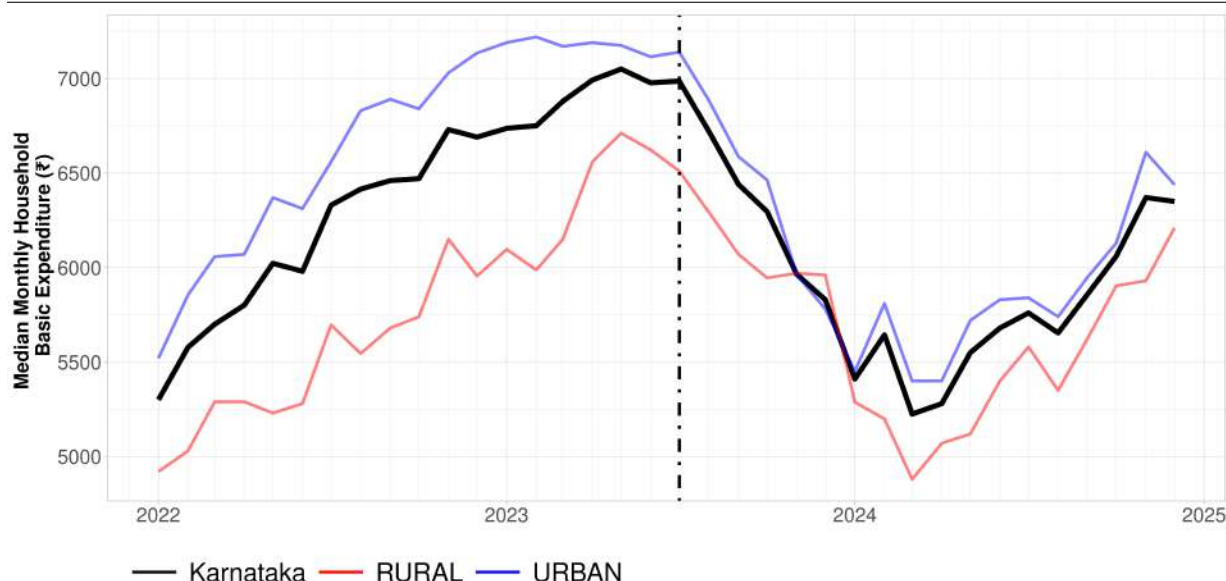
	Median monthly basic expenditure (Rs.)		
	2022	2023	2024
Rural	5417.5	6123.0	5375.0
Urban	6465.0	7127.5	5820.0
KA*	6176.2	6743.2	5667.0
Rural India	6050.0	6505.0	6927.5
Urban India	6667.0	7360.0	7955.0
India [§]	6485.0	7082.5	7560.0

*All Karnataka sample

[§]All India sample including Karnataka

Figure 5.2 also indicates a *drop* in household expenditure for both urban and rural households. There is a sharper drop in expenditure for urban households. This may be because there are fewer number of PHH candidates who benefit from the Annabhagya scheme in the CPHS sample, and a higher number of urban households who are AAY ration card holders. Figure 5.2 is also showing signs of a recovery in basic expenditure in the past two waves across rural and urban areas. However, it is important to note that basic household expenditure in Karnataka is still lower than the pre-scheme(2022) levels.

Figure 5.2 Patterns in monthly expenditure on food and cooking fuel by rural vs. urban households



Put together, this evidence provides some support to the thesis that the Annabhagya scheme is successful in delivering benefits to the households by reducing their need to spend on food grains (support for Hypothesis 1).

5.3.3 Median monthly household planned expenditure

Does the broader set of consumption expenses by the household change in response to the Annabhagya scheme benefits? If so, in what direction is the change?

Table 5.3 presents how much the household spends on a regular, planned set of consumption items each month. This includes expenses on rent, utilities, monthly travel expenses, school fees, telecommunication bills, all EMIs, and regular financial investments such as buying insurance.

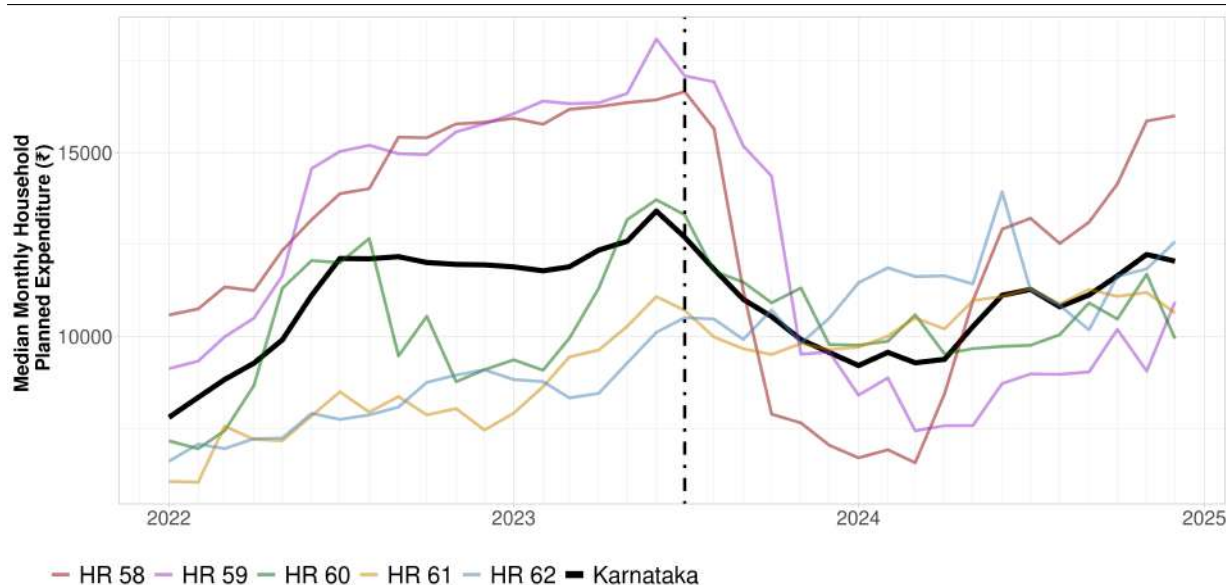
Table 5.3 Median monthly household planned expenditure, across regions

	Median monthly planned expenditure (Rs.)		
	2022	2023	2024
HR 58	13529.5	15853.0	12723.8
HR 59	14757.0	16343.0	8921.8
HR 60	9283.0	11315.8	9904.8
HR 61	7700.2	9654.5	10927.5
HR 62	7805.8	9868.5	11633.5
KA*	11531.8	11862.2	10964.0
India [§]	10330.0	11634.0	12773.5

*All Karnataka sample

[§]All India sample including Karnataka

Table 5.3 indicates a *decrease* in household expenditure on planned consumption items between the pre-scheme and the post-scheme period for the southern HRs (Bengaluru and Mysore). We observe an *increase* in planned expenditure for coastal and northern HRs (Dharwad and Bellary) in the post scheme period. The drop in planned expenses is only slightly larger than the drop in food and cooking fuel expenses observed in Table 5.1. Similar trends can be seen in the monthly patterns presented in Figure 5.1.

Figure 5.3 Patterns in monthly household planned expenditure, across regions

Further, the differences between households in different regions mimic the pattern of differences seen in Table 5.1. Households in the south Karnataka HRs spend *less* in the

post-scheme period compared to the pre-scheme period. This is the opposite of the expenditure pattern of households in HRs 60, 61 and 62.

In all cases, the change between the pre-scheme and post-scheme periods for planned household expenditure is *larger* than the change in household expenditure on food and cooking fuel. What drives the magnitude of the difference needs a deeper analysis of what are the differences in the consumption set of households in the northern and coastal HRs, compared to the southern HRs.

Is there a difference in expenditure for households of the urban and rural regions? Table 5.4 finds that households in urban regions report *decreased* expenditure on planned consumption items. Rural regions have slightly increased levels of expenditure on planned consumption items in the post-scheme period (2024).

Table 5.4 Median monthly planned expenditure for rural and urban households

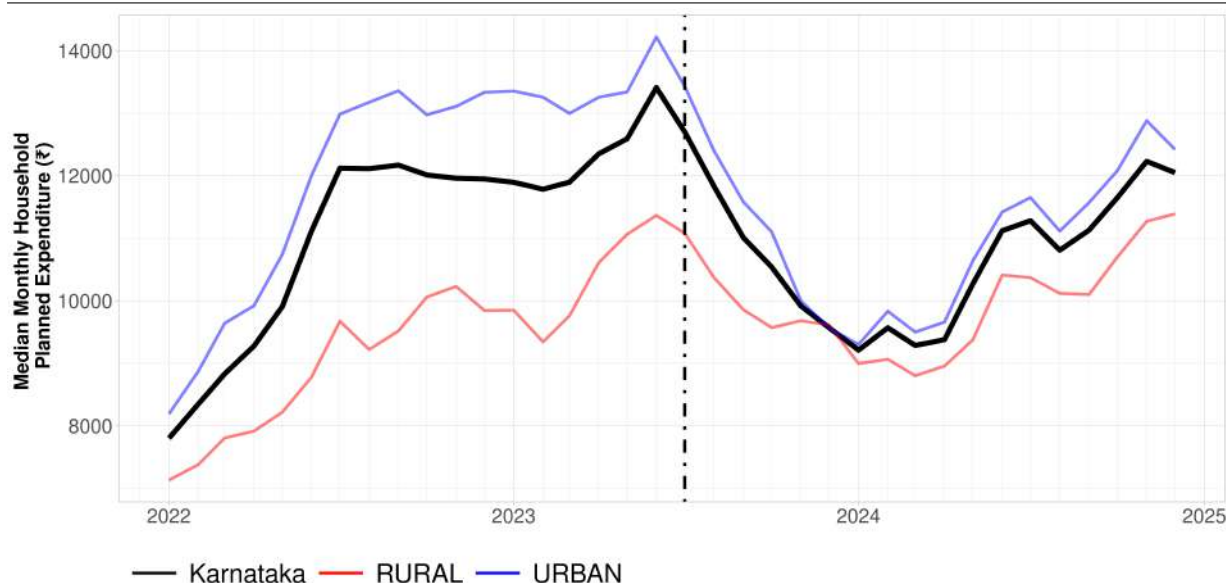
	Median monthly planned expenditure (Rs.)		
	2022	2023	2024
Rural	9000.0	9853.0	10108.5
Urban	12489.5	13126.0	11266.8
KA*	11531.8	11862.2	10964.0
Rural India	9245.0	10340.0	11136.2
Urban India	10817.0	12263.5	13793.5
India [§]	10330.0	11634.0	12773.5

*All Karnataka sample

[§]All India sample including Karnataka

The same pattern can be seen in Figure 5.4 showing the monthly patterns, and could reflect a higher decreased expenditure by urban households, which have a larger weight in the CPHS sample than rural households. However, it is important to note that both rural and urban areas are seeing a recovery in the past two waves.

Figure 5.4 Patterns of planned household expenditure for rural and urban households



5.3.4 Volatility of monthly household expenditure on food

As described in Section 5.3, if there is a significant change in monthly household expenditure, either as a decrease or an increase in household expenditure, it will manifest as higher volatility of expenditure. Table 5.5 shows that the volatility of food expenses rose from 20.3% in the pre-scheme period to 24.4% in the post-scheme period. This suggests that the changes by households in their consumption expenditure is significant.

Table 5.5 Monthly volatility of household expenditure on food and cooking fuel

	Volatility of monthly basic expenditure (%)		
	2022	2023	2024
HR 58	19.1	19.5	24.7
HR 59	14.2	14.7	32.9
HR 60	19.0	17.8	19.1
HR 61	21.5	22.1	23.3
HR 62	23.4	22.2	23.3
KA*	20.3	19.5	24.4
India [§]	16.4	17.3	16.6

* All Karnataka sample

[§] All India sample including Karnataka

Like in the analysis on the levels of expenditure, the southern HRs have different responses compared to the remaining three HRs. HR 58 and 59 have *increased* expenditure

volatility in the post-scheme period, while HRs 60, 61, and 62 all have similar volatility in the same period. One interpretation could be that household expenditure in the coastal and northern HRs have adjusted *more rapidly* to the scheme, while the household in the southern HRs are taking a longer time to adjust (as can be seen in Figure 5.1).

Figure 5.5 Wave-wise patterns in volatility of household expenditure on food and cooking fuel

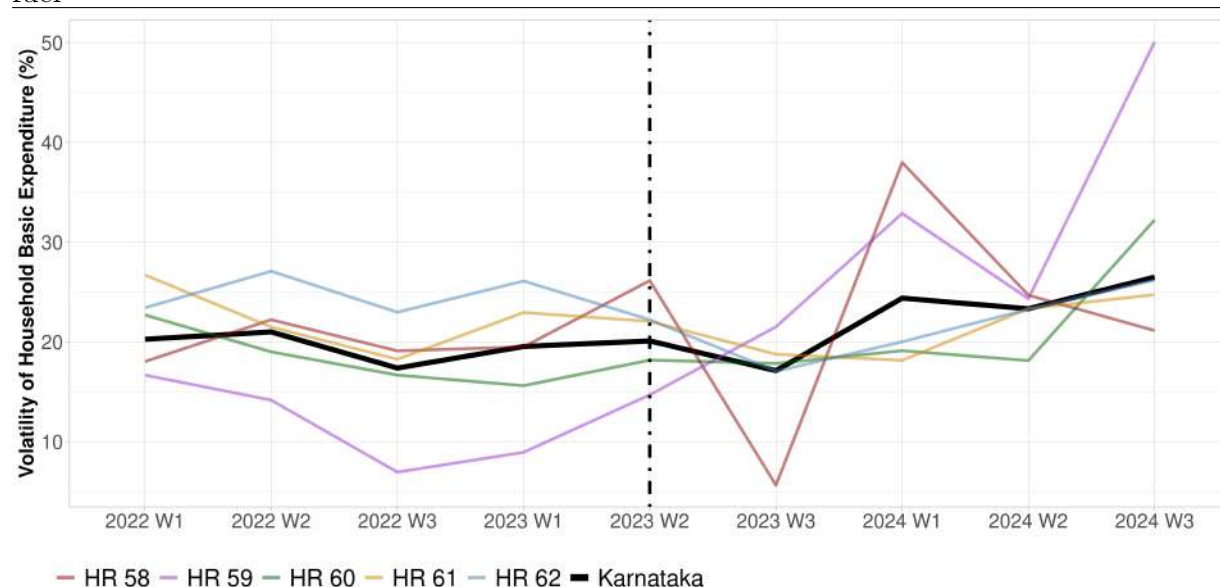


Figure 5.5 supports this interpretation. In this graph, the monthly volatility for HRs 58 and 59 are higher than the values for the other HRs.

The analysis shows rural and urban households have had a similar response expenditure on food and on planned consumption items. This is also true for volatility of household expenditure on these items as well as can be seen in Tables 5.6 and Figure 5.6. Table 5.6 shows that monthly volatility of household expenditure on food and cooking fuel increased for both rural and urban households. The increase in urban volatility is a little higher than the increase in rural volatility.

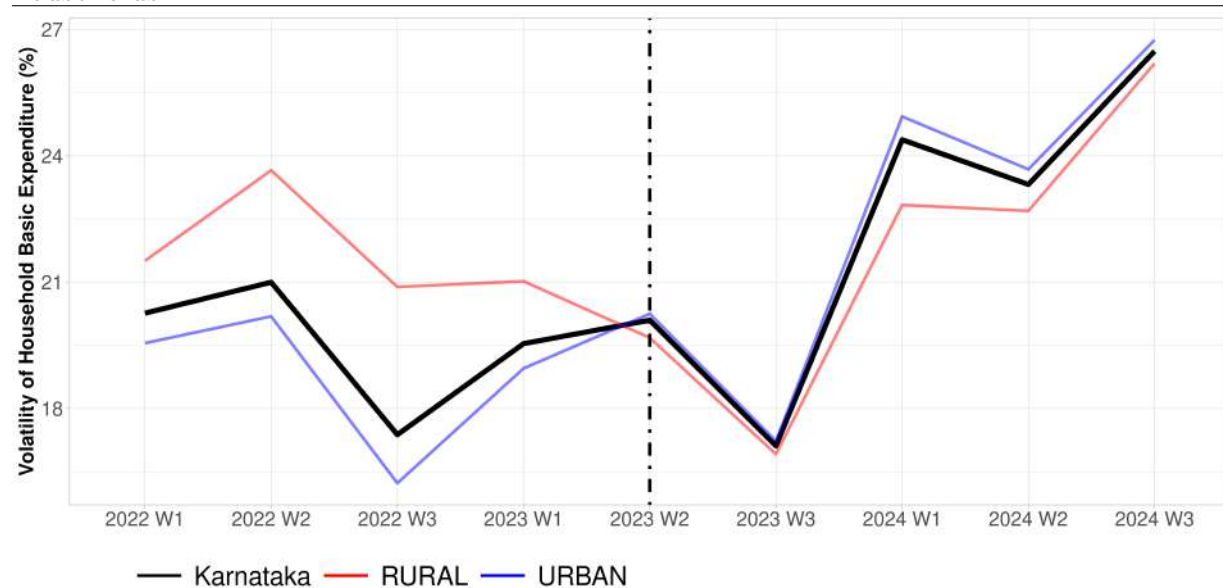
Table 5.6 Volatility of urban and rural household expenditure on food and cooking fuel

	Volatility of monthly basic expenditure (%)		
	2022	2023	2024
Rural	21.5	19.7	22.8
Urban	19.6	19.0	24.9
KA*	20.3	19.5	24.4
Rural India	15.9	17.7	17.0
Urban India	16.5	17.1	16.4
India [§]	16.4	17.3	16.6

*All Karnataka sample

[§]All India sample including Karnataka

Figure 5.6 Patterns in household food and cooking fuel expenditure, rural and urban households



5.4 Summary with key findings

This chapter focused on understanding the responses of households in Karnataka which are eligible for the Annabhagya scheme. Since the administrative information used to identify eligible households is not collected as part of the CPHS data, this scheme has been assessed using the sample of Karnataka households available in CPHS. Given the target benefit of ensuring food grains to eligible households, the measure of assessment used is how much the household has to spend on food.

In this examination, three observations emerge:

1. There has been a decrease in median monthly household expenditure on food for the full Karnataka sample, suggesting that the increased food supply from Annabhagya has helped lower the expenditure burden of the household (Table 5.1, Figure 5.1).
2. There are strong regional variations.

- (a) Household food expenditure has decreased for households in southern Karnataka, but increased for households in the coastal region, and increased significantly for households in the northern regions of Karnataka (Table 5.1, Figure 5.1).

Volatility of household expenditure on food has increased in Karnataka. However, variation in household response across HRs persist. Households in south Karnataka HRs (58 and 59) have higher volatility of food expenditure in the post-scheme period. Households in the coastal and northern regions have similar volatility in food expenditure in the post-scheme period (Table 5.5, Figure 5.5).

One potential explanation is that the scheme is better able to deliver transfers in kind (grain) to households in the southern regions, but can only deliver transfers in cash to households in the coastal and northern regions.¹ This requires data from micro-surveys within these regions before it can be tested and validated.

- (b) The decrease in rural household food expenditure is much smaller in magnitude than that for urban households.

The gap between levels of urban household expenditure on food and rural household expenditure on food has decreased after the introduction of the Annabhagya scheme (Table 5.2, Figure 5.2).

Volatility of urban and rural household expenditure has increased (Table 5.6, Figure 5.6).

3. Household expenditure on planned items reflects the expenditure on food items. In regions where food expenditure decreased in the post-scheme period, there is a decrease in planned expenditure. In regions where there is an increase in food expenditure, there is an increase in planned expenditure (Tables 5.1, 5.3; Figures 5.1, 5.3).

In all cases, the change in the median household planned expenditure is higher than the change in the median household food expenditure.

¹Some evidence of this is presented in Table 4.1.

Chapter 6

The Gruhajyothi scheme

Gruhajyothi scheme was announced on 1st July 2023, and disbursement started August 2023. The scheme provided fixed monthly electricity units to domestic installations. Beneficiaries have to mandatorily register on the Seva Sindhu portal and link their Aadhaar card to avail the benefits of the scheme. The scheme merges and supersedes the Bhagya Jyothi or Kuteera Jyothi Scheme, and the Amrut Jyothi Scheme.

6.1 Scheme design

Scheme eligibility - Inclusion criteria:

1. Domestic installations only

Scheme eligibility - Exclusion criteria:

1. Commercial/industrial installations

Benefits to eligible households:

- Free, fixed monthly allowance of ‘*entitled units*’ per month, per domestic installation.
- Fixed monthly allowance equals average monthly consumption per consumer for FY 2022-23¹ plus additional units of 10% of average consumption, subject to *maximum 200 units per month*.
- *New installations* are entitled to *40 units* per month.
- Amrut Jyothi beneficiaries (SC,ST, and BPL families) are entitled to *75 units* + 10% per month.
- Bhagya Jyothi and Kuteera Jyothi installations are entitled to *53 units* + 10% per month.

¹With effect from 1 Feb 2024, domestic consumers using less than 48 units per month in the LT-2 category under the Gruhajyothi scheme, will be entitled to additional 10 units instead of 10% of average monthly consumption

- Additional units above the entitled units have to be paid for, by each installation.
- However, if the monthly consumption is above 200 units, the beneficiary has to pay the bill fully for that month. The scheme benefit resumes from the following month.

Mode of delivery: reduction in electricity bill for the household.

6.2 Measures to assess responses to the scheme

Gruhajyothi scheme is targeted to deliver differential benefits to all domestic installations (households), with additional quantities of electricity to target households. This leads to the following hypotheses about household responses to the scheme:

Hypothesis 1: The scheme effectively delivers an additional 10% electricity allowance to households.

If households do not increase their use of electricity from their current levels, household electricity expenditure, up to the allowance, will be free. The household expenditure on electricity is likely to *decrease* after the introduction of the scheme.

Similarly, there is likely to be an *increase* in the number of households that have zero electricity expenditure.

Hypothesis 2: If electricity expenditure reduces, households might buy *more* appliances with their savings.

Some of these appliances include the washing machine, air-conditioners (ACs) and computers. Washing machines help reduce the household work of the women of the household, which frees up time for other activities (productive or leisure). Given recent changes in temperature and the climate, an electricity subsidy may be high enough to increase AC adoption. Personal computers are often used for productive/remunerative activities in the formal sector or increasingly for education, and electricity subsidies may encourage an increase in ownership of computers.

Hypothesis 3: Free electricity is likely to reduce the revenue of DISCOMs, and worsen their financial health. This stress might lead to increased power cuts and *reduced* power availability.

6.3 Gruhajyothi assessment

All the above variables are observed at the household level. The expenditure variables are observed on a monthly-basis while the others are observed at the wave level.

6.3.1 Fraction of households with zero electricity expenditure, across regions

Table 6.1 Fraction of households with zero electricity spend (%)

	Fraction of households with zero electricity spend (%)		
	2022	2023	2024
HR 58	0.5	25.3	75.2
HR 59	0.6	30.0	61.2
HR 60	0.6	24.1	41.4
HR 61	2.1	26.9	42.5
HR 62	3.4	32.0	90.0
KA*	1.3	28.0	63.1
India [§]	15.9	17.9	22.8

*All Karnataka sample

[§]All India sample including Karnataka

We examine the fraction of households with zero electricity expenditure in Table 6.1. We focus on the fraction for the pre-scheme period (2022) and the post-scheme introduction period (2024) since the 2023 period is a mix of observations before and after the introduction of the scheme. We follow a similar pattern of pre-post analyses for all the following variables.

Given that the scheme has increased every household's allocation by at least 10% of their average consumption, we see a sharp increase in the fraction of households who have zero electricity bills. Table 6.1 shows that the fraction of households with zero electricity expenditure has increased from 1.3% in 2022 to 63.1% in 2024 for the entire Karnataka household sample. However, we do see variation across HRs. HR 60 (Coastal Karnataka) and HR 61 (Dharwad) have slightly more than 40% of the households with zero electricity expenditure as compared to 75% of HR 58 and 61% of HR 59. Less than 10% of the households in HR 62 (Bellary) pay any electricity bills. The yearly average data in Table 6.1 shows that the households have not significantly changed their consumption behaviour - for the majority of households their new consumption is still less than the increased allowance, allowing them to have zero electricity expenditure. These findings are in-line with our hypothesis 1.

Figure 6.1 Fraction of households with zero electricity spend (%), across HRs

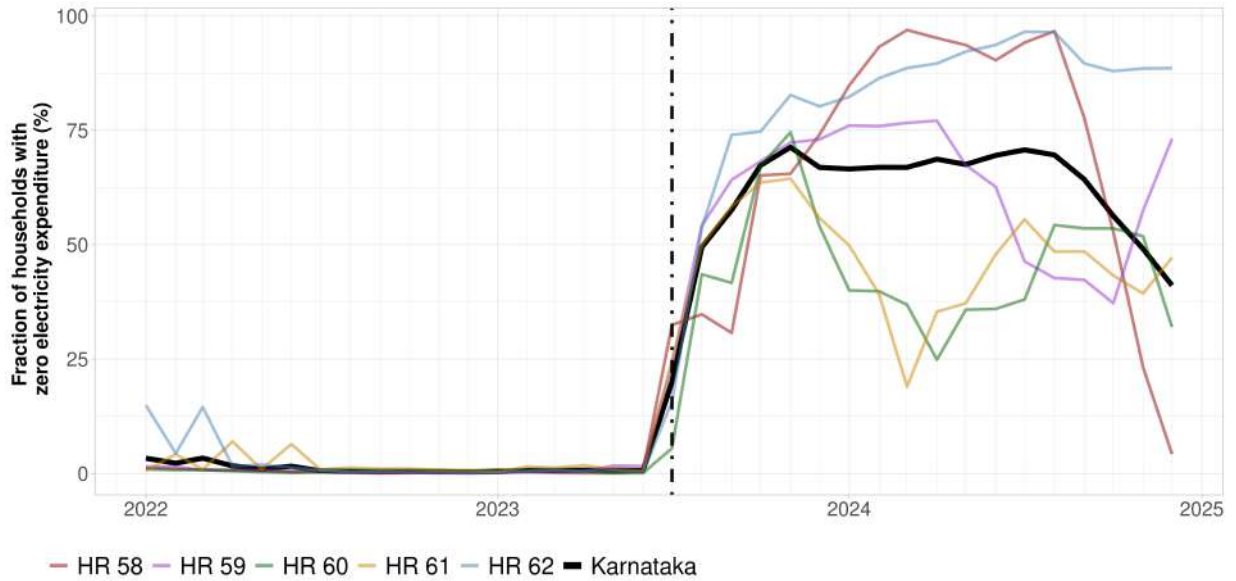


Figure 6.1 shows that the Gruhajyothi scheme has led to a sharp increase in the fraction of households with zero electricity expenditure. The sharp increase was seen across all HRs after scheme implementation. While HR 62 (Bellary) has been rising after scheme implementation, HRs 59, 60, and 61 are settling to a higher fraction (compared to pre-scheme) after some fluctuations. HR 58 (Bengaluru) saw a sharp decline in the latest period - we need to observe this over some more time to see if this is a structural change.

6.3.2 Fraction of rural vs. urban households with zero electricity expenditure

Table 6.2 Fraction of households with zero electricity spend (%)

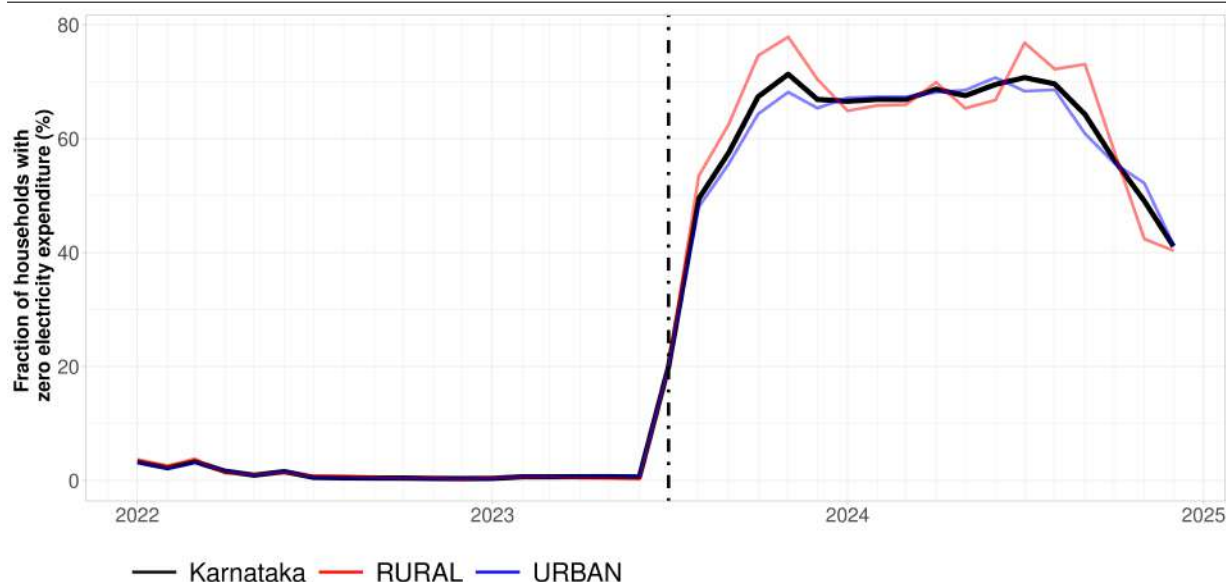
	Fraction of households with zero electricity spend (%)		
	2022	2023	2024
Rural	1.4	30.1	63.4
Urban	1.2	27.1	63.0
KA*	1.3	28.0	63.1
Rural India	16.0	17.8	22.4
Urban India	15.8	17.9	23.0
India [§]	15.9	17.9	22.8

* All Karnataka sample

[§] All India sample including Karnataka

Table 6.2 shows that fraction of households with zero electricity expenditure was around $\sim 1\%$ in both rural and urban areas in 2022 and has increased to $\sim 63\%$ in 2024. We see a decline in the fraction in the latest wave (Figure 6.2) across both rural and urban areas, but we should observe this over another wave to draw any conclusions.

Figure 6.2 Fraction of households with zero electricity spend (%), Rural vs Urban



6.3.3 Median monthly household electricity expenditure: Rs., across HRs

Table 6.3 Median monthly household electricity expenditure, across regions

	Median monthly household electricity expenditure (Rs.)		
	2022	2023	2024
HR 58	350.0	270.0	0.0
HR 59	300.0	209.2	0.0
HR 60	240.0	250.0	40.0
HR 61	231.2	215.0	40.0
HR 62	254.2	250.0	0.0
KA*	290.0	255.0	0.0
India [§]	350.0	350.0	405.0

*All Karnataka sample

[§]All India sample including Karnataka

Table 6.3 shows that the median monthly household electricity expenditure in Karnataka was Rs. 290 in 2022, which has dropped to zero in 2024. The drop in expenditure is seen across all HRs. The drop is the highest in HR 58 (Bengaluru) and HR 59 (Mysore) where the median expenditure has dropped from Rs. 350 and Rs. 300 respectively in 2022 to zero in 2024. The drop in expenditure is the lowest in HR 61 (Dharwad) where the median expenditure has dropped from Rs. 231.2 in 2022 to Rs. 40 in 2024.

Figure 6.3 Median monthly household electricity expenditure, across HRs

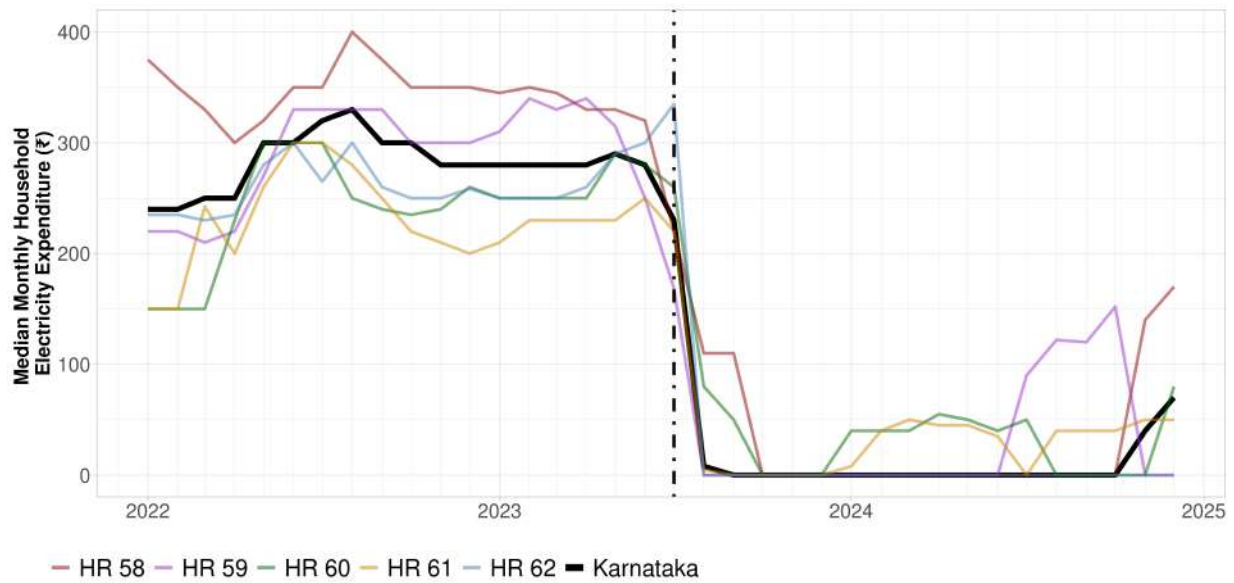


Figure 6.3 shows a sharp drop in median monthly electricity expenditure across all HRs, immediately after the scheme implementation. We need on-ground investigation for HRs 60 and 61 to understand different magnitudes of expenditure reduction as compared to other HRs. This drop can be attributed to the increased free electricity allowance provided by the scheme (hypothesis 1).

6.3.4 Median monthly electricity expenditure in rural vs. urban households

Table 6.4 shows that while urban households had a higher median electricity expenditure in 2022, the median electricity expenditure has fallen to zero in 2024 for both rural and urban households.

Table 6.4 Median monthly household electricity expenditure, rural vs. urban

	Median monthly household electricity expenditure (Rs.)		
	2022	2023	2024
Rural	250.0	239.5	0.0
Urban	300.0	262.5	0.0
KA*	290.0	255.0	0.0
Rural India	300.0	300.0	350.0
Urban India	377.5	382.5	475.0
India§	350.0	350.0	405.0

*All Karnataka sample

§All India sample including Karnataka

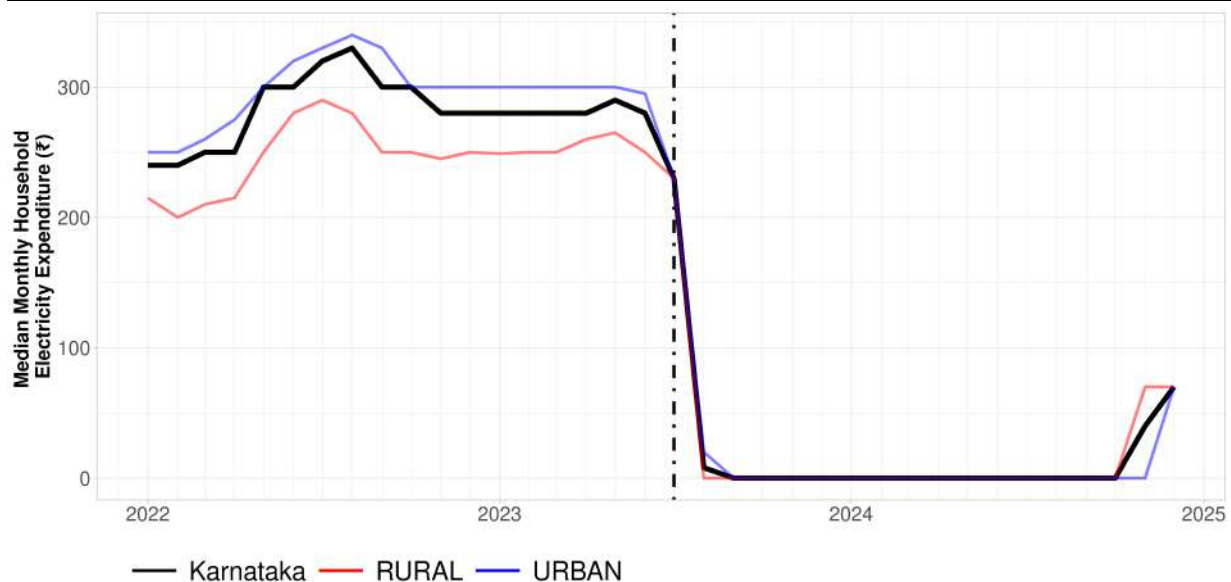
Figure 6.4 Median monthly electricity expenditure for rural vs. urban households

Figure 6.4 shows that patterns in median expenditure on electricity has fallen sharply to zero across both rural and urban areas, after scheme implementation. However, we are seeing some increase in the latest wave.

6.3.5 Fraction of households with washing machines across HRs

Table 6.5 Fraction of households with washing machines, across regions

	Fraction of households with washing machines (%)		
	2022	2023	2024
HR 58	45.3	44.2	88.6
HR 59	16.8	38.7	60.5
HR 60	18.0	23.0	39.5
HR 61	12.3	12.6	20.4
HR 62	1.6	7.2	11.3
Rural	4.0	6.3	25.4
Urban	27.1	31.6	54.1
KA*	20.9	24.6	45.5
Rural India	18.1	22.6	31.9
Urban India	39.2	49.7	63.6
India§	32.9	41.4	52.5

* All Karnataka sample

§ All India sample including Karnataka

Table 6.5 shows that the fraction of households with washing machines more than doubled to 45.5% from 2022 to 2024 for the entire Karnataka sample. We see a higher ownership in HR 58 (Bengaluru) and HR 59 (Mysore) as compared to other HRs in the post-scheme introduction period.

Figure 6.5 Patterns in fraction of households with washing machines, across HRs

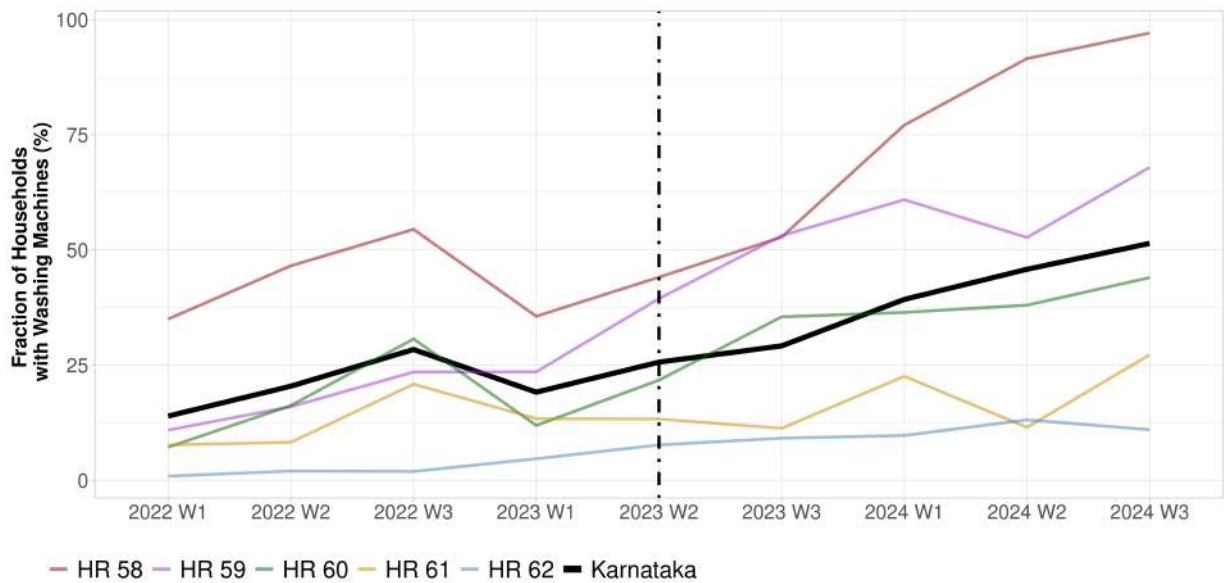


Figure 6.5 shows an increasing trend in washing machine ownership across all HRs. There is a difference between rural and urban households with every other household in urban areas having a washing machine. The difference in rural and urban areas reflect the priorities of households - urban areas have higher demands for the time of women - possibly making the households invest in washing machines to reduce female household labour load (links to hypothesis 2 in Section 6.2).

6.3.6 Fraction of households with air-conditioners, across regions

Table 6.6 Fraction of households with ACs (%)

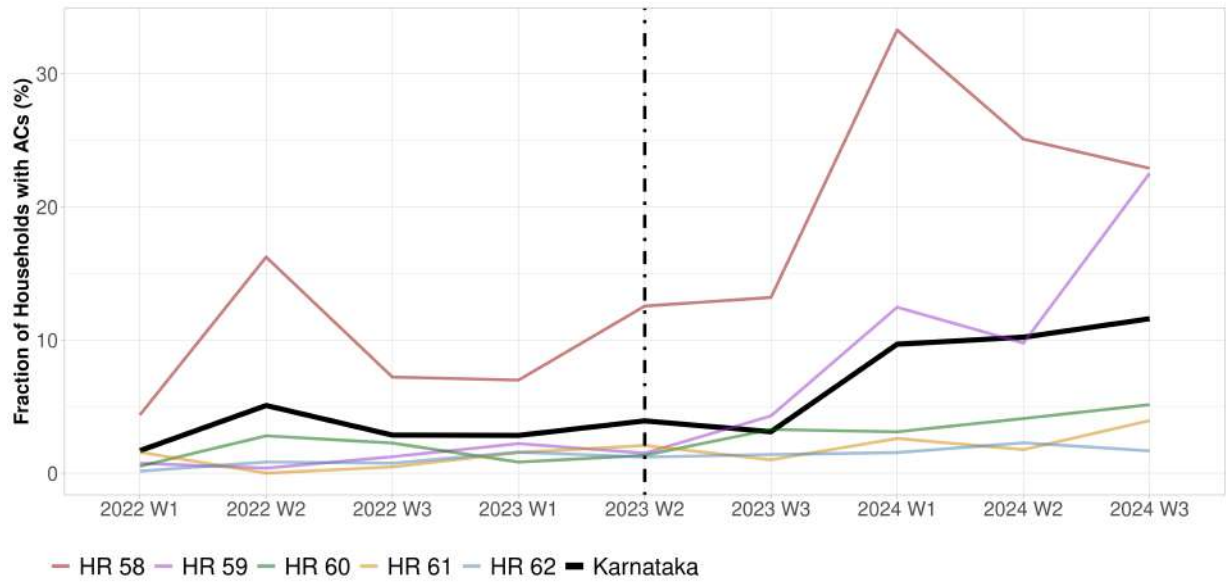
	Fraction of households with ACs (%)		
	2022	2023	2024
HR 58	9.3	10.9	27.1
HR 59	0.8	2.7	14.9
HR 60	1.9	1.8	4.1
HR 61	0.7	1.5	2.8
HR 62	0.6	1.4	1.8
Rural	0.7	0.9	2.4
Urban	4.2	4.2	14.0
KA*	3.2	3.3	10.5
Rural India	2.8	4.3	8.5
Urban India	15.8	19.8	32.1
India [§]	11.9	15.1	23.8

*All Karnataka sample

[§]All India sample including Karnataka

Table 6.6 shows that less than 10% of households own an AC even after the scheme was introduced. The table also shows that households owning an AC tripled between the pre-scheme period and the post-scheme period for the entire Karnataka sample. There is wide cross-regional variation with almost one in three households in the HR 58 (Bengaluru) owning ACs, as compared to very low ownership across HRs 60, 61, and 62 in the post-scheme period (2024). Households in urban areas have over 6 times higher AC ownership as compared to rural areas in 2024. The household ownership of ACs is lower in Karnataka as compared to India.

Figure 6.6 Patterns in fraction of households with ACs across regions



From Figure 6.6, we don't see any increase in ownership that can be easily attributed to the Gruhajyothi scheme.

6.3.7 Fraction of households owning computers, across regions

Table 6.7 Fraction of households with computers

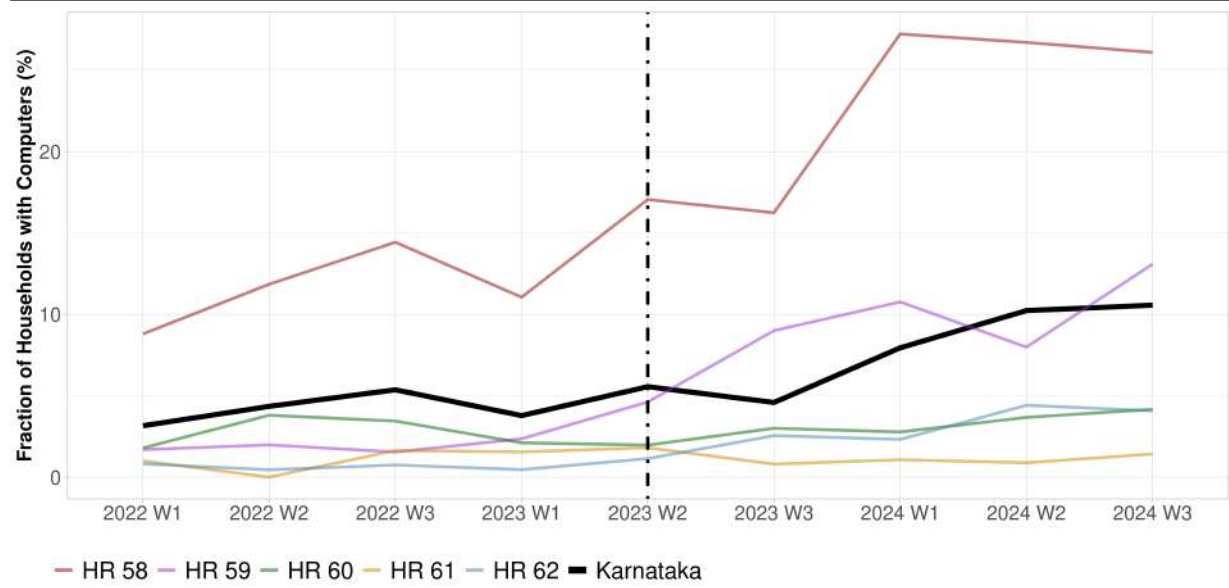
	Fraction of households with computers (%)		
	2022	2023	2024
HR 58	11.7	14.8	26.7
HR 59	1.7	5.3	10.6
HR 60	3.0	2.4	3.5
HR 61	0.9	1.4	1.1
HR 62	0.7	1.4	3.6
Rural	0.9	1.2	1.9
Urban	5.6	6.0	12.9
KA*	4.3	4.6	9.6
Rural India	1.4	2.5	5.8
Urban India	8.6	12.4	23.0
India [§]	6.5	9.4	17.0

*All Karnataka sample

[§]All India sample including Karnataka

Table 6.7 shows that the fraction of households with computers has increased slowly from 4.3% in 2022 to 9.6% in 2024 in the state. Since computers are required in many formal sector jobs and for education, it possibly explains higher ownership of computers by households in technology hubs like Bengaluru (HR 58) and Mysore (HR 59) as compared to other HRs, even in the post-scheme period (2024). Urban areas tend to have higher formal employment opportunities, which is reflected in the higher computer ownership in urban areas. Figure 6.7 validates the observations made in the Table 6.7.

Figure 6.7 Patterns in the fraction of households owning computers, across HRs



6.3.8 Average availability of electricity per day, across regions

Table 6.8 Average availability of electricity in a day

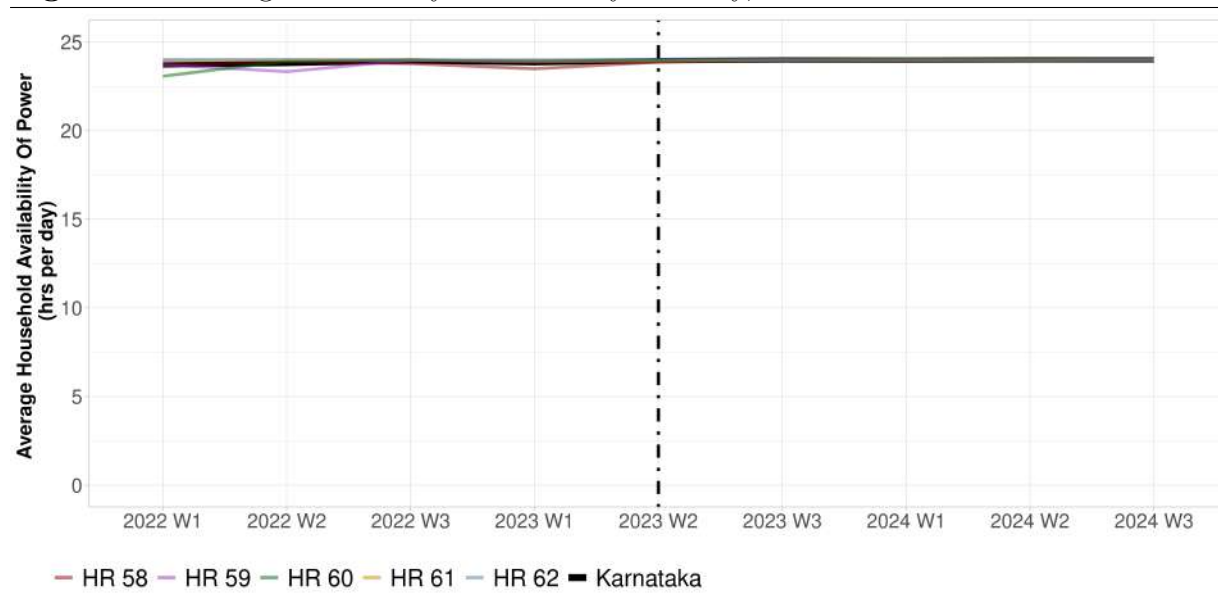
	Average availability of electricity (hrs per day)		
	2022	2023	2024
HR 58	23.8	23.8	24.0
HR 59	23.7	24.0	24.0
HR 60	23.7	24.0	24.0
HR 61	24.0	23.9	24.0
HR 62	24.0	24.0	24.0
KA*	23.8	23.9	24.0
India [§]	22.9	23.0	23.1

*All Karnataka sample

[§]All India sample including Karnataka

Table 6.8 shows that the average availability of electricity in a day for the overall Karnataka sample has *increased* from 23.8 hours in the pre-scheme period to 24 hours in the post-scheme period. This suggests that the households are now experiencing universal availability of electricity.

Figure 6.8 Average availability of electricity in a day, across HRs



The wave-by-wave pattern in Figure 6.8 also shows a consistent high availability of electricity in all HRs. This disproves our hypothesis 3 that the scheme will lead to reduced power availability. However, the scheme has only been in place for a short period, and it is useful to track power availability over time, as one of the possible indicators of the financial stress on DISCOMs.

6.3.9 Average availability of electricity: hrs per day, Rural vs Urban

In Table 6.9, we observe that the rural areas had 23.5 hours of electricity in a day in 2022, which has increased to 24 hours in 2024. Urban areas had 23.9 hours of electricity in a day in 2022, which has increased to 24 hours in a day in 2024. Rural availability of electricity has converged with the urban areas now. Karnataka does better in terms of electricity availability as compared to India.

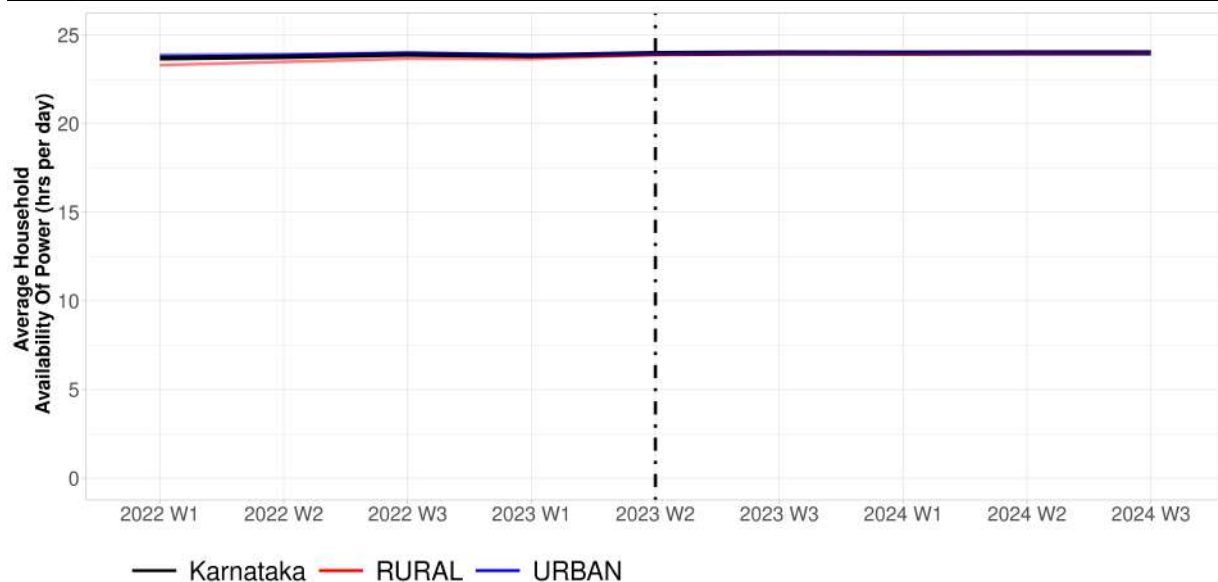
Table 6.9 Average availability of electricity in a day

	Average availability of electricity (hrs per day)		
	2022	2023	2024
Rural	23.5	23.8	24.0
Urban	23.9	24.0	24.0
KA*	23.8	23.9	24.0
Rural India	22.5	22.5	22.6
Urban India	23.0	23.2	23.4
India [§]	22.9	23.0	23.1

*All Karnataka sample

[§]All India sample including Karnataka

Figure 6.9 Average availability of electricity in a day, Rural vs Urban



6.4 Summary with key findings

Gruhajyothi scheme provided free, fixed monthly electricity units to domestic installations. Every household received, at least, an additional 10% electricity allowance to their previous year's average consumption. We have three observations basis the hypotheses laid out in section 6.2:

1. The scheme has led to a sharp increase in the fraction of households with zero electricity expenditure. 63% of the households in Karnataka had zero electricity expenditure in 2024. This is because households did not increase their consumption at a similar rate to the increased allowance (Tables 6.1, 6.2, Figures 6.1, 6.2).

The scheme has also led to a corresponding sharp drop in median monthly electricity expenditure across all HRs (Table 6.3, Figure 6.3). This reduction in monthly expenditure is similar in rural and urban areas (Table 6.4, Figure 6.4).

2. We also hypothesised that decrease in electricity expenditure would lead to households investing in more appliances with their savings. We tracked ownership of washing machines, ACs, and computers.

The scheme has led to an increasing trend in washing machine ownership across HRs with Bengaluru leading the pack at 88.6% ownership in 2024. Urban households have over 50% washing machine ownership in 2024 as compared to only 25.4% in rural households. Washing machines reduce the cleaning load on the women of the household, thus, freeing up time for other productive activities (Table 6.5, Figure 6.5).

Households with AC ownership have also been slowly increasing across HRs with 10.5% of Karnataka households owning an AC. Here also, we see higher ownership patterns in urban areas and in southern HRs (Bengaluru and Mysore) (Table 6.6, Figure 6.6).

There's also been a slow increase in computer ownership across HRs, with IT hubs Bengaluru and Mysore leading. Urban areas have higher computer ownership as compared to rural areas. Computers are often used for productive/remunerative activities in the formal sector or for education - thus, potentially increasing future incomes for the household (Table 6.7, Figure 6.7).

3. We observe an increase in the average availability of electricity in a day for the overall Karnataka sample over the years. We also observe that the rural availability of electricity has converged with the urban areas now and is at 24 hours (Tables 6.8, 6.9, Figures 6.8, 6.9).

Overall, the Gruhajyothi scheme has reached many households across Karnataka and has led to a significant reduction in electricity expenditure for households. We will see long term impact of the scheme as we see more household data over time.

Chapter 7

The Gruhalakshmi scheme

The Gruhalakshmi scheme involves a cash transfer to the women or trans-gender head of families in Karnataka, with eligibility criteria that is related to their income. This scheme was announced on 15th June 2023, and disbursement started on 15th August 2023.

7.1 Scheme design

Scheme eligibility - Inclusion criteria:

1. Women who are mentioned as head of the family with Antyodaya or BPL or APL ration cards, or
2. Trans-gender person who are similarly mentioned in any of the listed ration cards.

Scheme eligibility – Exclusion criteria:

1. Head of the family, or the husband of the head, is an income tax-payer/files GST.

Benefits to eligible households:

- Rs.2000 per month to the female/trans-gender head of family and listed in the ration card.

Mode of delivery: an AADHAAR linked bank account/IPPB account of the female/trans-gender head of the family, whether the account is a joint account or not.

(For the remainder of the report, we will refer to the Gruhalakshmi scheme as benefiting women, for ease of discussion.)

7.2 Measures to assess responses to the scheme

Unlike the Annabhagya and Gruhajyothi schemes which targets the overall household, Gruhalakshmi is specifically targeted towards placing income in the hands of the woman head of the household. This leads to the selection of two sets of measures: those for the efficiency of the targeted delivery (different from the overall delivery of benefits presented in Chapter 4), and those that measure responses of the household. These are listed as follows:

Hypothesis 1: Since the scheme involves an income transfer to all households with a woman registered on the household ration card, we expect that households with an eligible woman who is above 18 years of age receive this income.¹

Since the delivery mechanism for the cash transfer is through a bank account, we additionally examine what fraction of the women in the household have a bank account.

Hypothesis 2: We expect that household expenditure on women-related goods and services may see an increase, if earlier there were household budget constraints preventing these.

This includes expenditure on sanitary napkins, cosmetics, beauty and hygiene products as well as spending on services offered by parlors, salons and spas, and spending on daycare. This also includes health and education expenditures as it is likely that women will choose to spend on education of their children, or to spend on medical care if the household receives excess income because of their presence.

We measure both the level of expenditure and its volatility, expecting that the level may be higher after the scheme introduction, and the volatility may drop, if there were constraints before the scheme introduction.

As with the previous scheme assessment, the above measures are observed at the level of the family or household. The income and expenditure variables are observed on a monthly-basis.

¹This is the eligibility to be registered on the household ration card.

7.3 Gruhalakshmi assessment

In the following assessment, the changes as listed in the above hypotheses will be examined using household features and household expenditure for the CPHS sample, for the overall state as well as for the HR samples. This will be done for the sample of rural and urban households as well.

7.3.1 Households with woman heads in Karnataka, across regions

The CPHS sample does not track which members of the household are registered as part of the household ration card. However, there is information about the individual members in the household and their age, which allows a count of women who are above 18 years of age. Table 7.1 presents the fraction of households in Karnataka that have at least one such woman.

Table 7.1 Fraction on households with a woman head, across regions

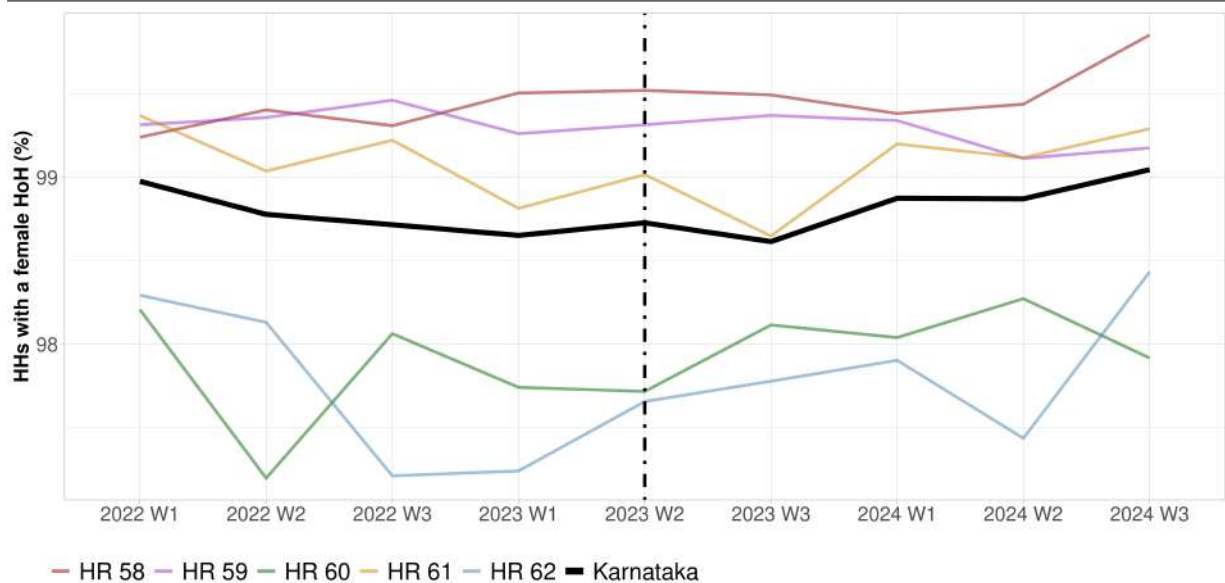
	Fraction of households with a woman (%)		
	2022	2023	2024
HR 58	99.3	99.5	99.6
HR 59	99.4	99.3	99.2
HR 60	97.8	97.9	98.1
HR 61	99.2	98.8	99.2
HR 62	97.9	97.6	97.9
Rural	98.5	98.5	99.0
Urban	99.0	98.7	98.9
KA*	98.8	98.7	98.9
Rural India	98.1	98.7	99.0
Urban India	98.2	98.7	98.7
India [§]	98.2	98.7	98.8

* All Karnataka sample

[§] All India sample including Karnataka

The fraction of households with a woman that fits the Gruhalakshmi criteria has been at more than 98%, even in 2022, before the scheme introduction. This level has remained the same, or increased slightly to 98.9%, in the post-scheme introduction period. There has been a modest increase between these two periods for rural households, while urban households, broadly, remain the same.

Figure 7.1 Wave-wise patterns of households with a woman head, across regions



This pattern is visible in Figure 7.1. The vertical line marks the start of Wave 2 of 2023, which covers the months of May to August of the year. The scheme was operationalised in August 2023, after which there is a positive change in the fraction of households reporting a woman head. The patterns show a greater wave to wave variation in this fraction, than the change in the overall sample. The southern HRs (58 and 59) has a marginally higher fraction of households with women who are older than 18 years, compared to the coastal northern HRs (60 and 62).

Table 7.2 Fraction of households with women holding bank accounts, across regions

	Fraction of households with women holding bank accounts (%)		
	2022	2023	2024
HR 58	99.4	99.5	99.7
HR 59	99.4	99.2	99.0
HR 60	97.8	97.4	98.0
HR 61	99.1	98.4	99.2
HR 62	97.9	97.5	97.9
Rural	98.5	97.9	99.0
Urban	98.9	98.7	98.9
KA*	98.8	98.5	98.9
Rural India	98.2	97.9	98.0
Urban India	98.3	98.1	97.8
India [§]	98.3	98.1	97.9

*All Karnataka sample

[§]All India sample including Karnataka

Table 7.2 presents the fraction of households with a woman member holding a bank account. This data similarly indicates that in more than 98% of households, women have their own bank accounts. Both rural and urban areas have high banking penetration. We observe a small increase relative to the pre-scheme level in rural areas.

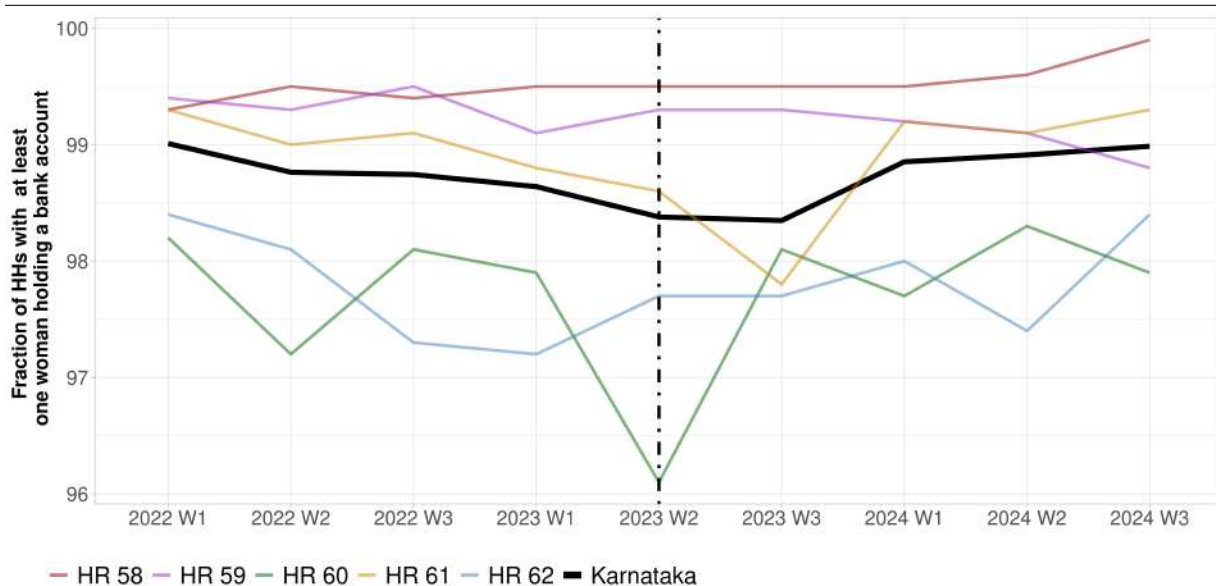
Figure 7.2 Wave-wise patterns in the fraction of households with a woman member holding a bank account, across regions

Figure 7.2 shows that the fraction of households with women holding bank account has

been stable, with a slight drop in the pre-scheme period covering Wave 1 2022 to Wave 2 2023. After the scheme was operationalised, there is a perceptible increase in the positive change in the fraction of households where the woman has her own bank account. There is a much higher cross-HR variation in this fraction, both compared to the overall Karnataka sample and for the trends in each HR. However, this variation is small compared to the overall level of households where women have their own accounts.

Table 7.3 Median monthly government transfers to woman headed household, across regions

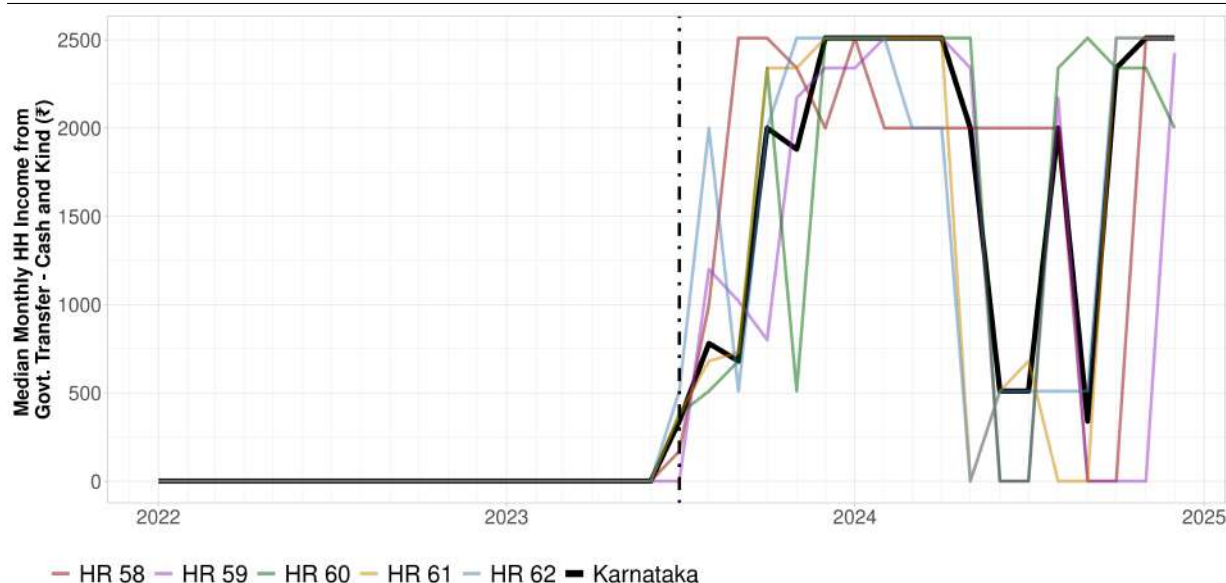
	Monthly income from government transfers to the median woman headed household (Rs.)		
	2022	2023	2024
HR 58	0.0	85.0	2000.0
HR 59	0.0	0.0	2255.0
HR 60	0.0	190.0	2425.0
HR 61	0.0	190.0	2510.0
HR 62	0.0	255.0	2000.0
KA*	0.0	170.0	2425.0
India [§]	0.0	195.0	325.0

* All woman headed households in Karnataka

[§]All woman headed households in India including Karnataka

Given that more than 98% of the households in the sample have a woman member with a bank account who is eligible for the scheme, we anticipate that the median household in the sample is likely to receive the income transfer. Table 7.3 presents the amount of government transfers that the median woman headed household in the sample received in the pre-scheme, and in the post-scheme introduction period. This was the amount received across both cash and kind, which the median household reported as Rs. 0 in the pre-scheme period. In the post-scheme introduction period, the median household with a woman head reported receiving monthly government transfers of Rs. 2425.

Figure 7.3 Patterns of monthly government transfers to the median woman headed household, across regions



This change is observed systematically for the median woman headed household across all the HRs. Figure 7.3 shows that the median woman headed household was reporting no income from government transfer received in the months of the pre-sample introduction period. These are not reported differently for each scheme, and is an aggregate income transfer from all the schemes. The monthly pattern shows that there was variation in the scheme roll-out across the different HRs. However, for all HRs, the monthly transfer to the median household was substantially higher by more than Rs.2000 in the post-scheme introduction period. From Figure 7.3, we are also seeing sudden drops and surges in government transfers across all HRs post May 2024. This may be because of lags in government transfers and/or bunching of monthly transfers across few months. While the transfers may sometimes be delayed by a few months in different regions, from Table 7.3, it appears that the HRs are receiving the transfers fairly regularly.

Table 7.4 Median monthly government transfers to woman headed household, rural vs. urban regions

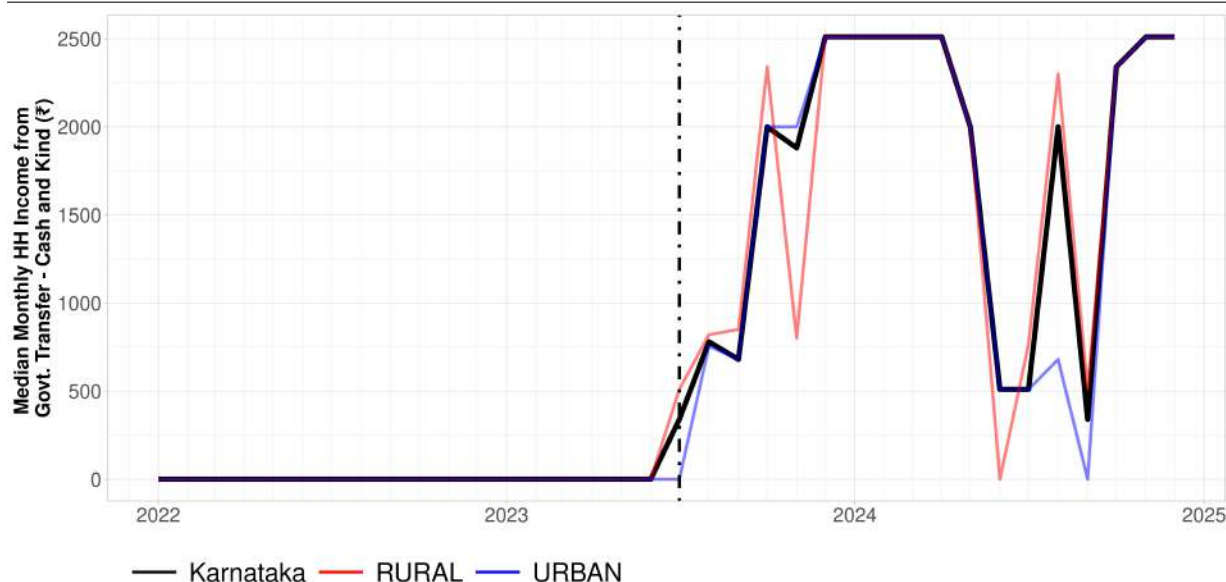
	Median monthly household income from government transfers - cash & kind (Rs)		
	2022	2023	2024
Rural	0.0	255.0	2425.0
Urban	0.0	0.0	2425.0
KA*	0.0	170.0	2425.0
Rural India	80.0	307.5	475.0
Urban India	0.0	125.0	257.5
India [§]	0.0	195.0	325.0

* All woman headed households in Karnataka

[§]All woman headed households in India including Karnataka

Table 7.4 and Figure 7.4 finds that these changes are reflected in the examination of the median woman headed household, whether these are in the rural or the urban regions. There was some variation in the roll-out in the rural regions. However, these appear to have stabilised by the end of our sample period of observation. There were drops and surges post May 2024 that seem to have affected both rural and urban regions.

Figure 7.4 Patterns in monthly government transfers to the median woman headed rural and urban households



Similarly, the monthly patterns of fraction of households receiving government transfers in Figure 7.5 show that this benefit to woman headed households across all regions holds for all the HRs, even though there is some variation across the regions. HR 59 has a lower

fraction of woman headed households receiving income transfers by the end of the period of observation. What drives such variation requires a deeper examination of local area data.

Table 7.5 Fraction of woman headed households receiving government transfers, across regions

	Fraction of woman headed households receiving government transfers (%)		
	2022	2023	2024
HR 58	10.5	48.7	69.9
HR 59	8.8	39.5	64.7
HR 60	7.3	43.9	73.9
HR 61	7.9	43.8	73.0
HR 62	11.9	40.1	69.6
KA [*]	9.6	43.1	71.7
India [§]	28.8	32.4	44.1

^{*}All woman headed households in Karnataka

[§]All woman headed households in India including Karnataka

What about the fraction of woman headed households that are getting any income transfers from the government at all? Table 7.5 shows that around 10% of the sample of woman headed households reported receiving some income transfers from the government in the pre-scheme introduction period (2022). This increased to nearly 71% in the post-scheme introduction period. This is systematic change, with the substantial increase reflecting across all the HRs. The fraction is much higher for Karnataka as compared to India.

Figure 7.5 Patterns in fraction of woman headed households receiving government transfers, across regions

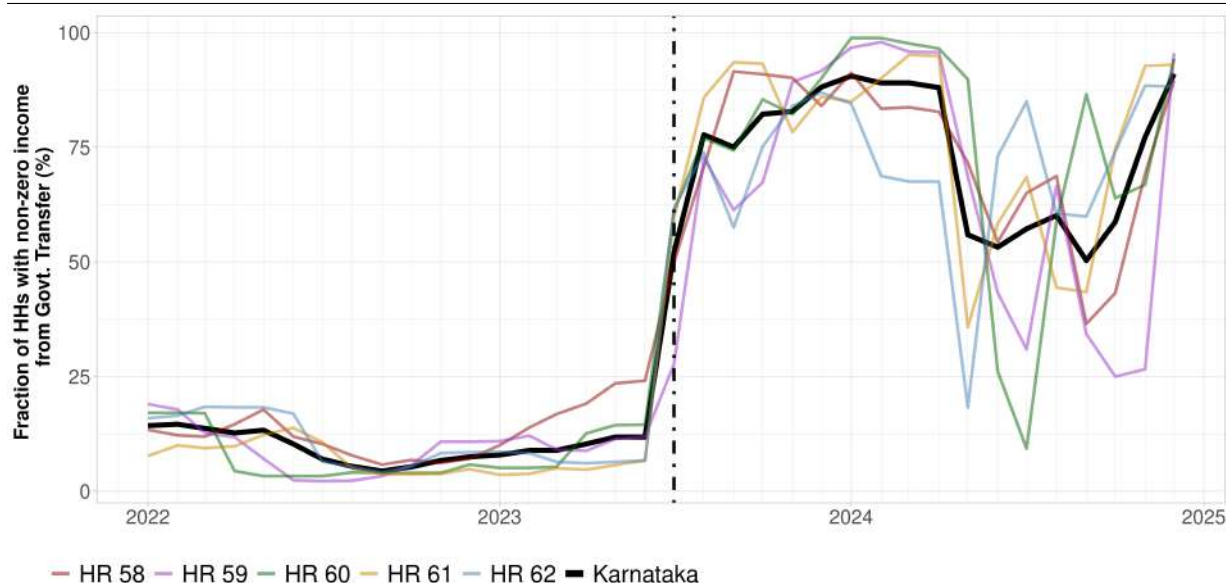


Figure 7.5 validates the observations in Table 7.5. One inference from this can be that there are administrative gaps in identifying eligible women headed households in the HR 59 region. This cross-sectional variation can be a useful indicator on where further effort is required to close any information gaps in delivering scheme benefits. Figure 7.5 also reflects the sudden drop in transfers post May 2024. We are observing an uneven recovery across HRs starting July 2024.

Therefore, in combination with the changes in values received by the median woman headed household from the pre-scheme period to the post-scheme introduction period, this table suggests that the scheme transfers are effective. As pointed out earlier, it is difficult to link the transfers received by the household to a specific scheme, since the CPHS survey does not differentiate income received by scheme, but rather as the aggregate amount received by the household.

Table 7.6 Fraction of woman headed rural and urban households receiving government transfers

	Fraction of woman-headed households receiving income transfers (%)		
	2022	2023	2024
Rural	13.7	46.8	75.0
Urban	8.2	41.8	70.4
KA*	9.6	43.1	71.7
Rural India	31.3	33.1	47.5
Urban India	27.7	32.1	42.3
India [§]	28.8	32.4	44.1

*All woman headed households in Karnataka

[§]All woman headed households in India including Karnataka

A similar examination of the coverage of the sample households in rural regions and urban regions show that more eligible woman headed households receive government transfers in the post-scheme introduction period compared to the pre-scheme period. Table 7.6 presents the regional comparison. Even in the pre-scheme region, a higher fraction of rural region households with eligible women were recipients of government income transfers compared to those in urban regions. Both saw a substantial increase in coverage: the fraction of rural sample households with scheme eligible women receiving an income transfer increased from 13.7% in the pre-scheme period to 75% in the post-scheme period. For similar urban households, the coverage increased from 8.2% to 70.4%.

Figure 7.6 Patterns in fraction of woman headed rural and urban households receiving government transfers

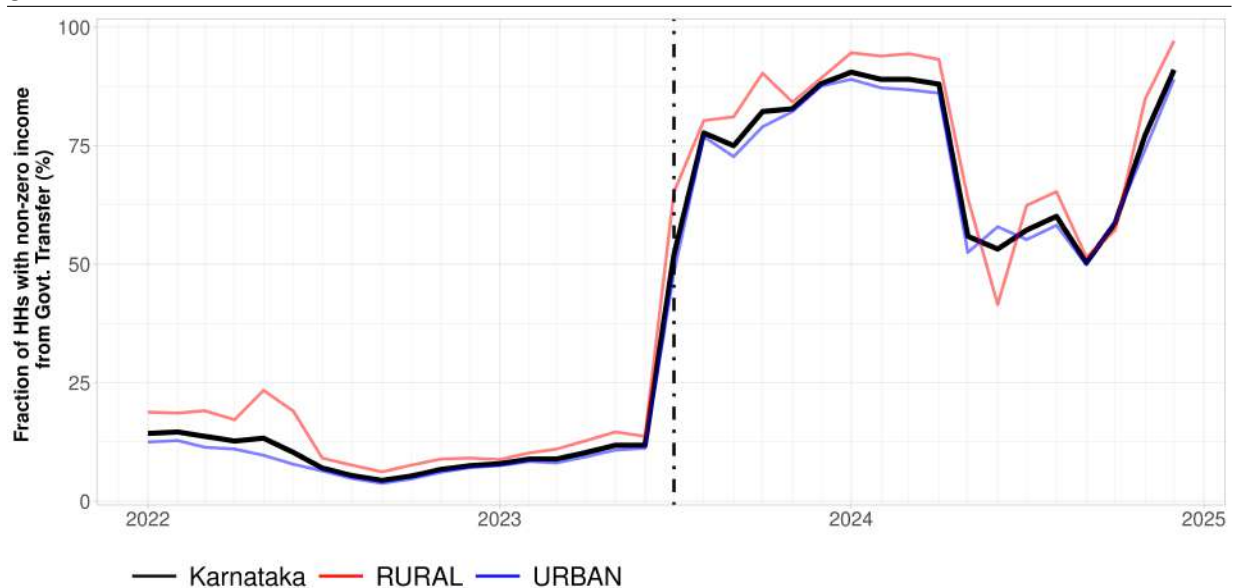


Figure 7.6 presents the monthly patterns of changes in the fraction of households in the urban and rural sample, and finds a significant increase in the fraction of households receiving government transfers immediately around the time that the scheme was implemented.

7.3.2 Median monthly household women-centric expenditure, across regions

One hypothesis is that a household where the female head receives income transfers will likely spend on expenditure items that are more likely to benefit women more than the overall household. Among the various items of expenditure in the CPHS, is the amount spent on clothing and footwear, cosmetics and toiletries, education, health, kitchen appliances, parlour and spa expenses, sanitary napkins, domestic help and laundry, diapers, utensils, creche and house-nurse for children, and elderly respectively. In addition, we include the expenditure by the household on education and health. We expect that these would be higher when female heads receive income over which they can exercise agency to spend.

Table 7.7 Median monthly women-centric household expenditure, across regions

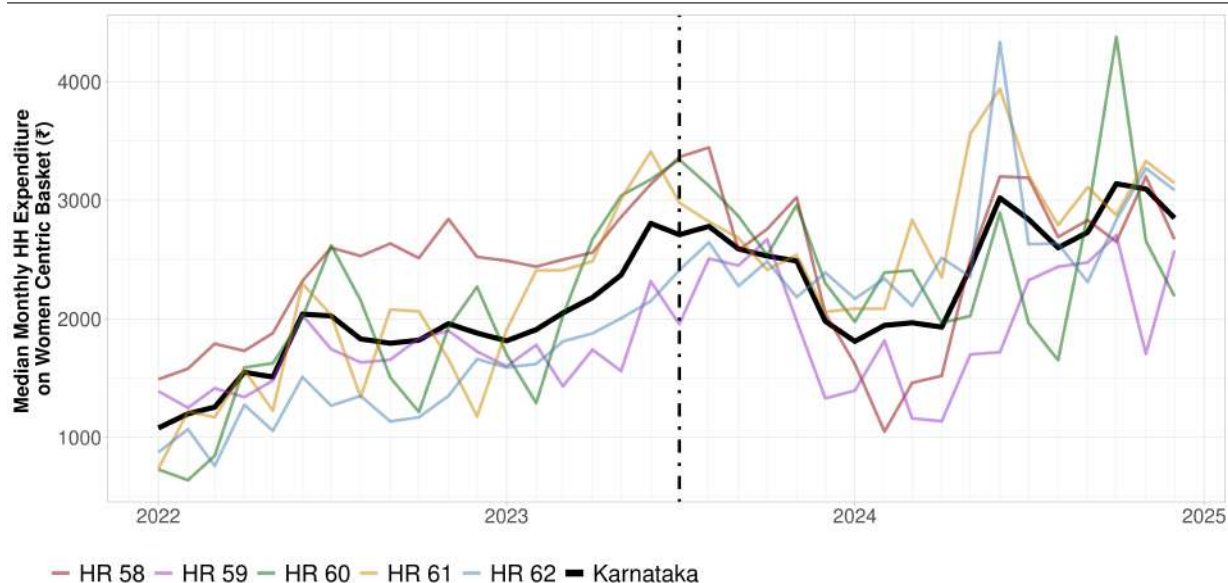
	Median monthly expenditure on women-centric basket (Rs.)		
	2022	2023	2024
HR 58	2416.2	2667.5	2660.0
HR 59	1643.8	1867.5	1768.5
HR 60	1608.0	2768.8	2290.0
HR 61	1454.0	2512.5	2992.5
HR 62	1219.2	2166.2	2572.0
KA*	1807.5	2429.5	2665.0
India [§]	1685.0	2185.0	2675.0

*All woman headed households in Karnataka

[§]All woman headed households in India including Karnataka

Table 7.7 shows the median monthly household expenditure when there is a female head of the household went up in the CPHS Karnataka sample, from Rs. 1807.5 in the pre-scheme period to Rs. 2665.0 in the post-scheme introduction period, a change of around 47%.

Figure 7.7 Patterns of women-centric household expenditure, across regions



When we examine the monthly patterns of these expenditures in Figure 7.7, we find a positive trend in these household expenditure across all the HRs. However, there have been drops and surges across all HRs post the scheme implementation.

Table 7.8 Median monthly women-centric expenditure in rural and urban households

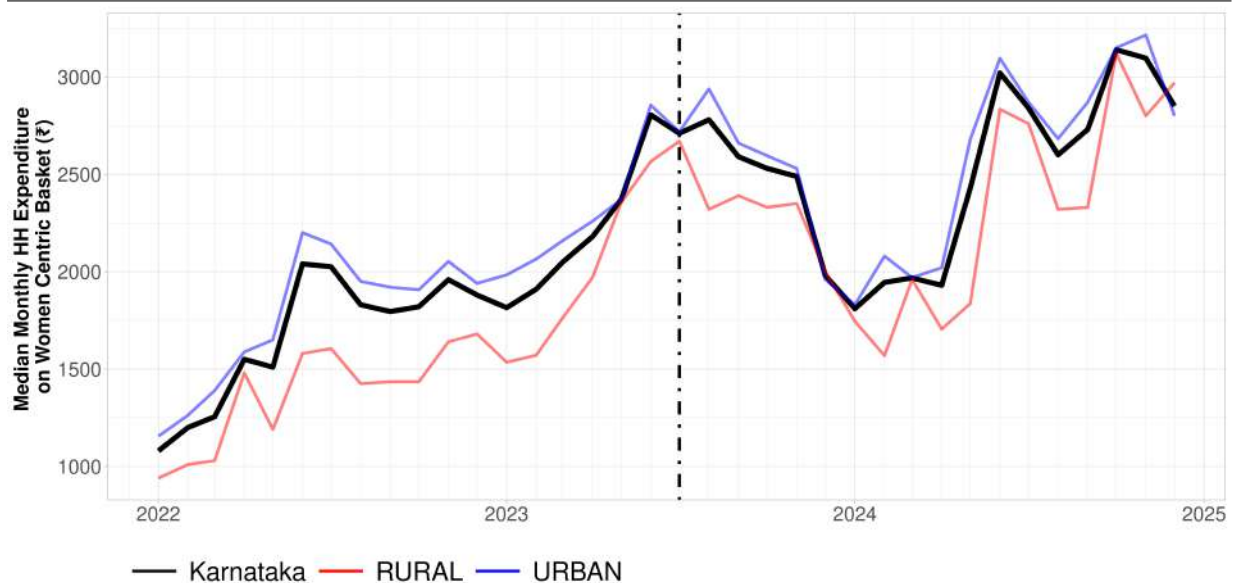
	Median monthly expenditure on women-centric basket (Rs)		
	2022	2023	2024
Rural	1435.0	2325.0	2325.0
Urban	1913.5	2450.0	2741.2
KA*	1807.5	2429.5	2665.0
Rural India	1470.0	1866.5	2135.5
Urban India	1780.0	2337.0	3055.0
India [§]	1685.0	2185.0	2675.0

* All woman headed households in Karnataka

[§]All woman headed households in India including Karnataka

Both rural and urban households in the CPHS sample, saw changes in the median monthly women-centric expenditure between the pre-scheme and the post-scheme introduction periods, as seen in Table 7.8. The rural household expenditure increased from Rs.1435 in 2022 to Rs.2325 in 2024. For urban households, the change is from Rs.1913.5 in 2022 to Rs.2741.2 in 2024.

Figure 7.8 Patterns in women-centric expenditure in rural and urban households



The monthly patterns in these expenditures support a similar inference. During the pre-scheme period, the monthly expenditure on women-centric baskets is increasing for both rural and urban households, but it grows faster for the rural households compared to urban households. After the introduction of the scheme, there is a flattening of these expenditures in both rural and urban households, followed by a drop. Post 2024, rural and urban areas have followed a similar increasing trend.

7.4 Summary with key findings

The Gruhalakshmi scheme transfers cash benefits into the account of a woman head of a household. Here, the identification of eligible households is done without access to administrative data, but using the proxy of households with a woman above 18 years of age and who has a bank account. Since the target benefit is the woman, the analysis is done on measures that reveal preferences and agency of the woman, as opposed to the family or a man head of the household. One such measure available in the CPHS data is household expenditure on items that are woman-centric.

The examination asks whether there has been an increase in the median monthly expenditure on such items in the post-scheme period. The findings are that

1. There is an increase in the post-scheme period expenditure on women-centric items compared to the pre-scheme period (Tables 7.7, 7.8, Figures 7.7, 7.8).
2. However, this expenditure has regional variation in households of the different HRs. These variations are similar to that observed in household food, and planned expenditure, observed in the assessment of the Annabhagya scheme in Sections 5.3.1 and 5.3.3.

There are two barriers to conduct an assessment of the Gruhalakshmi scheme: identification of households with a woman head of household, and variables to measure the agency of the woman in the use of income transferred to her account. This will be better done with (a) access to administrative data on households that are identified as having a woman head of household and those without, and (b) data from micro-surveys on a sample of both these households that better capture agency of the woman on how to spend the income received from government transfers.

Another approach to assess this scheme is to identify households with a woman head of household where the scheme is not operational, or it is not available. The first can be done if the roll-out of the Gruhalakshmi scheme was done in a phased manner across the different Karnataka HRs. The alternative is to identify households from the CPHS data in neighbouring states, and which are across the border from Karnataka. This increases the chances that the household in Karnataka and in the neighbouring border state have similar backgrounds and preferences, making it a source for a benchmark household response without the effect of the scheme.

Chapter 8

The Shakti scheme

The Shakti scheme was announced on 11th June 2023 and was effective immediately. The scheme provides free rides to domicile women/transgender-persons on state-run non-premium buses within Karnataka. A Shakti card is supposed to be issued to every beneficiary to avail the benefit.

8.1 Scheme design

Scheme eligibility - Inclusion criteria:

1. Domicile women/transgender persons (includes current members of households and persons who have migrated intra-state) in Karnataka.

Scheme eligibility - Exclusion criteria:

1. Men
2. Women who have migrated from other states

Benefits to eligible individuals:

- Free bus rides on state-run non-premium buses within Karnataka.

Mode of delivery: zero-value ticket against a valid Government of Karnataka/India identity card (containing the photograph and residential address of the passenger) until a Shakti card is issued.

8.2 Measures to assess responses to the scheme

The scheme offers women free intra-city and intra-state travel within Karnataka. This leads to some hypotheses about travelling patterns within households:

Hypothesis 1: Since the scheme is targeted towards enabling travel by women, the fraction of women who spend non-zero time on travel is likely to *increase*. This change will capture first-time travellers.

Given free bus travel, women are likely to travel more and farther. Thus, the average time spent on travel per day is likely to *increase*. Higher ability to travel also potentially expands the labour market available to women as they are able to travel farther for work.

Hypothesis 2: With the greater access to transport free bus rides provide, the fraction of working-age women participating in the labour force is likely to *increase*. While the ability to travel expands the types of labour available to them, it also allows more women to consider working in the first place.

Hypothesis 3: Given free bus rides for women, the household expenditure on bus, train, and ferry fare is likely to *decrease* after the scheme implementation for households where women have the agency to travel.

If more women travel on buses instead of private vehicles, the household expenditure on petrol and diesel is likely to *decrease* after the scheme implementation. However, if there is a higher congestion in bus travel, the household expenditure on petrol and diesel is likely to *increase* as more people shift to private vehicles.

Counterfactual 1: Given that the scheme is targeted only to women, a counterfactual to changes in women's travel pattern is to observe changes in men's travel patterns, around the same time of the scheme introduction. The hypothesis is that men's travel patterns do not change, but that women's travel patterns do.

Counterfactual 2: If travel is an integral input to enabling women's participation in the labour force, a counterfactual to any change in women's labour force participation is against the labour force participation of men, around the same time of the scheme introduction. The hypothesis is that men's travel patterns do not change in response to the Shakti scheme, but that of women do.

8.3 Shakti assessment

In the following assessment, the above hypotheses can be monitored using CPHS data on the time use of individual members of the household and elements of the reported household expenditure. CPHS tracks the number of hours spent on travel by individual members, and monthly household expenditure on bus/train/ferry and on petrol and diesel. The time travel variables are observed at the individual level per wave. The expenditure variables are observed at the household level on a monthly-basis.

8.3.1 Fraction of women with non-zero travel times, across regions

Table 8.1 Fraction of women with non-zero travel time

	Fraction of women with non-zero travel times (%)		
	2022	2023	2024
HR 58	26.9	27.0	65.7
HR 59	79.5	75.1	67.8
HR 60	56.0	75.1	45.3
HR 61	20.2	55.2	39.9
HR 62	13.3	21.3	25.3
KA*	39.6	51.7	50.6
India [§]	53.5	68.8	72.4

*All women in Karnataka

[§]All women in India including Karnataka

Table 8.1 presents the fraction of women with non-zero travel times in Karnataka, for the pre-scheme period (2022) and the post-scheme introduction period (2024). The fraction of women with at least some travel (as characterised by non-zero travel times) increased from 39.6% in 2022 to 50.6% in 2024 for the entire Karnataka women sample. This suggests that more women who were not travelling in the period before the scheme, are travelling in the period after the introduction of the scheme.

The table shows that this increase is not symmetric for all Karnataka regions. The highest increase in this fraction are for women in households of the HR 58 (Bengaluru) and HR 61 (Dharwad) regions. There is a decrease for women in households of the HR 59 (Mysore) and HR 60 (Coastal Karnataka), where the fraction of women travelling were higher (pre-scheme) than for the rest of Karnataka. HR 62 (Bellary) only sees a small increase post-scheme implementation. Women report taking more frequent, but short trips. The results align with hypothesis 1 presented in Section 8.2 that the scheme encouraged many more first-time travellers to travel.

Figure 8.1 Patterns in the fraction of women with non-zero travel times, by region

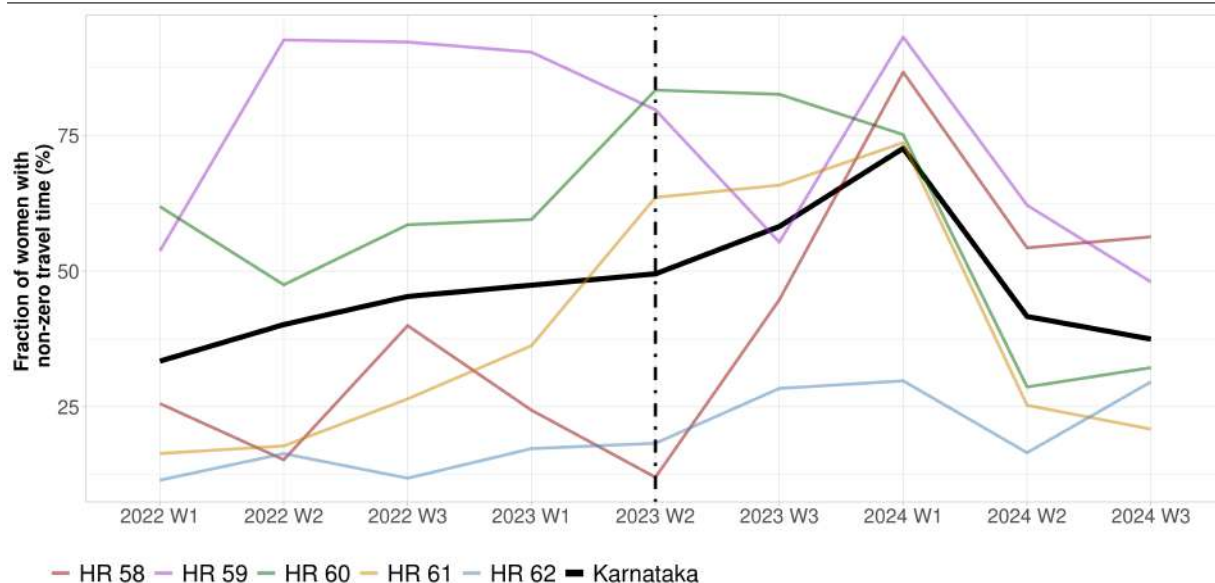


Figure 8.1 shows an increase in fraction of women with non-zero travel times across all HRs except HR 59 (Mysore), immediately after the scheme implementation. While we see an increase in the fraction after scheme implementation, we are observing a moderation post 2024 W1. There are many possible reasons for this - increased congestion in buses has led to private vehicle adoption and/or women have reduced taking leisure trips.

8.3.2 Fraction of women with non-zero travel times in rural vs. urban households

Table 8.2 Fraction of women with non-zero travel times

	Fraction of women with non-zero travel times (%)		
	2022	2023	2024
Rural	34.7	52.3	52.7
Urban	41.0	51.5	49.7
KA*	39.6	51.7	50.6
Rural India	53.3	67.3	71.5
Urban India	53.6	69.4	72.9
India [§]	53.5	68.8	72.4

*All women in Karnataka

[§]All women in India including Karnataka

Table 8.2 shows that the fraction of women with non-zero travel times has increased across rural and urban areas, after scheme implementation. While women in rural areas travelled

less than women in urban areas in 2022, they now travel more than urban women.

Figure 8.2 Fraction of women with non-zero travel times, by Rural vs. Urban (%)

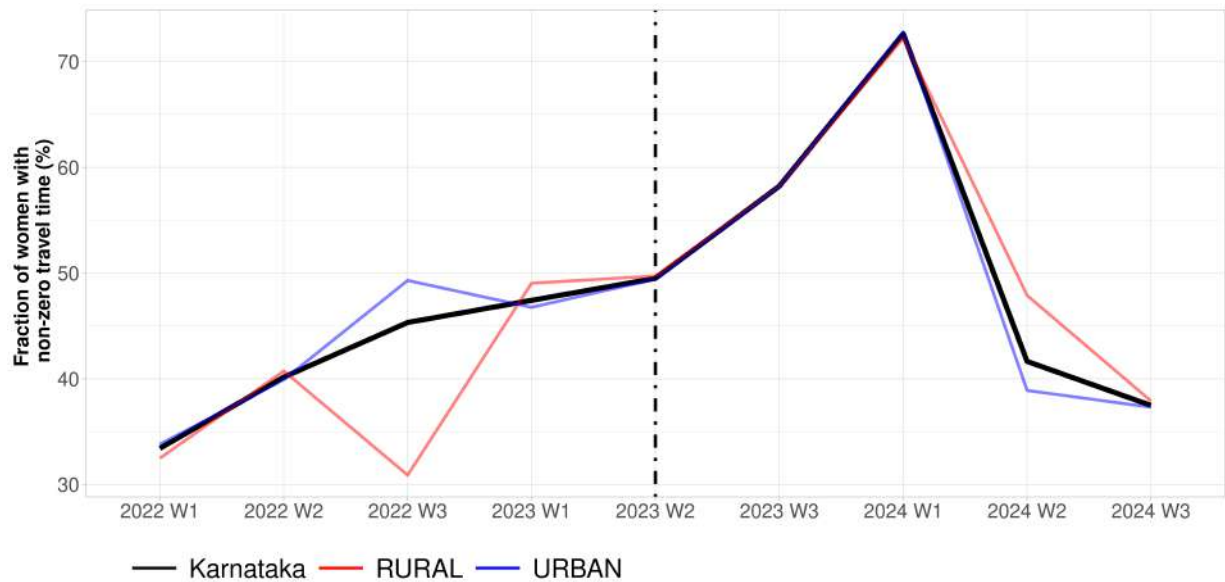


Figure 8.2 also shows that there is a convergence between rural and urban areas immediately after the scheme implementation. We observe a sharp decline post 2024 W1 in both rural and urban areas. This needs to be investigated further using qualitative insights.

8.3.3 Average travel time per day by women, across regions

Table 8.3 Average travel time per day by women, across regions

	Average time spent on travel per day (hrs.)		
	2022	2023	2024
HR 58	0.2	0.2	0.5
HR 59	0.8	0.6	0.4
HR 60	0.5	0.6	0.2
HR 61	0.2	0.4	0.2
HR 62	0.1	0.1	0.1
KA*	0.4	0.4	0.3
India [§]	0.5	0.6	0.6

*All women in Karnataka

[§]All women in India including Karnataka

Table 8.3 indicates that the average time spent on travel per day by women has marginally decreased after the introduction of the scheme. It went from 0.4 hours in the pre-scheme

period (2022) to 0.3 hours in the post-scheme period (2024) for all women in Karnataka. This was contrary to expectation in hypothesis 1 (Section 8.2).

However, we observe varied behaviour across HRs. Time travelled by women increased in Bengaluru HR while falling or remaining constant for other HRs. Mysore HR has had the highest average travel time per day by women through time - further analysis is required to determine why is it falling. Similarly, more micro-data is required to explain travel times in coastal HR.

Figure 8.3 Average travel time per day, by women: Hrs., across HRs

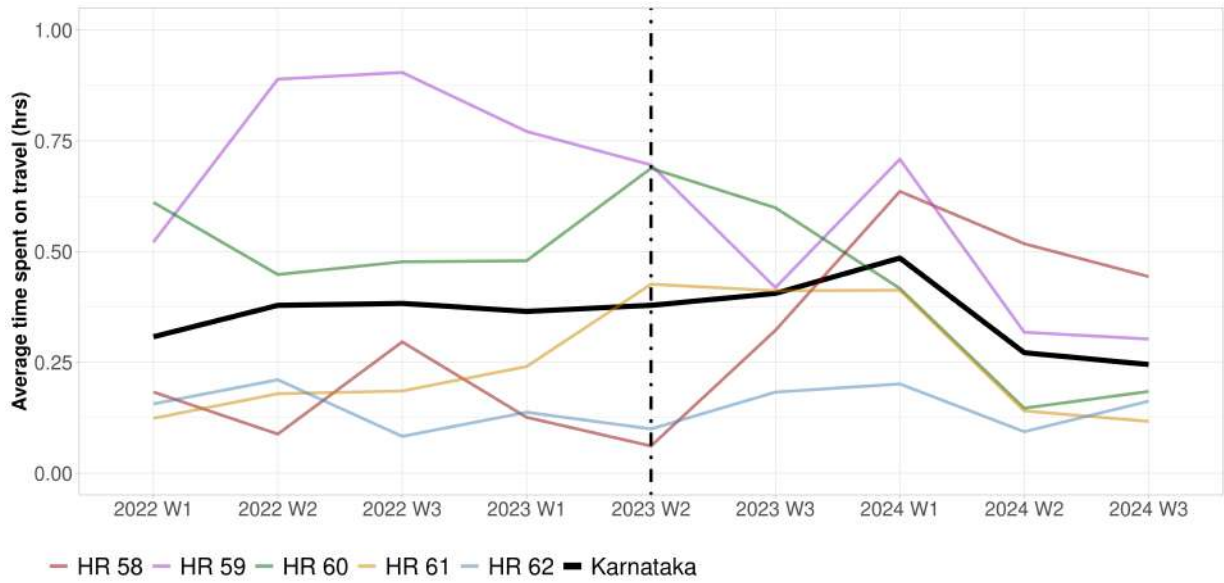


Figure 8.3 shows an increase in average women travel times in Bengaluru HR, immediately after the scheme implementation. We find a reduction in average travel times in all HRs post 2024 W1. This secular decline needs to be observed for a few more waves to draw conclusions.

8.3.4 Average travel time per day by employed women, across regions

Table 8.4 Average travel time per day, by employed women: Hrs.

	Average time spent on travel per day (hrs.)		
	2022	2023	2024
HR 58	0.3	0.3	0.7
HR 59	0.9	0.8	0.5
HR 60	0.8	0.7	0.5
HR 61	0.3	0.4	0.4
HR 62	0.6	0.2	0.3
KA*	0.6	0.5	0.5
India [§]	0.7	0.8	0.7

*All employed women in Karnataka

[§]All employed women in India including Karnataka

Table 8.4 shows that average travel times for employed women is more than all women averages in Karnataka (Table 8.3). Post the scheme, average travel time for employed women has decreased from 0.6 hours to 0.5 hours per day.

However, the trends are similar to the all women sample. There is an increase in the average travel time for employed women in HR 58 (Bengaluru).

Figure 8.4 Average travel time per day, by employed women: Hrs., across HRs

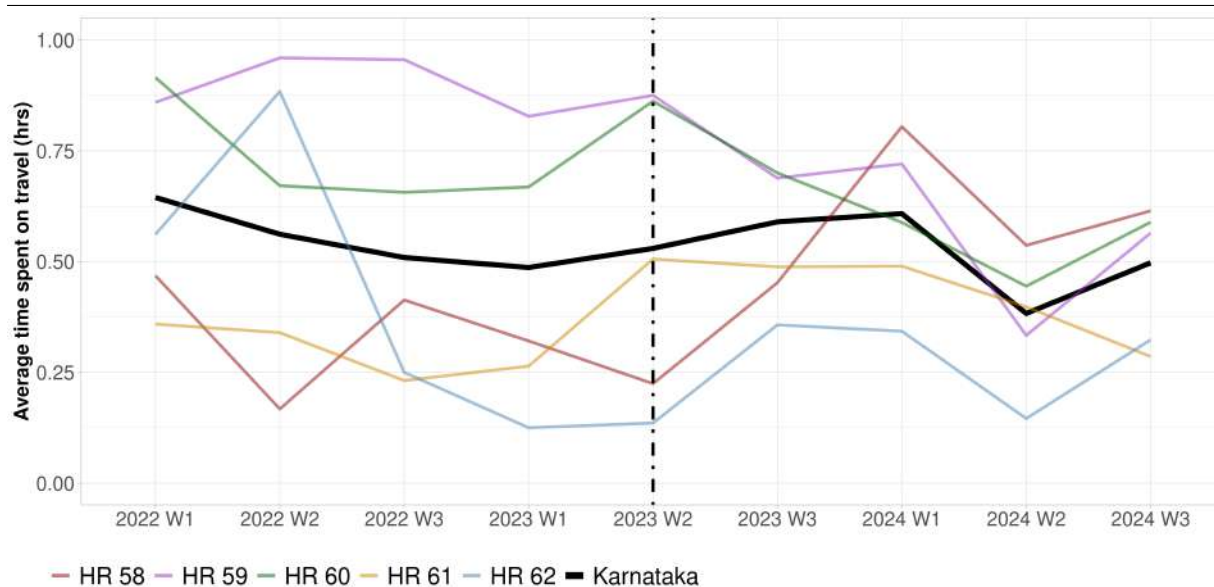


Figure 8.4 substantiates these cross-regional variations in travel time of women within the trends from the pre-scheme period to the post-scheme introduction periods. All HRs report a drop in average travel times for employed women post 2024 W1.

8.3.5 Average travel time per day by women in rural vs. urban regions

Table 8.5 Average travel time per day, by women in rural vs. urban regions

	Average time spent on travel per day (hrs.)		
	2022	2023	2024
Rural	0.3	0.4	0.3
Urban	0.4	0.4	0.3
KA*	0.4	0.4	0.3
Rural India	0.5	0.6	0.6
Urban India	0.5	0.6	0.6
India [§]	0.5	0.6	0.6

*All women in Karnataka

[§]All women in India including Karnataka

Table 8.5 shows that the average time spent on travel by women in rural areas has remained constant at 0.3 hours per day in 2022 and 2024 respectively. For urban areas this has decreased from 0.4 hours per day in 2022 to 0.3 hours in 2024.

Figure 8.5 Average travel time per day, by women, Rural vs. Urban

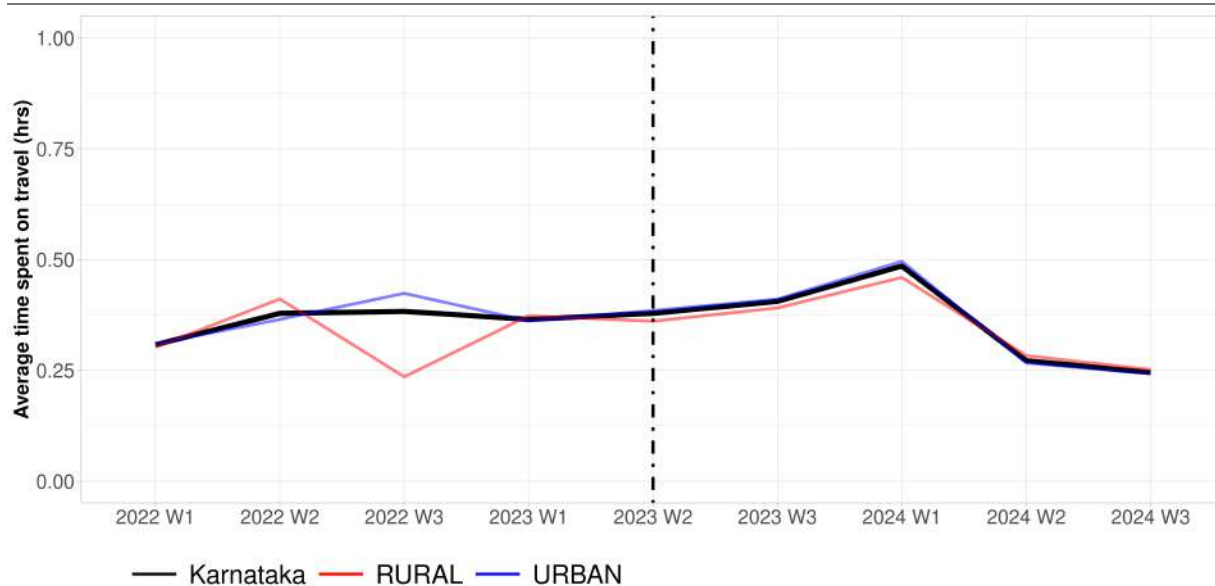


Figure 8.5 shows similar trends and levels in rural and urban areas with slightly more variation in rural areas. We also observe a drop in travel times post 2024 W1.

8.3.6 Average travel time per day, by employed rural vs. urban women

Table 8.6 Average travel time per day, by employed women in rural vs. urban regions

	Average time spent on travel per day (hrs.)		
	2022	2023	2024
Rural	0.5	0.4	0.5
Urban	0.6	0.6	0.5
KA*	0.6	0.5	0.5
Rural India	0.7	0.8	0.8
Urban India	0.7	0.8	0.7
India [§]	0.7	0.8	0.7

* All employed women in Karnataka

[§] All employed women in India including Karnataka

Table 8.6 shows that the average time spent on travel by employed women has remained constant at 0.5 hours per day in rural areas. In urban areas, the average time spent on travel per day by employed women has declined marginally from 0.6 hours to 0.5 hours.

Figure 8.6 Average travel time per day, by employed women: Hrs., Rural vs. Urban

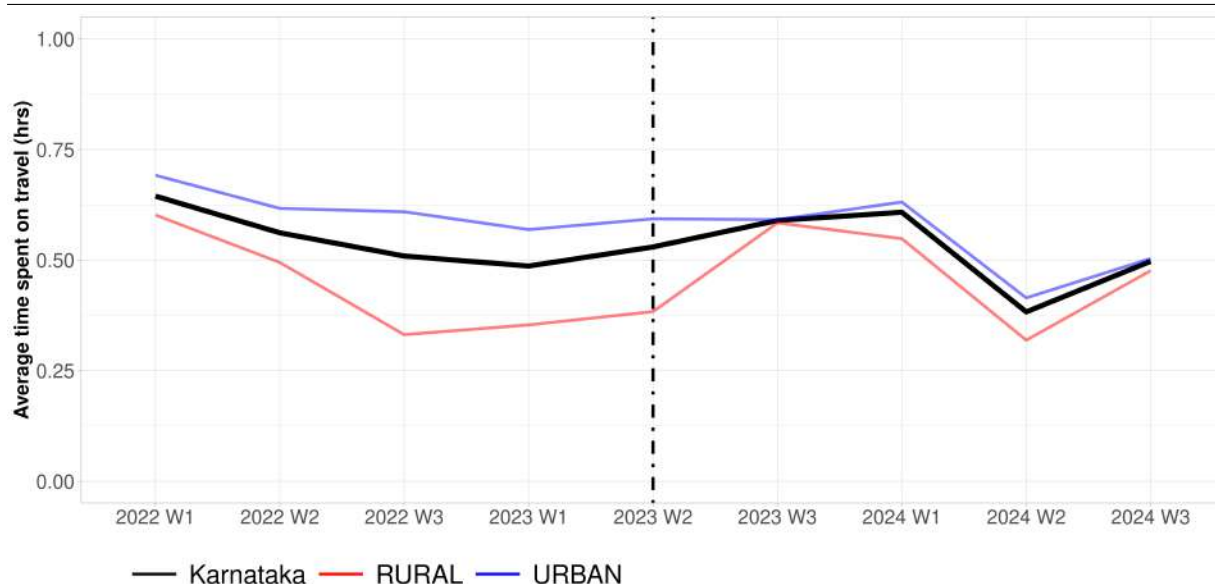


Figure 8.6 shows that there is more variation in the amount of time travelled by rural women. We also observe a decline post 2024 W1 in both rural and urban areas.

8.3.7 Median monthly travel expense (using bus, train and ferry), across regions

Table 8.7 Median monthly household travel expense (bus, train and ferry), across regions

	Median monthly household expenditure (Rs)		
	2022	2023	2024
HR 58	200.0	242.5	68.5
HR 59	200.0	146.0	0.0
HR 60	155.0	215.0	0.0
HR 61	150.0	200.0	155.0
HR 62	141.0	130.0	0.0
Rural	160.0	190.0	0.0
Urban	182.5	200.0	0.0
KA*	175.0	195.0	0.0
Rural India	0.0	0.0	0.0
Urban India	0.0	0.0	0.0
India [§]	0.0	0.0	0.0

*All Karnataka sample

[§]All India sample including Karnataka

Table 8.7 indicates a drop in the median monthly household expenditure on travel by bus, ferry and train, from Rs. 175 in the pre-scheme period to Rs.0 in the post-scheme period for all households in Karnataka. The reduction is similar for both rural and urban areas. This is consistent with hypothesis 2 in Section 8.2, which is that with free travel for women, travel expenditure reduces for the entire household.

However, this is not a symmetric observation for all the regions. The median monthly household expenditure on travel in HRs 61 reports an *increase*.

Figure 8.7 Median monthly household travel expenditure (bus, train and ferry), across HRs

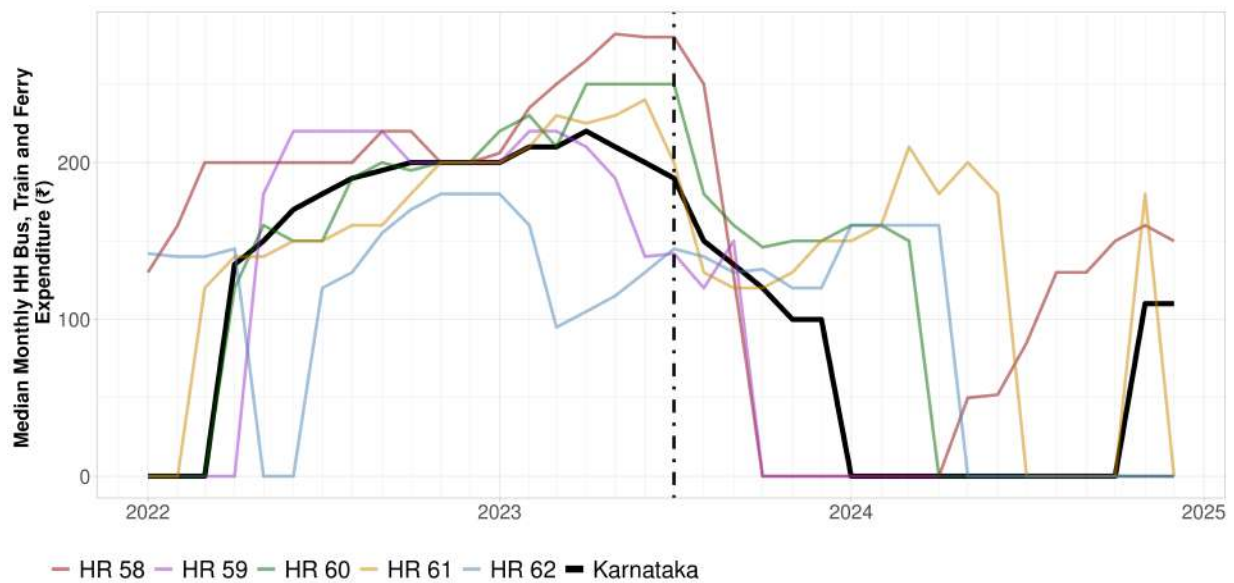


Figure 8.7 shows a decrease in median monthly household expenditure on bus, train and ferry across all HRs, immediately after the scheme implementation.

8.3.8 Median monthly household travel expense (petrol and diesel), across HRs

Table 8.8 Median monthly travel expense (petrol and diesel), across regions

	Median monthly household expenditure (Rs)		
	2022	2023	2024
HR 58	1300.0	1340.0	1500.0
HR 59	1112.5	1200.0	900.0
HR 60	775.0	1000.0	1275.0
HR 61	640.0	955.0	1300.0
HR 62	725.0	960.0	1250.0
Rural	800.0	955.0	1275.0
Urban	1000.0	1065.0	1300.0
KA*	952.5	1015.0	1300.0
Rural India	970.0	1175.0	1255.0
Urban India	1200.0	1351.0	1500.0
India [§]	1200.0	1300.0	1450.0

*All Karnataka sample

[§]All India sample including Karnataka

Table 8.8 shows that the median monthly household expenditure on petrol and diesel increased from Rs.952.5 in the pre-scheme period, to Rs.1300 in the post-scheme period. This increase also took place for households in both the rural and urban areas.

Figure 8.8 Median monthly travel expense (petrol and diesel), across regions

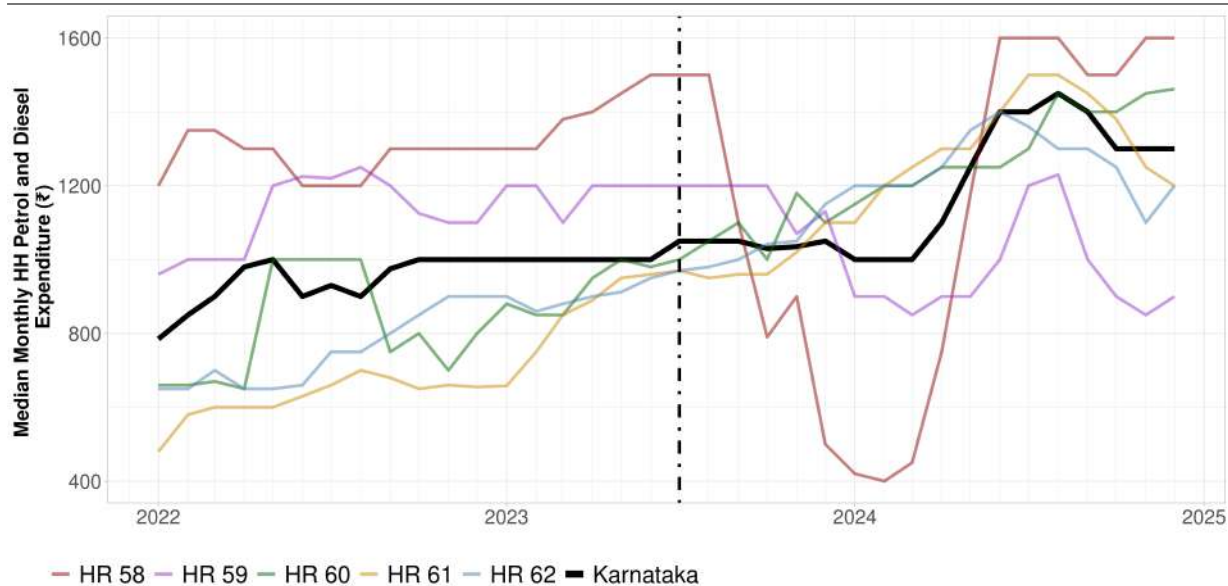


Figure 8.8 shows similar varying patterns across different regions in Karnataka. The median monthly household expenditure on petrol and diesel *decreased* for HRs 59, 61, and 62 while it stabilised for households in Bengaluru and coastal HRs (HRs 58 and 60 respectively), in the latest wave of the data.

An increase in expenditure on petrol and diesel indicates a likely increase in the use of private vehicles for travel. One reason that this could be a response to free public transport travel for women is congestion in the public transport systems. Congestion in public transport could lead to households substituting public transport with private vehicles. Thus, the cross-sectional variation in household expenditure on petrol and diesel could indicate regional variation in sufficient availability of existing public transport and access to this utility.

8.3.9 Labour Force Participation Rate (LFPR) of women, across regions

Table 8.9 Fraction of employable women who are in the Labour Force, cross-region variation

	Fraction of employable women in the Labour Force (%)		
	2022	2023	2024
HR 58	9.8	7.1	6.2
HR 59	7.7	6.5	5.2
HR 60	6.5	5.9	5.6
HR 61	3.8	4.3	3.7
HR 62	2.7	3.7	3.4
KA*	6.7	5.8	4.9
India [§]	7.8	9.4	11.2

*All Karnataka sample
[§]All India including Karnataka

Table 8.9 indicates that FLFPR (Female Labour Force Participation Rate) has gone down at the state level post-schemes from 6.7% to 4.9%. This was contrary to expectation in hypothesis 2 (Section 8.2). However, there is some HR level variation. The southern and coastal HRs have seen a similar decline, but the northern HRs of 61 and 62 have remained stable and slightly increased respectively.

Figure 8.9 Fraction of employable women who are in the Labour Force, cross-region variation

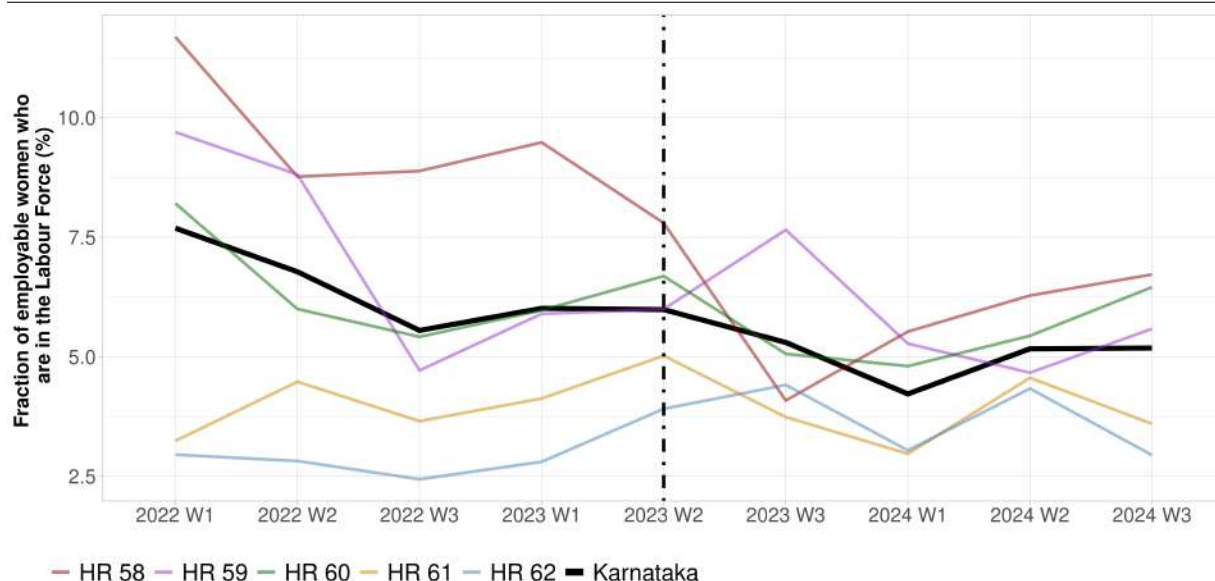


Figure 8.9 shows a slight dip in FLPR after scheme implementation, but this slowly picks back up through 2024. This trend is similar for most HRs except Bengaluru (HR 58) which saw an increase post-scheme that has continued through 2024.

8.3.10 Labour Force Participation Rate (LFPR) of rural vs. urban women

Table 8.10 Fraction of employable women who are in the Labour Force, urban vs. rural

	Fraction of employable women in the Labour Force (%)		
	2022	2023	2024
Rural	10.7	6.2	3.9
Urban	5.1	5.6	5.2
KA*	6.7	5.8	4.9
Rural India	10.6	12.1	13.4
Urban India	6.6	8.2	10.1
India [§]	7.8	9.4	11.2

*All Karnataka sample

[§]All India including Karnataka

Table 8.10 shows that LFPR has fallen sharply for rural women from 10.7% in 2022 to 3.9% in 2024. Urban LFPR has largely stayed stable, and close to the state level average.

Figure 8.10 Fraction of employable women who are in the Labour Force, urban vs. rural

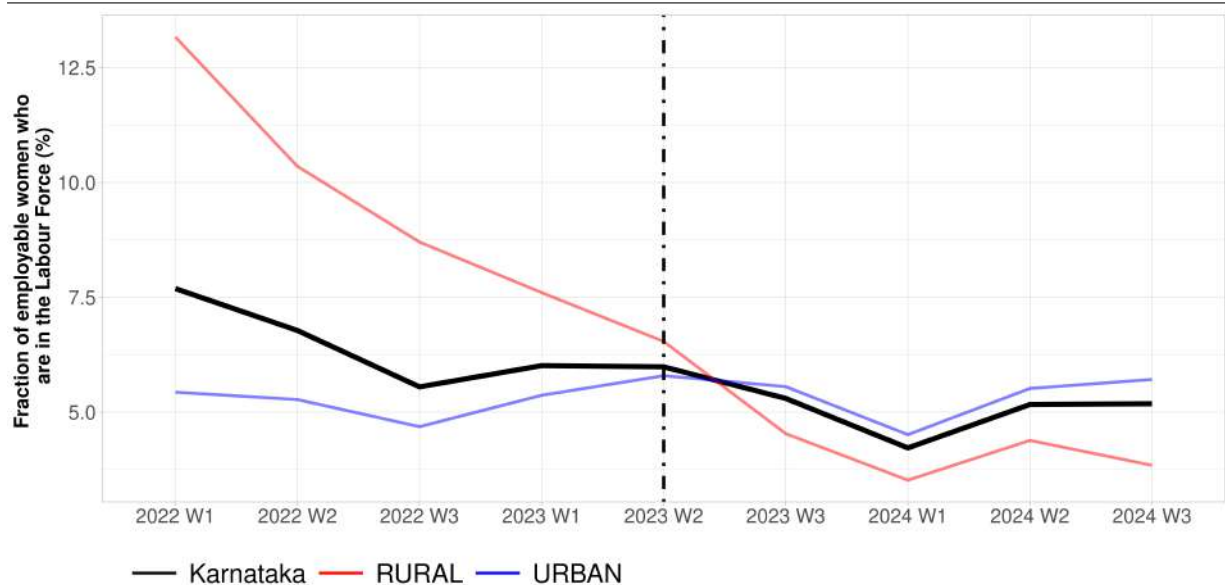


Figure 8.10 shows that the decreasing trend in rural LFPR started long before scheme implementation. Post the schemes this fall has stabilised more and the gap between rural and urban LFPR has greatly reduced.

8.3.11 Counterfactual 1 – average travel time per day of women vs. men

The approach used to attribute an observed change in response to any externally imposed intervention is to identify a ‘counterfactual’ – the response of those who are not targeted by the intervention. For example, in the case of Shakti as the externally imposed intervention, the targeted response is the travel time of women who are the target of the intervention. A counterfactual may be the travel time of *men* who are not the target of the intervention. If there is a change in the women’s travel patterns, while none is observed in the same for men, the change may be attributable to the scheme.

Table 8.11 Average travel time per day for women vs. men

	Average time spent on travel per day (hrs.)		
	2022	2023	2024
Men	0.6	0.6	0.6
Women	0.4	0.4	0.3
KA*	0.5	0.5	0.4

*All members in Karnataka

Table 8.11 shows that women spent less time travelling than men, in the pre-scheme period (2022). The average time spent on travel per day by women was 0.4 hours vs. 0.6 hours by men in 2022. This is expected since men are likely to travel further and longer for work related reasons, while women are likely to not do so.

The table also shows that the average time spent on travel per day in the post-scheme period has remained constant for men while marginally declining for women. The reasons for this observation need to be studied further.

Figure 8.11 Patterns in average travel time per day for women vs. men

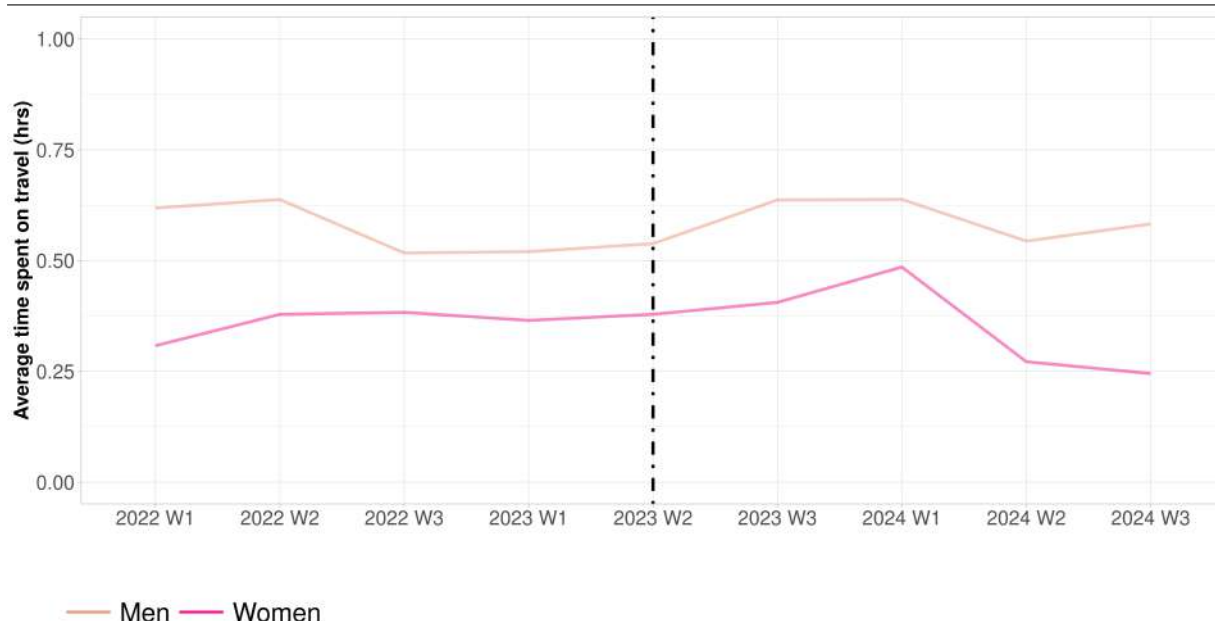


Figure 8.11 indicates an increasing trend for an increase in both men and women, immediately after scheme implementation. Post 2024 W1, we observe a decline, but at a sharper rate for women.

8.3.12 Counterfactual 1 – average travel time per day, employed women vs. employed men

Table 8.12 Average time travelled per day by employed women and employed men

	Average time spent on travel per day (hrs.)		
	2022	2023	2024
Men	0.6	0.6	0.6
Women	0.6	0.5	0.5
KA*	0.6	0.6	0.6

*Employed people in Karnataka

Table 8.12 indicates that employed women and employed men travel similar amounts of time per day on average. From Figure 8.12, there is an increasing trend in average travel time per day for employed men and women in the immediate aftermath of the scheme implementation. Post 2024 W1, we observe a decline in average travel times, albeit with a sharper rate for employed women.

Figure 8.12 Patterns in average travel time per day for employed women vs. employed men

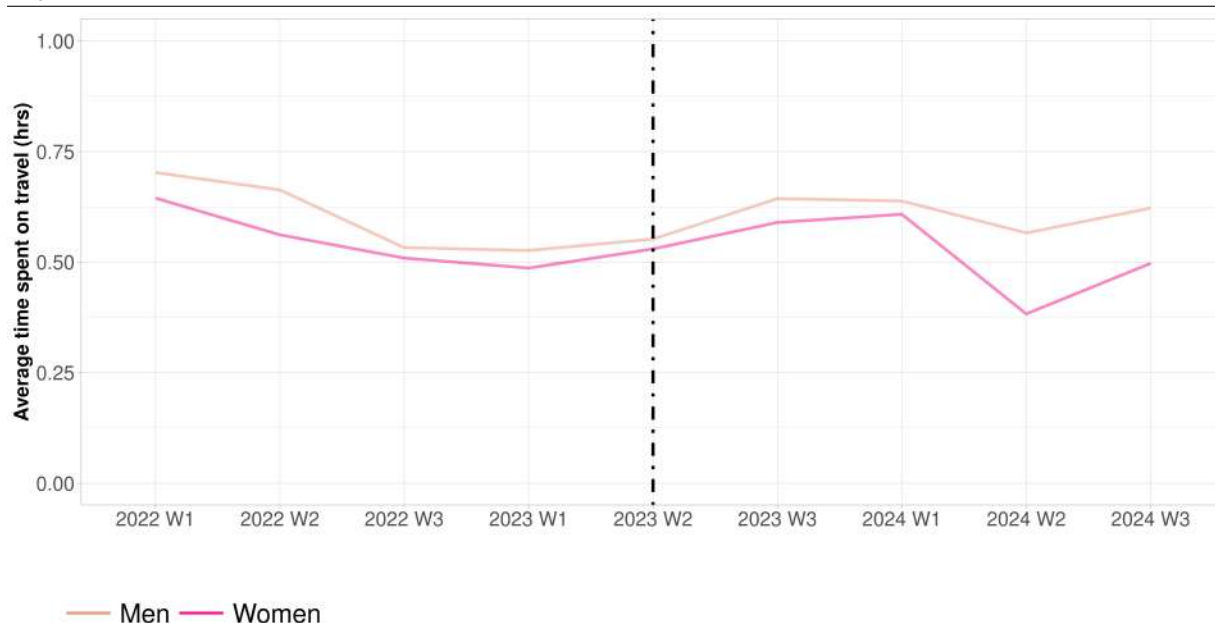


Figure 8.12 shows that while there were increased travel times for employed women immediately in the post scheme period, these gains seem to have moderated over time.

8.3.13 Counterfactual 2 – Labour Force Participation Rate (LFPR), women vs. men

Table 8.13 Fraction of employable individuals in the labour force, women and men

	Fraction of employable individuals in the Labour Force (%)		
	2022	2023	2024
Men	63.8	63.7	65.2
Women	6.7	5.8	4.9
KA*	37.5	37.3	37.6

*All members in Karnataka

Table 8.13 shows that LFPR for women is consistently lower than LFPR for men. Male LFPR has slightly gone up from 63.8% in the pre-scheme period to 65.2%, while female LFPR has dropped from 6.7% to 4.9%.

While the gap between male and female LFPR is expected, we would have hoped to see a rise in FLFPR post-schemes.

Figure 8.13 Patterns in LFPR for women vs. men

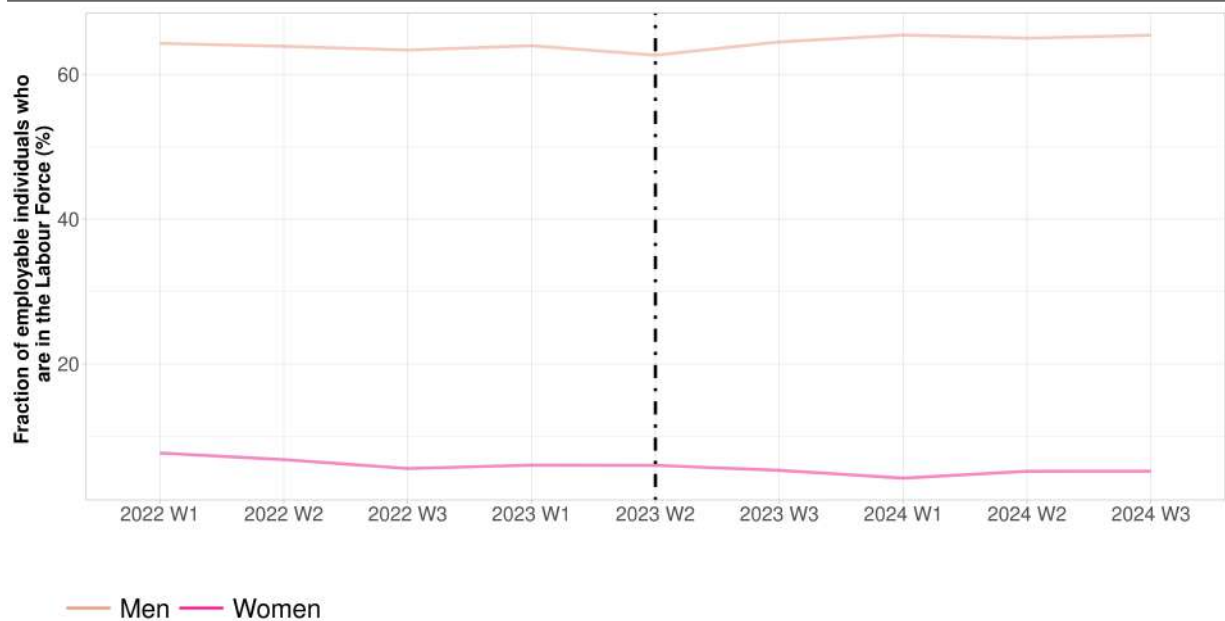


Figure 8.13 indicates an continuously stable trend for both men and women, both before and after scheme implementation. Post 2023 W2, there is a slight decline for women and slight increase for men.

8.4 Summary with key findings

The Shakti scheme provides free bus rides to domicile women within Karnataka. We observed the following responses to the scheme:

1. There is an increase in first-time travellers, measured as an increase in the fraction of women with non-zero travel times, after the introduction of the scheme. 50.6% women in Karnataka had some travel in 2024 as compared to only 39.6% in 2022 (Table 8.1, Figure 8.1).

We also see that the average time spent on travel per day by women decreased moderately from 0.4 hours in 2022 to 0.3 hours in 2024 (Table 8.3, Figure 8.3). This could be happening because women often take shorter but more frequent trips instead of long trips. The increase in fraction of women travelling suggests that new women are being encouraged to travel - for work/leisure/education.

2. The highest increases in travel time are reported by women in the Bengaluru HR. This could be due to higher availability of buses in the public transport utility. This increase is observed in both rural and urban households, with rural travel times nearly converging with urban travel times (Table 8.5, Figure 8.5).
3. We observe near consistent travel times for employed women, especially rural employed women (Tables 8.4, 8.6, Figures 8.4, 8.6). While the scheme is intended towards women, we see stable average travel times of men around the scheme as well (Table 8.11, Figure 8.11). Both employed women and employed men have similar average travel times (Table 8.12, Figure 8.12).
4. There is a drop in median monthly travel expenditure on bus, train and ferry after the scheme implementation, as expected in hypothesis 2 (Table 8.7, Figure 8.7). But there is an increase in the median monthly travel expenditure on petrol and diesel (Table 8.8, Figure 8.8).

In both cases, there is significant and large variation in the reported expenditure on both these items. These have changed differently for households in different HRs. The drivers for these changes require further study, with possibly different sources of data.

Chapter 9

The Yuvanidhi scheme

The Yuvanidhi scheme was announced in July 2023, and disbursement started in January 2024. The scheme is for the unemployed youth of Karnataka. The scheme provides an unemployment allowance of Rs 3000 per month for graduates and Rs 1500 per month for diploma holders of the state.

9.1 Scheme design

Scheme eligibility - Inclusion criteria:

1. 2023 degree holders (first cohort) who are unemployed for 180 days or more upon completion of the degree.
2. Graduates or diploma holders.
3. Domicile students (who have studied in Karnataka for at least 6 years).

Scheme eligibility - Exclusion criteria:

1. Employed youth
2. Youth receiving apprenticeship grants
3. Youth enrolled for higher education
4. Self-employed youth (have existing business loans through banks/government)

Benefits to eligible individuals:

- Unemployment allowance of Rs 3000 per month for graduates.
- Unemployment allowance of Rs 1500 per month for diploma holders.
- The allowance is paid till the candidate gets employment or for a maximum period of 2 years.
- Unemployment declaration needs to be submitted by the candidate and independent verification of unemployment status is done each month.

Mode of delivery: direct benefit transfer into the bank account of the beneficiary.

9.2 Measures to assess responses to the scheme

We expect the following responses of the individuals and households to the scheme:

Hypothesis 1: Since the scheme is intended towards graduates/diploma holders, the fraction of households with a diploma or graduate degree holder who get non-zero cash transfers from government is likely to *increase* after the scheme implementation.

Hypothesis 2: Since the scheme is providing financial support to graduates/diploma holders, the incentives for Karnataka youth to find work will become lower, potentially *reducing* labour force participation rates.

Educational characteristics are observed at the individual level per wave. The income variables are observed at the household level on a monthly-basis.

9.3 Yuvanidhi assessment

The actual disbursement for Yuvanidhi scheme started in January 2024 (6 months after 2023 graduates passed out). Since the period of assessment is up to Wave 2 of 2024 in the CPHS dataset, there is limited data to evaluate the scheme’s impact. The dataset does not have identifiers for household members who are 2023 graduates, but there are variables for members who hold a diploma or a graduate degree. These are used as a proxy to identify potential beneficiaries of the scheme.

9.3.1 Fraction of households with a diploma or graduate degree, across regions

Table 9.1 Fraction of households with at least one diploma or graduate degree, across regions

	Fraction of households with at least one diploma-holder or above		
	2022	2023	2024
HR 58	34.6	34.8	36.0
HR 59	30.1	31.5	31.2
HR 60	28.4	29.0	34.6
HR 61	21.2	21.9	27.3
HR 62	20.3	23.7	27.2
Rural	14.9	15.2	19.0
Urban	32.4	32.8	36.2
KA*	27.5	27.9	31.0
Rural India	14.2	14.7	16.0
Urban India	33.4	33.6	37.5
India [§]	27.7	27.8	29.9

*All Karnataka sample

[§]All India sample including Karnataka

Table 9.1 shows that 27.5% of households in Karnataka have at least one member with a diploma or graduate degree in 2022. This fraction has increased to 31% in 2024. Urban areas have almost double the fraction of households (36.2%) with at least one diploma or higher degree holder compared to rural areas (19%) in 2024. The table also shows that HR 58 (Bengaluru region) and HR 60 (the coastal region) have the highest proportion of diploma or graduate degree holder families. HR 61 and HR 62 (the Dharwad and Bellary regions) have the lowest proportion of diploma or graduate degree holder families.

Figure 9.1 Patterns in the fraction of households with a diploma or graduate degree, across HRs

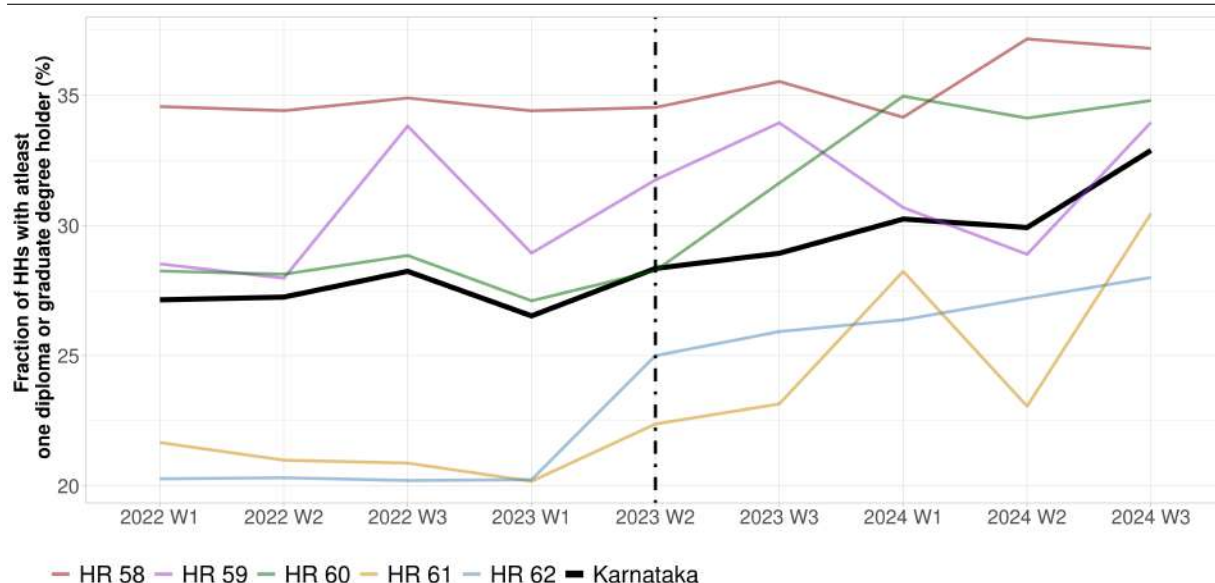


Figure 9.1 validates the observation of cross-regional variation in the fraction of families holding degrees. But it shows that there is a steady increase in the fraction of degree holder households in HR 61 and 62 as well.

9.3.2 Fraction of diploma or graduate degree households receiving government transfers, across regions

Table 9.2 Fraction of households with graduates/diplomas receiving government transfers, across regions

	Fraction of households with some cash transfers from government (Rs.)		
	2022	2023	2024
HR 58	5.7	42.5	64.5
HR 59	7.5	34.5	61.9
HR 60	6.7	40.1	71.1
HR 61	2.5	40.4	63.1
HR 62	7.7	28.7	50.1
KA*	6.2	37.1	63.7
India [§]	21.9	24.5	32.8

*All households with diploma/graduates in Karnataka

[§]All households with diploma/graduates in India including Karnataka

Table 9.2 show that government transfers to households with diploma or graduate-degree holders increased. This went from 6.2% in the pre-scheme period (2022) to 63.7% in the post-scheme period (2024) for the households in the overall Karnataka sample. This fraction includes cash transfers from *all* sources including the Yuvanidhi scheme. While we cannot dis-aggregate Yuvanidhi specific cash transfer, the data shows that households with highly qualified members also receive benefits from the government.

Figure 9.2 Fraction of households with graduates/diplomas with non-zero cash transfers from government: %, across HRs

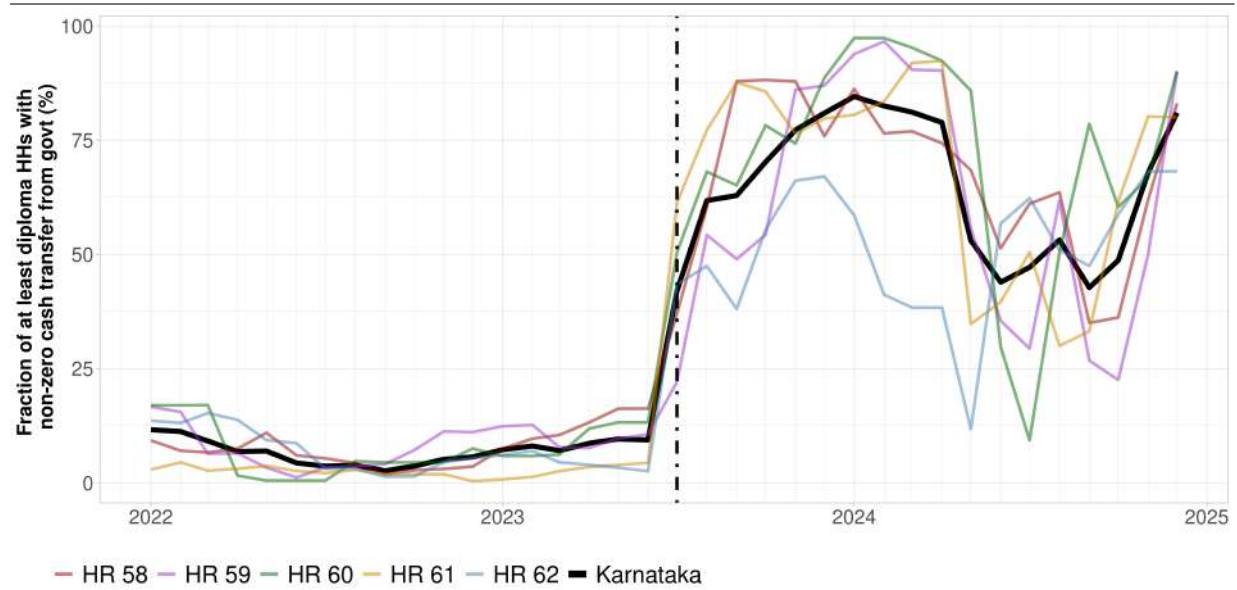


Figure 9.2 shows that there is an increase in government transfers to households with diploma or graduate degree holders, immediately after the scheme implementation, although this cannot be attributed to the Yuvanidhi scheme. We also observe a sharp decline in the fraction post May 2024 followed by a recovery from September 2024.

9.3.3 Fraction of rural vs. urban households with a diploma or graduate degree receiving government transfers

Table 9.3 Fraction of rural vs. urban households with graduates/diplomas, receiving government transfers

	Fraction of households with some cash transfers from government (Rs.)		
	2022	2023	2024
Rural	11.3	42.9	71.5
Urban	5.5	36.1	62.1
KA*	6.2	37.1	63.7
Rural India	25.4	26.3	39.3
Urban India	21.3	24.2	31.4
India [§]	21.9	24.5	32.8

* All households with diploma/graduates in Karnataka

[§] All households with diploma/graduates in India including Karnataka

Table 9.3 finds that the fraction of households with diploma or graduate degree holders receiving government transfers is slightly higher in rural areas (~72%), compared to urban areas (~62%) in the post-scheme period.

Figure 9.3 Fraction of households with graduates/diplomas with non-zero cash transfers from government: %, Rural vs. Urban

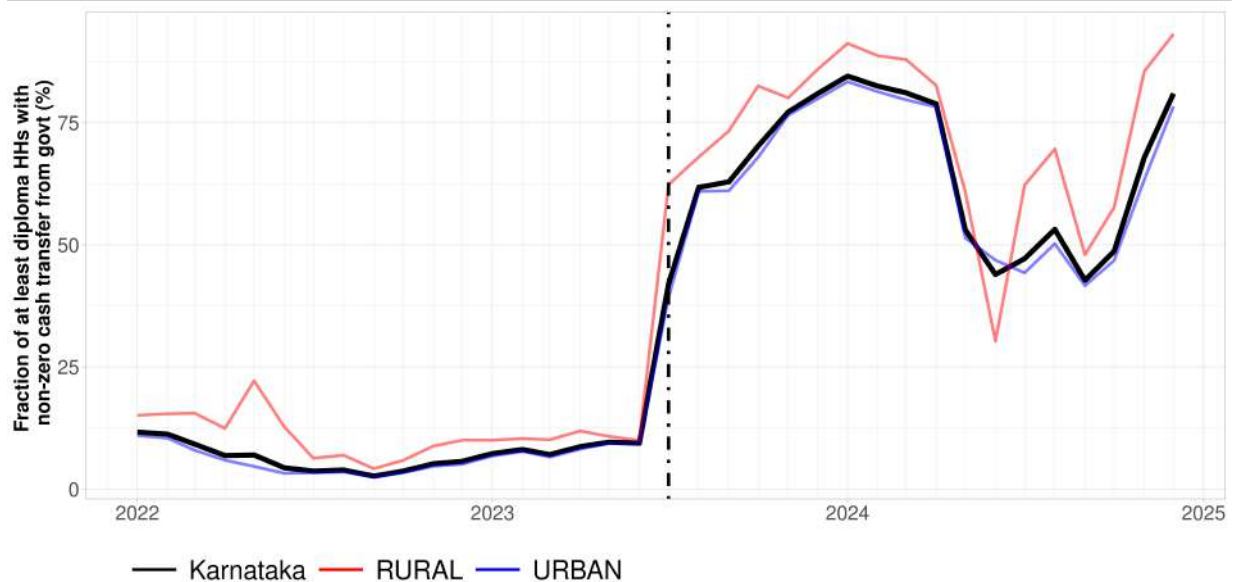


Figure 9.3 also points to a rise, after July 2023. This implies that the increase may be attributed to other schemes, such as like Gruhalakshmi.

9.3.4 Labour Force Participation Rate (LFPR) of graduates (diploma +), across regions

Table 9.4 Fraction of employable graduates who are in the Labour Force, cross-region variation

	Fraction of employable graduates in the Labour Force (%)		
	2022	2023	2024
HR 58	63.4	62.9	67.6
HR 59	72.2	70.7	71.3
HR 60	61.3	59.2	54.0
HR 61	67.4	66.7	58.5
HR 62	65.1	63.8	61.6
KA*	65.9	65.0	63.6
India [§]	60.5	61.3	61.0

* All Karnataka sample

[§] All India including Karnataka

Table 9.4 shows that the labour force participation rate of diploma or graduate-degree holders slightly decreased. This went from 65.9% in the pre-scheme period (2022) to 63.6% in the post-scheme period (2024) for individuals in the overall Karnataka sample. However there is some HR variation, with HR 58 (Bengaluru) increasing from 63.4% in 2022 to 67.6% in 2024.

Figure 9.4 Fraction of employable graduates who are in the Labour Force, cross-region variation

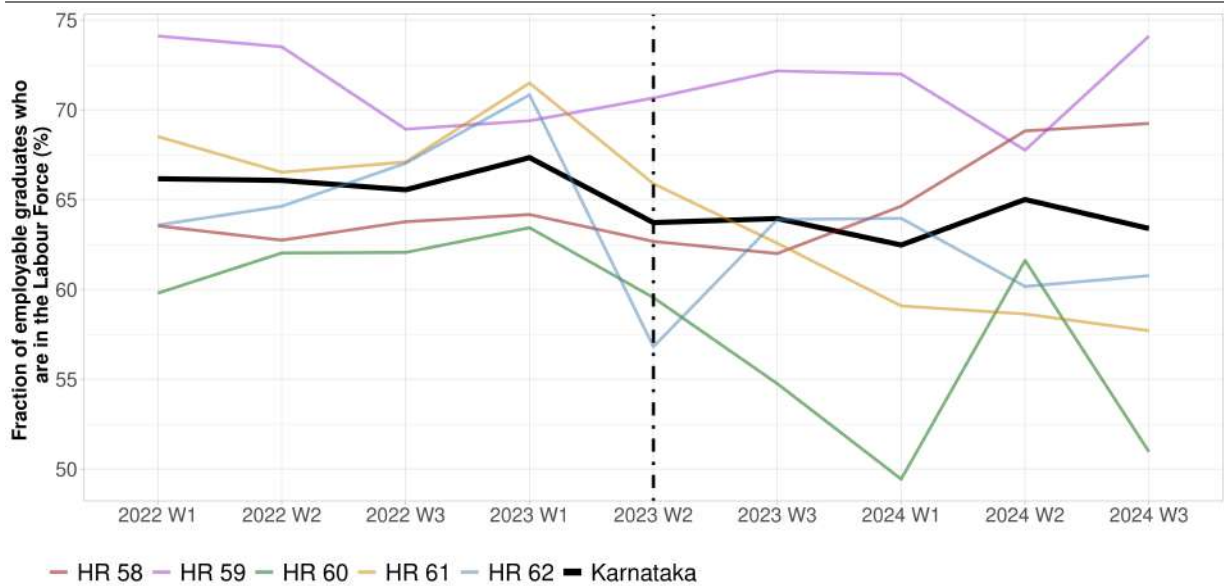


Figure 9.2 shows that LFPR for diploma or graduate-degree holders largely stayed stable at the state level. At the HR level there is more variation, with sudden upward/downward spikes for HRs 59 and 60 especially in 2024. As the state level LFPR is largely stable, it is difficult to attribute any changes to the Yuvanidhi scheme yet.

9.3.5 Labour Force Participation Rate (LFPR) of rural vs. urban graduates (diploma +)

Table 9.5 Fraction of employable graduates who are in the Labour Force, urban vs. rural

	Fraction of employable graduates in the Labour Force (%)		
	2022	2023	2024
Rural	63.1	58.3	55.1
Urban	66.3	66.0	65.3
KA*	65.9	65.0	63.6
Rural India	64.8	65.5	64.5
Urban India	59.9	60.6	60.3
India [§]	60.5	61.3	61.0

*All Karnataka sample

[§]All India including Karnataka

Table 9.5 finds that LFPR for all Karnataka diploma or graduate degree holders reduced very slightly from 2022 (~65.9%) to 2024 (~63.6%). While the urban figures match this pattern, rural has dropped sharply from 63.1% to 55.1%.

Figure 9.5 Fraction of employable graduates who are in the Labour Force, urban vs. rural

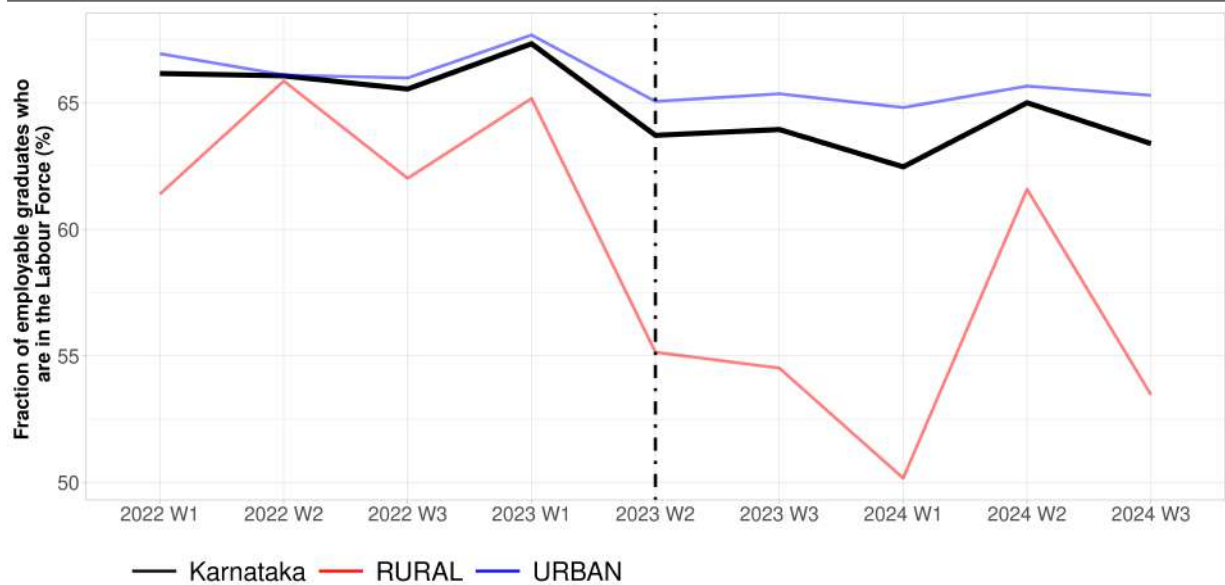


Figure 9.5 shows higher LFPR volatility for rural diploma or graduate degree holders when compared to urban/state level. While there is a large spike in 2024 W2 that brings it close the overall state level, there is a sharp fall again in 2024 W3.

9.4 Summary with key findings

The Yuvanidhi scheme was announced in July 2023 and disbursement started in January 2024. For the first cohort (2023 degree holders), the scheme provides an unemployment allowance of Rs 3000 per month for graduates and Rs 1500 per month for diploma holders of the state. Our main observations, made within the caveat that the disbursement started only in January 2024, and that limited data is available to assess the scheme’s effect, are:

1. CPHS data until December 2024 shows that there are 31% households in Karnataka who have at least one diploma/graduate degree holder.

This fraction varies across HRs with northern HRs lagging, but we see an increasing trend over the years. Urban households are approximately twice as likely to have at least one diploma/graduate degree holder compared to rural households.

2. In 2024, 63.7% of the households with diploma/graduates received some cash transfers from the government, although this cannot be attributed to the Yuvanidhi scheme.
3. The labour force participation rate in Karnataka for diploma and graduate degree holders is seeing a slight decline from the pre-scheme period. This trend varies across HRs and rural-urban regions, with rural especially deviating much lower than the state average.